Chapter 4
Intelligent Storage Systems

KEY CONCEPTS

- Intelligent Storage Systems
- Cache Mirroring and Vaulting
- Logical Unit Number
- LUN Masking
- Meta LUN
- Virtual Storage Provisioning
- High-End Storage Systems
- Midrange Storage Systems

Business-critical applications require high levels of performance, availability, security, and scalability. A disk drive is a core element of storage that governs the performance of any storage system. Some of the older disk-array technologies could not overcome performance constraints due to the limitations of disk drives and their mechanical components. RAID technology made an important contribution to enhancing storage performance and reliability, but disk drives, even with a RAID implementation, could not meet the performance requirements of today's applications.

With advancements in technology, a new breed of storage solutions, known as *intelligent storage systems*, has evolved. These intelligent storage systems are feature-rich RAID arrays that provide highly optimized I/O processing capabilities. These storage systems are configured with a large amount of memory (called *cache*) and multiple I/O paths and use sophisticated algorithms to meet the requirements of performance-sensitive applications. These arrays have an operating environment that intelligently and optimally handles the management, allocation, and utilization of storage resources. Support for flash drives and other modern-day technologies, such as virtual storage provisioning and automated storage tiering, has added a new dimension to storage system performance, scalability, and availability.

This chapter covers components of intelligent storage systems along with storage provisioning to applications.

4.1 Components of an Intelligent Storage System

An intelligent storage system consists of four key components: *front end*, *cache*, *back end*, and *physical disks*. Figure 4-1 illustrates these components and their interconnections. An I/O request received from the host at the front-end port is processed through cache and back end, to enable storage and retrieval of data from the physical disk. A read request can be serviced directly from cache if the requested data is found in the cache. In modern intelligent storage systems, front end, cache, and back end are typically integrated on a single board (referred to as a *storage processor* or *storage controller*).

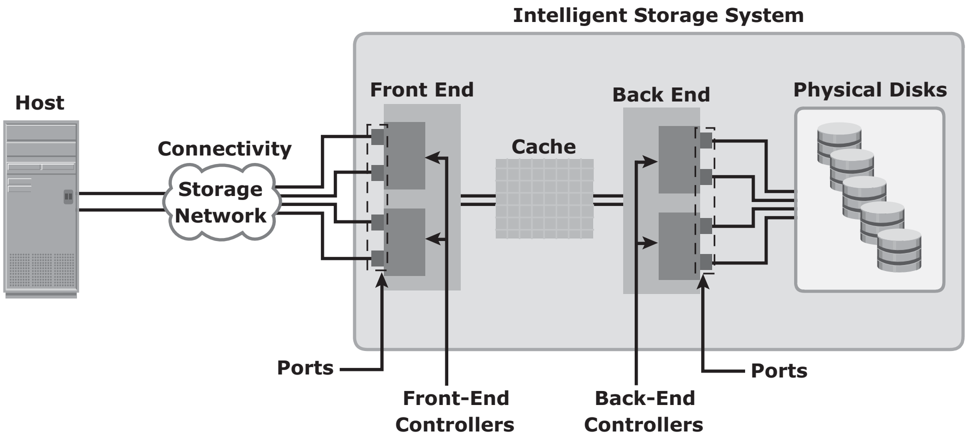


Figure 4-1: Components of an intelligent storage system

4.1.1 Front End

The front end provides the interface between the storage system and the host. It consists of two components: front-end ports and front-end controllers. Typically, a front end has redundant controllers for high availability, and each controller contains multiple ports that enable large numbers of hosts to connect to the intelligent storage system. Each front-end controller has processing logic that executes the appropriate transport protocol, such as Fibre Channel, iSCSI, FICON, or FCoE for storage connections.

Front-end controllers route data to and from cache via the internal data bus. When the cache receives the write data, the controller sends an acknowledgment message back to the host.

4.1.2 Cache

Cache is semiconductor memory where data is placed temporarily to reduce the time required to service I/O requests from the host.

Cache improves storage system performance by isolating hosts from the mechanical delays associated with rotating disks or hard disk drives (HDD). Rotating disks are the slowest component of an intelligent storage system. Data access on rotating disks usually takes several millisecond because of seek time and rotational latency. Accessing data from cache is fast and typically takes less than a millisecond. On intelligent arrays, write data is first placed in cache and then written to disk.

Structure of Cache

Cache is organized into pages, which is the smallest unit of cache allocation. The size of a cache page is configured according to the application I/O size. Cache consists of the *data store* and *tag RAM*. The data store holds the data whereas the tag RAM tracks the location of the data in the data store (see Figure 4-2) and in the disk.

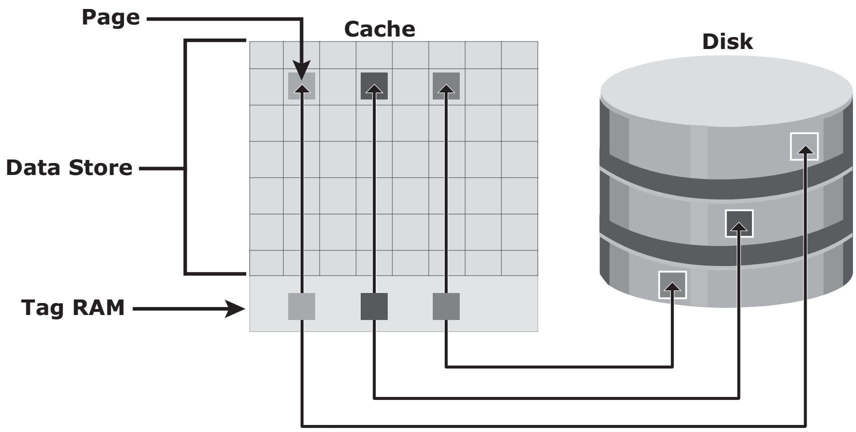


Figure 4-2: Structure of cache

Entries in tag RAM indicate where data is found in cache and where the data belongs on the disk. Tag RAM includes a *dirty bit* flag, which indicates whether the data in cache has been committed to the disk. It also contains time-based information, such as the time of last access, which is used to identify cached information that has not been accessed for a long period and may be freed up.

Read Operation with Cache

When a host issues a read request, the storage controller reads the tag RAM to determine whether the required data is available in cache. If the requested data is found in the cache, it is called a *read cache hit* or *read hit* and data is sent directly to the host, without any disk operation (see Figure 4-3 [a]). This provides a fast response time to the host (about a millisecond). If the requested data is not found in cache, it is called a *cache miss* and the data must be read from the disk (see Figure 4-3 [b]). The back end accesses the appropriate disk and retrieves the requested data. Data is then placed in cache and finally sent to the host through the front end. Cache misses increase the I/O response time.

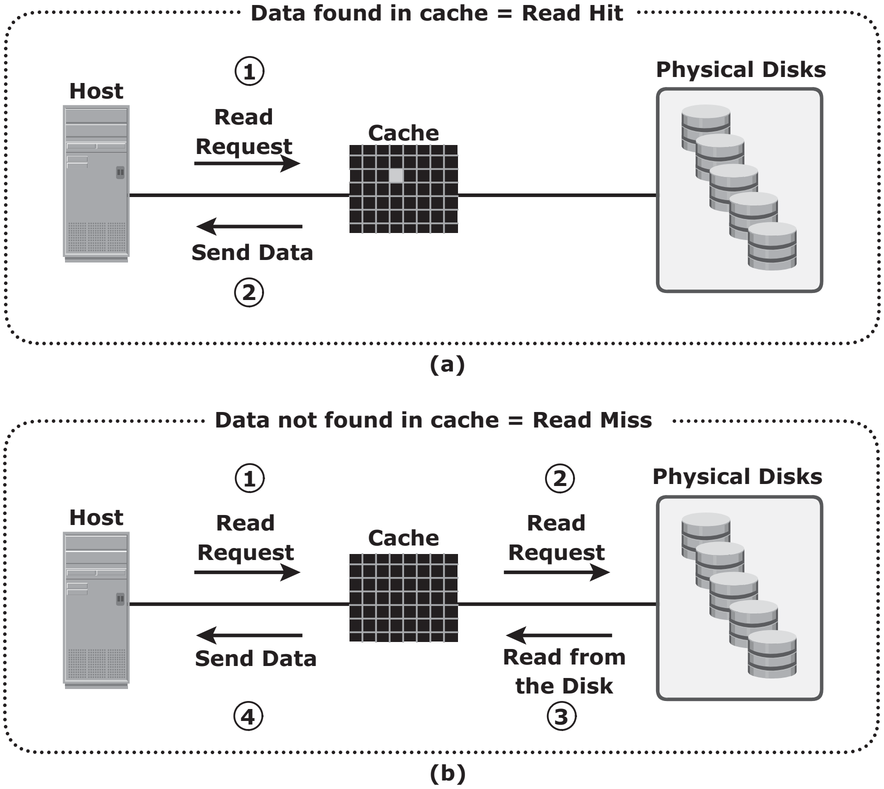


Figure 4-3: Read hit and read miss

A *prefetch* or *read-ahead* algorithm is used when read requests are sequential. In a sequential read request, a contiguous set of associated blocks is retrieved. Several other blocks that have not yet been requested by the host can be read from the disk and placed into cache in advance. When the host subsequently requests these blocks, the read operations will be read hits. This process significantly improves the response time experienced by the host. The intelligent storage system offers fixed and variable prefetch sizes. In *fixed prefetch*, the intelligent storage system prefetches a fixed amount of data. It is most suitable when host I/O sizes are uniform. In *variable prefetch*, the storage system prefetches an amount of data in multiples of the size of the host request. *Maximum prefetch* limits the number of data blocks that can be prefetched to prevent the disks from being rendered busy with prefetch at the expense of other I/Os.

Read performance is measured in terms of the *read hit ratio*, or the *hit rate*, usually expressed as a percentage. This ratio is the number of read hits with respect to the total number of read requests. A higher read hit ratio improves the read performance.

Write Operation with Cache

Write operations with cache provide performance advantages over writing directly to disks. When an I/O is written to cache and acknowledged, it is completed in far less time (from the host's perspective) than it would take to write directly to disk. Sequential writes also offer opportunities for optimization because many smaller writes can be coalesced for larger transfers to disk drives with the use of cache.

A write operation with cache is implemented in the following ways:

- **Write-back cache:** Data is placed in cache and an acknowledgment is sent to the host immediately. Later, data from several writes are committed (de-staged) to the disk. Write response times are much faster because the write operations are isolated from the mechanical delays of the disk. However, uncommitted data is at risk of loss if cache failures occur.
- **Write-through cache:** Data is placed in the cache and immediately written to the disk, and an acknowledgment is sent to the host. Because data is committed to disk as it arrives, the risks of data loss are low, but the write-response time is longer because of the disk operations.

Cache can be bypassed under certain conditions, such as large size write I/O. In this implementation, if the size of an I/O request exceeds the predefined size, called *write aside size*, writes are sent to the disk directly to reduce the impact of large writes consuming a large cache space. This is particularly useful in an environment where cache resources are constrained and cache is required for small random I/Os.

Cache Implementation

Cache can be implemented as either dedicated cache or global cache. With dedicated cache, separate sets of memory locations are reserved for reads and writes. In global cache, both reads and writes can use any of the available memory addresses. Cache management is more efficient in a global cache implementation because only one global set of addresses has to be managed.

Global cache allows users to specify the percentages of cache available for reads and writes for cache management. Typically, the read cache is small, but it should be increased if the application being used is read-intensive. In other global cache implementations, the ratio of cache available for reads versus writes is dynamically adjusted based on the workloads.

Cache Management

Cache is a finite and expensive resource that needs proper management. Even though modern intelligent storage systems come with a large amount of cache, when all cache pages are filled, some pages have to be freed up to accommodate new data and avoid performance degradation. Various cache management algorithms are implemented in intelligent storage systems to proactively maintain a set of free pages and a list of pages that can be potentially freed up whenever required. The most commonly used algorithms are discussed in the following list:

- **Least Recently Used (LRU):** An algorithm that continuously monitors data access in cache and identifies the cache pages that have not been accessed for a long time. LRU either frees up these pages or marks them for reuse. This algorithm is based on the assumption that data that has not been accessed for a while will not be requested by the host. However, if a page contains write data that has not yet been committed to disk, the data is first written to disk before the page is reused.
- **Most Recently Used (MRU):** This algorithm is the opposite of LRU, where the pages that have been accessed most recently are freed up or marked for reuse. This algorithm is based on the assumption that recently accessed data may not be required for a while.

As cache fills, the storage system must take action to flush dirty pages (data written into the cache but not yet written to the disk) to manage space availability. *Flushing* is the process that commits data from cache to the disk. On the basis of the I/O access rate and pattern, high and low levels called *watermarks* are set in cache to manage the flushing process. *High watermark* (*HWM*) is the cache utilization level at which the storage system starts high-speed flushing of cache data. *Low watermark* (*LWM*) is the point at which the storage system stops flushing data to the disks. The cache utilization level, as shown in Figure 4-4, drives the mode of flushing to be used:

- **Idle flushing:** Occurs continuously, at a modest rate, when the cache utilization level is between the high and low watermark.
- **High watermark flushing:** Activated when cache utilization hits the high watermark. The storage system dedicates some additional resources for flushing. This type of flushing has some impact on I/O processing.
- **Forced flushing:** Occurs in the event of a large I/O burst when cache reaches 100 percent of its capacity, which significantly affects the I/O response time. In forced flushing, system flushes the cache on priority by allocating more resources.

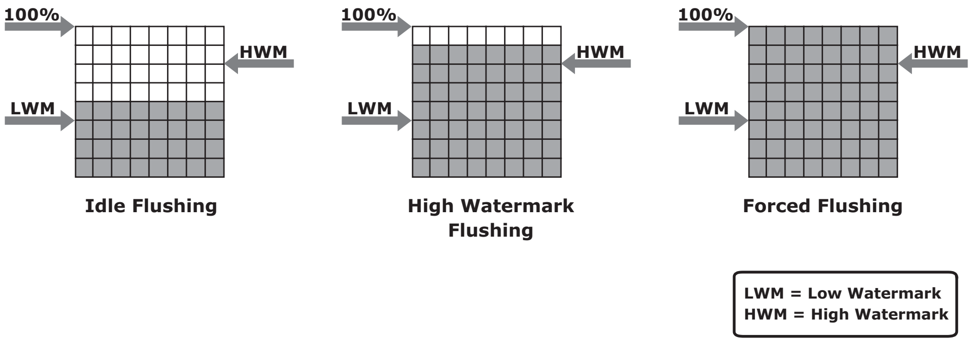


Figure 4-4: Types of flushing

Cache Data Protection

Cache is volatile memory, so a power failure or any kind of cache failure will cause loss of the data that is not yet committed to the disk. This risk of losing uncommitted data held in cache can be mitigated using *cache mirroring* and *cache vaulting*:

- **Cache mirroring:** Each write to cache is held in two different memory locations on two independent memory cards. If a cache failure occurs, the write data will still be safe in the mirrored location and can be committed to the disk. Reads are staged from the disk to the cache; therefore, if a cache failure occurs, the data can still be accessed from the disk. Because only writes are mirrored, this method results in better utilization of the available cache.

  In cache mirroring approaches, the problem of maintaining *cache coherency* is introduced. Cache coherency means that data in two different cache locations must be identical at all times. It is the responsibility of the array operating environment to ensure coherency.

- **Cache vaulting:** The risk of data loss due to power failure can be addressed in various ways: powering the memory with a battery until the AC power is restored or using battery power to write the cache content to the disk. If an extended power failure occurs, using batteries is not a viable option. This is because in intelligent storage systems, large amounts of data might need to be committed to numerous disks, and batteries might not provide power for sufficient time to write each piece of data to its intended disk. Therefore, storage vendors use a set of physical disks to dump the contents of cache during power failure. This is called *cache vaulting* and the disks are called *vault drives*. When power is restored, data from these disks is written back to write cache and then written to the intended disks.

SERVER FLASH-CACHING TECHNOLOGY

Server flash-caching technology uses intelligent caching software and a PCI Express (PCIe) flash card on the host. This dramatically improves application performance by reducing latency, and accelerates throughput. Server flash-caching technology works in both physical and virtual environments and provides performance acceleration for read-intensive workloads. This technology uses minimal CPU and memory resources from the server by offloading flash management onto the PCIe card.

It intelligently determines which data would benefit by sitting in the server on PCIe flash and closer to the application. This avoids the latencies associated with I/O access over the network to the storage array. With this, the processing power required for an application's most frequently referenced data is offloaded from the back-end storage to the PCIe card. Therefore, the storage array can allocate greater processing power to other applications.

4.1.3 Back End

The *back end* provides an interface between cache and the physical disks. It consists of two components: back-end ports and back-end controllers. The back-end controls data transfers between cache and the physical disks. From cache, data is sent to the back end and then routed to the destination disk. Physical disks are connected to ports on the back end. The back-end controller communicates with the disks when performing reads and writes and also provides additional, but limited, temporary data storage. The algorithms implemented on back-end controllers provide error detection and correction, along with RAID functionality.

For high data protection and high availability, storage systems are configured with dual controllers with multiple ports. Such configurations provide an alternative path to physical disks if a controller or port failure occurs. This reliability is further enhanced if the disks are also dual-ported. In that case, each disk port can connect to a separate controller. Multiple controllers also facilitate load balancing.

4.1.4 Physical Disk

Physical disks are connected to the back-end storage controller and provide persistent data storage. Modern intelligent storage systems provide support to a variety of disk drives with different speeds and types, such as FC, SATA, SAS, and flash drives. They also support the use of a mix of flash, FC, or SATA within the same array.

4.2 Storage Provisioning

Storage provisioning is the process of assigning storage resources to hosts based on capacity, availability, and performance requirements of applications running on the hosts. Storage provisioning can be performed in two ways: traditional and virtual. *Virtual provisioning* leverages virtualization technology for provisioning storage for applications. This section details both traditional and virtual storage provisioning.

4.2.1 Traditional Storage Provisioning

In traditional storage provisioning, physical disks are logically grouped together and a required RAID level is applied to form a set, called a RAID set. The number of drives in the RAID set and the RAID level determine the availability, capacity, and performance of the RAID set. It is highly recommend that the RAID set be created from drives of the same type, speed, and capacity to ensure maximum usable capacity, reliability, and consistency in performance. For example, if drives of different capacities are mixed in a RAID set, the capacity of the smallest drive is used from each disk in the set to make up the RAID set's overall capacity. The remaining capacity of the larger drives remains unused. Likewise, mixing higher revolutions per minute (RPM) drives with lower RPM drives lowers the overall performance of the RAID set.

RAID sets usually have a large capacity because they combine the total capacity of individual drives in the set. *Logical units* are created from the RAID sets by partitioning (seen as slices of the RAID set) the available capacity into smaller units. These units are then assigned to the host based on their storage requirements.

Logical units are spread across all the physical disks that belong to that set. Each logical unit created from the RAID set is assigned a unique ID, called a *logical unit number* (LUN). LUNs hide the organization and composition of the RAID set from the hosts. LUNs created by traditional storage provisioning methods are also referred to as *thick LUNs* to distinguish them from the LUNs created by virtual provisioning methods.

Figure 4-5 shows a RAID set consisting of five disks that have been sliced, or partitioned, into two LUNs: LUN 0 and LUN 1. These LUNs are then assigned to Host1 and Host 2 for their storage requirements.

When a LUN is configured and assigned to a non-virtualized host, a bus scan is required to identify the LUN. This LUN appears as a raw disk to the operating system. To make this disk usable, it is formatted with a file system and then the file system is mounted.

In a virtualized host environment, the LUN is assigned to the hypervisor, which recognizes it as a raw disk. This disk is configured with the hypervisor file system, and then virtual disks are created on it. *Virtual disks* are files on the hypervisor file system. The virtual disks are then assigned to virtual machines and appear as raw disks to them. To make the virtual disk usable to the virtual machine, similar steps are followed as in a non-virtualized environment. Here, the LUN space may be shared and accessed simultaneously by multiple virtual machines.

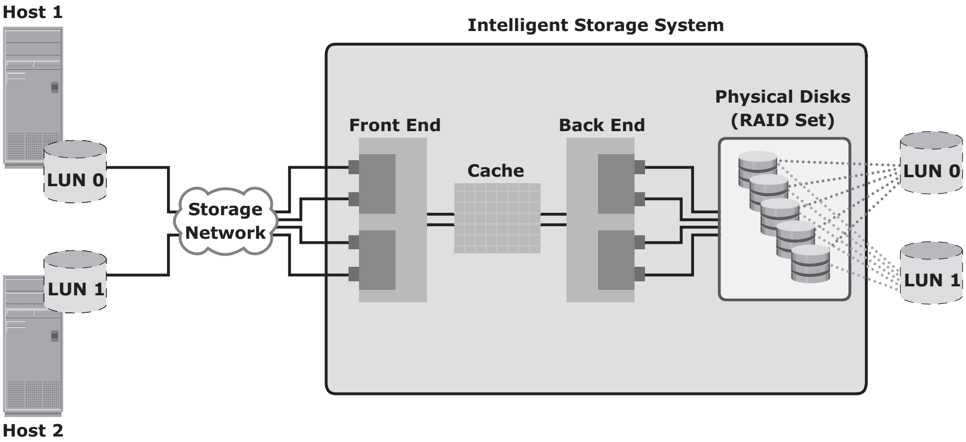


Figure 4-5: RAID set and LUNs

Virtual machines can also access a LUN directly on the storage system. In this method the entire LUN is allocated to a single virtual machine. Storing data in this way is recommended when the applications running on the virtual machine are response-time sensitive, and sharing storage with other virtual machines may impact their response time. The direct access method is also used when a virtual machine is clustered with a physical machine. In this case, the virtual machine is required to access the LUN that is being accessed by the physical machine.

LUN Expansion: MetaLUN

MetaLUN is a method to expand LUNs that require additional capacity or performance. A metaLUN can be created by combining two or more LUNs. A metaLUN consists of a base LUN and one or more component LUNs. MetaLUNs can be either *concatenated* or *striped*.

Concatenated expansion simply adds additional capacity to the base LUN. In this expansion, the component LUNs are not required to be of the same capacity as the base LUN. All LUNs in a concatenated metaLUN must be either protected (parity or mirrored) or unprotected (RAID 0). RAID types within a metaLUN can be mixed. For example, a RAID 1/0 LUN can be concatenated with a RAID 5 LUN. However, a RAID 0 LUN can be concatenated only with another RAID 0 LUN.

Concatenated expansion is quick but does not provide any performance benefit. (See Figure 4-6.)

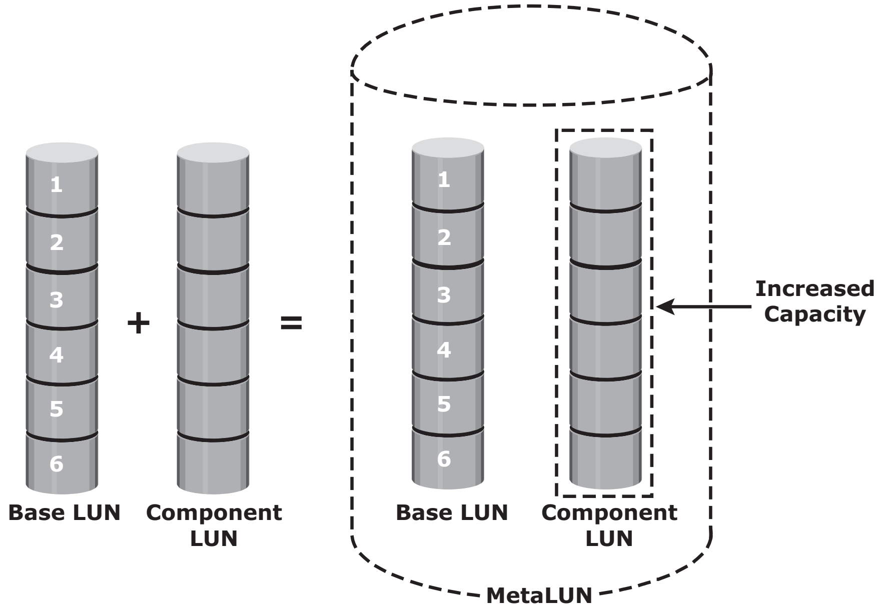


Figure 4-6: Concatenated metaLUN

Striped expansion restripes the base LUN's data across the base LUN and component LUNs. In striped expansion, all LUNs must be of the same capacity and RAID level. Striped expansion provides improved performance due to the increased number of drives being striped (see Figure 4-7).

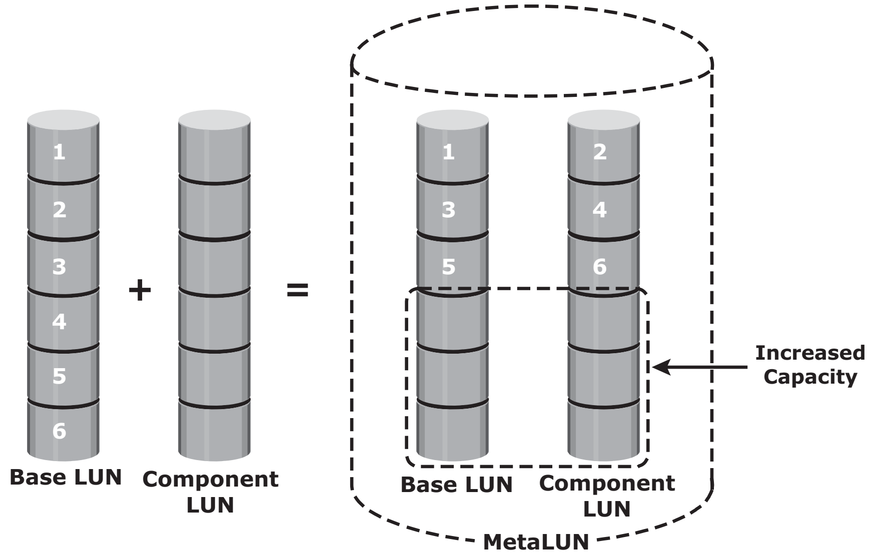


Figure 4-7: Striped metaLUN

All LUNs in both concatenated and striped expansion must reside on the same disk-drive type: either all Fibre Channel or all ATA.

4.2.2 Virtual Storage Provisioning

Virtual provisioning enables creating and presenting a LUN with more capacity than is physically allocated to it on the storage array. The LUN created using virtual provisioning is called a *thin LUN* to distinguish it from the traditional LUN.

Thin LUNs do not require physical storage to be completely allocated to them at the time they are created and presented to a host. Physical storage is allocated to the host "on-demand" from a shared pool of physical capacity. A *shared pool* consists of physical disks. A shared pool in virtual provisioning is analogous to a RAID group, which is a collection of drives on which LUNs are created. Similar to a RAID group, a shared pool supports a single RAID protection level. However, unlike a RAID group, a shared pool might contain large numbers of drives. Shared pools can be homogeneous (containing a single drive type) or heterogeneous (containing mixed drive types, such as flash, FC, SAS, and SATA drives).

Virtual provisioning enables more efficient allocation of storage to hosts. Virtual provisioning also enables oversubscription, where more capacity is presented to the hosts than is actually available on the storage array. Both shared pool and thin LUN can be expanded nondisruptively as the storage requirements of the hosts grow. Multiple shared pools can be created within a storage array, and a shared pool may be shared by multiple thin LUNs. Figure 4-8 illustrates the provisioning of thin LUNs.

Comparison between Virtual and Traditional Storage Provisioning

Administrators typically allocate storage capacity based on anticipated storage requirements. This generally results in the over provisioning of storage capacity, which then leads to higher costs and lower capacity utilization. Administrators often over-provision storage to an application for various reasons, such as, to avoid frequent provisioning of storage if the LUN capacity is exhausted, and to reduce disruption to application availability. Over provisioning of storage often leads to additional storage acquisition and operational costs.

Virtual provisioning addresses these challenges. Virtual provisioning improves storage capacity utilization and simplifies storage management. Figure 4-9 shows an example, comparing virtual provisioning with traditional storage provisioning.

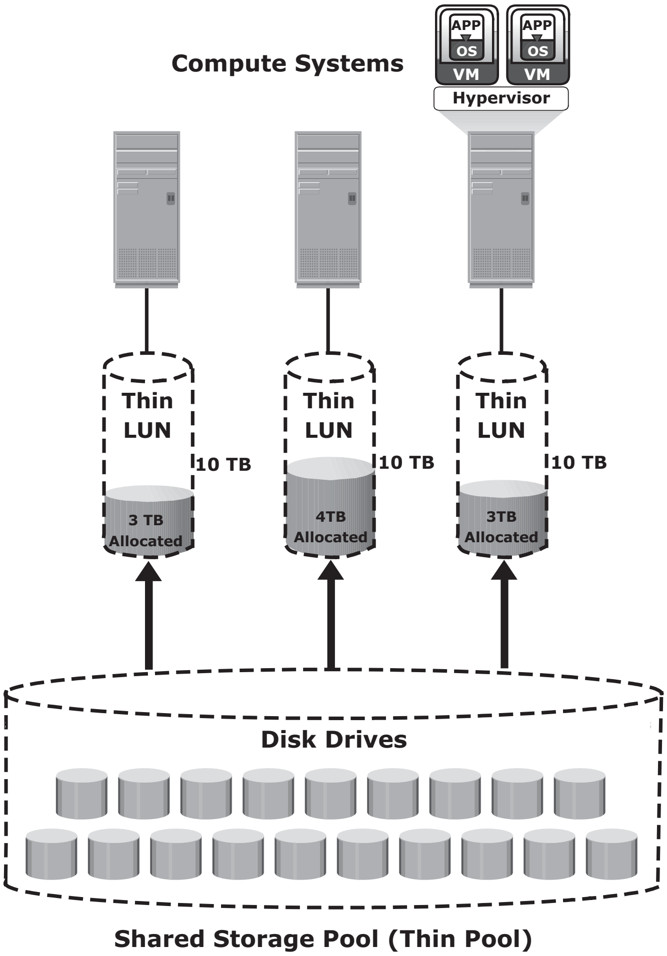


Figure 4-8: Virtual provisioning

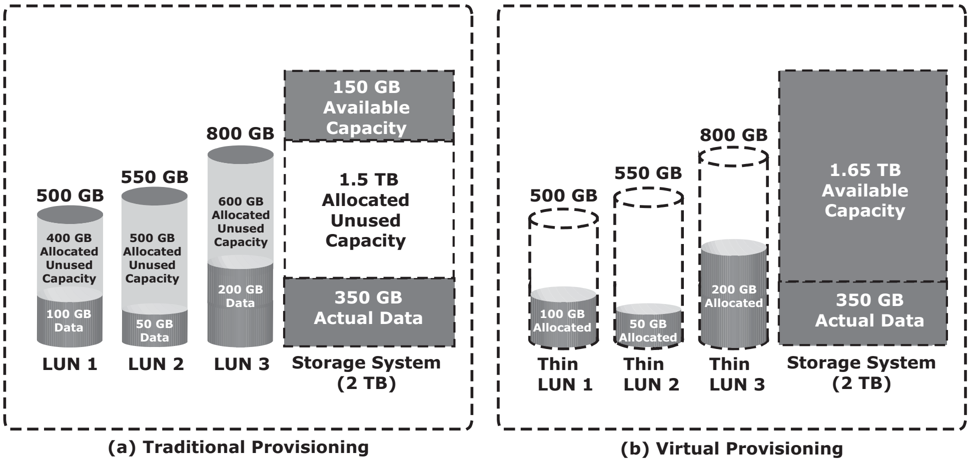


Figure 4-9: Traditional versus virtual provisioning

With traditional provisioning, three LUNs are created and presented to one or more hosts (see Figure 4-9 [a]). The total storage capacity of the storage system is 2 TB. The allocated capacity of LUN 1 is 500 GB, of which only 100 GB is consumed, and the remaining 400 GB is unused. The size of LUN 2 is 550 GB, of which 50 GB is consumed, and 500 GB is unused. The size of LUN 3 is 800 GB, of which 200 GB is consumed, and 600 GB is unused. In total, the storage system has 350 GB of data, 1.5 TB of allocated but unused capacity, and only 150 GB of remaining capacity available for other applications.

Now consider the same 2 TB storage system with virtual provisioning (see Figure 4-9 [b]). Here, three thin LUNs of the same sizes are created. However, there is no allocated unused capacity. In total, the storage system with virtual provisioning has the same 350 GB of data, but 1.65 TB of capacity is available for other applications, whereas only 150 GB is available in traditional storage provisioning.

Use Cases for Thin and Traditional LUNs

Virtual provisioning and thin LUN offer many benefits, although in some cases traditional LUN is better suited for an application. Thin LUNs are appropriate for applications that can tolerate performance variations. In some cases, performance improvement is perceived when using a thin LUN, due to striping across a large number of drives in the pool. However, when multiple thin LUNs contend for shared storage resources in a given pool, and when utilization reaches higher levels, the performance can degrade. Thin LUNs provide the best storage space efficiency and are suitable for applications where space consumption is difficult to forecast. Using thin LUNs benefits organizations in reducing power and acquisition costs and in simplifying their storage management.

Traditional LUNs are suited for applications that require predictable performance. Traditional LUNs provide full control for precise data placement and allow an administrator to create LUNs on different RAID groups if there is any workload contention. Organizations that are not highly concerned about storage space efficiency may still use traditional LUNs.

Both traditional and thin LUNs can coexist in the same storage array. Based on the requirement, an administrator may migrate data between thin and traditional LUNs.

4.2.3 LUN Masking

LUN masking is a process that provides data access control by defining which LUNs a host can access. The LUN masking function is implemented on the storage array. This ensures that volume access by hosts is controlled appropriately, preventing unauthorized or accidental use in a shared environment.

For example, consider a storage array with two LUNs that store data of the sales and finance departments. Without LUN masking, both departments can easily see and modify each other's data, posing a high risk to data integrity and security. With LUN masking, LUNs are accessible only to the designated hosts.

4.3 Types of Intelligent Storage Systems

Intelligent storage systems generally fall into one of the following two categories:

- High-end storage systems
- Midrange storage systems

Traditionally, high-end storage systems have been implemented with *active-active configuration*, whereas midrange storage systems have been implemented with *active-passive configuration*. The distinctions between these two implementations are becoming increasingly insignificant.

4.3.1 High-End Storage Systems

High-end storage systems, referred to as *active-active arrays*, are generally aimed at large enterprise applications. These systems are designed with a large number of controllers and cache memory. An active-active array implies that the host can perform I/Os to its LUNs through any of the available controllers (see Figure 4-10).

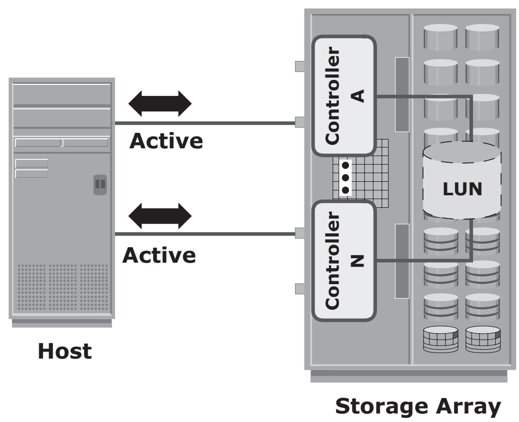


Figure 4-10: Active-active configuration

To address enterprise storage needs, these arrays provide the following capabilities:

- Large storage capacity
- Large amounts of cache to service host I/Os optimally

- Fault tolerance architecture to improve data availability
- Connectivity to mainframe computers and open systems hosts
- Availability of multiple front-end ports and interface protocols to serve a large number of hosts
- Availability of multiple back-end controllers to manage disk processing
- Scalability to support increased connectivity, performance, and storage capacity requirements
- Ability to handle large amounts of concurrent I/Os from a number of hosts and applications
- Support for array-based local and remote data replication

In addition to these features, high-end systems possess some unique features that are required for mission-critical applications.

4.3.2 Midrange Storage Systems

Midrange storage systems are also referred to as *active-passive arrays* and are best suited for small- and medium-sized enterprise applications. They also provide optimal storage solutions at a lower cost. In an active-passive array, a host can perform I/Os to a LUN only through the controller that owns the LUN. As shown in Figure 4-11, the host can perform reads or writes to the LUN only through the path to controller A because controller A is the owner of that LUN. The path to controller B remains passive and no I/O activity is performed through this path.

Midrange storage systems are typically designed with two controllers, each of which contains host interfaces, cache, RAID controllers, and interface to disk drives.

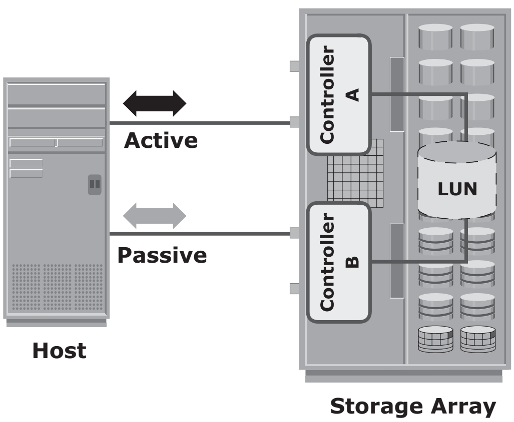


Figure 4-11: Active-passive configuration

Midrange arrays are designed to meet the requirements of small and medium enterprise applications; therefore, they host less storage capacity and cache than high-end storage arrays. There are also fewer front-end ports for connection to hosts. However, they ensure high redundancy and high performance for applications with predictable workloads. They also support array-based local and remote replication.

4.4 Concepts in Practice: EMC Symmetrix and VNX

To illustrate the concepts discussed in this chapter, this section covers the EMC implementation of intelligent storage arrays.

The EMC Symmetrix storage array is an active-active array implementation. Symmetrix is a solution for customers who require an uncompromising level of service, performance, and the most advanced business continuity solution to support large and unpredictable application workloads. Symmetrix also provides built-in, advanced-level security features and offers the most efficient use of power and cooling to support enterprise-level data storage requirements.

The EMC VNX storage array is an active-passive array implementation. It is EMC's midrange storage offering that delivers enterprise-quality features and functionalities. EMC VNX is a unified storage platform that offers storage for block, file, and object-based data within the same array. It is ideally suited for applications with predictable workloads that require moderate-to-high throughput. Details of unified storage and EMC VNX are covered in Chapter 8.

For the latest information on Symmetrix and VNX, visit `www.emc.com`.

4.4.1 EMC Symmetrix Storage Array

EMC Symmetrix establishes the highest standards for performance and capacity for an enterprise information storage solution and is recognized as the industry's most trusted storage platform. Symmetrix offers the highest level of scalability and performance to meet even unpredictable I/O workload requirements. The EMC Symmetrix offering includes the Symmetrix Virtual Matrix (VMAX) series.

The EMC Symmetrix VMAX series is an innovative platform built around a scalable Virtual Matrix architecture to support the future storage growth demands of virtual IT environments. Figure 4-12 shows the Symmetrix VMAX storage array. The key features supported by Symmetrix VMAX follows:

- Incrementally scalable to 2,400 disks
- Supports up to 8 VMAX engines (Each VMAX engine contains a pair of directors.)
- Supports flash drives, fully automated storage tiering (FAST), virtual provisioning, and Cloud computing

- Supports up to 1 TB of global cache memory
- Supports FC, iSCSI, GigE, and FICON for host connectivity
- Supports RAID levels 1, 1+0, 5, and 6
- Supports storage-based replication through EMC TimeFinder and EMC SRDF
- Highly fault-tolerant design that allows nondisruptive upgrades and full component-level redundancy with hot-swappable replacements

Figure 4-12: EMC Symmetrix VMAX

4.4.2 EMC Symmetrix VMAX Component

EMC Symmetrix VMAX contains one system bay and up to ten storage bays. A storage bay supports up to 16 drive array enclosures (DAEs), and each drive enclosure can house up to 15 drives. The system bay houses the system components, which include VMAX Engines, Matrix Interface Board Enclosure (MIBE), standby power supply (SPS) modules, and service processor:

- **VMAX Engine:** Consists of a pair of directors that contains four quad-core Intel processors, up to 128 GB of memory, and up to16 front-end ports for host access or SRDF channels.

- **Matrix Interface Board Enclosure (MIBE):** Contains two independent matrix switches that provide point-to point communication between directors. Each director has two connections to the V-Max Matrix Interface Board Enclosure. Because every director has two separate physical paths to every other director via the Virtual Matrix, this is a highly available interconnect with no single point of failure. This design eliminates the need for separate interconnects for data, control, messaging, and environmental and system tests. A single highly available interconnect suffices for all communications between the directors, which reduces complexity.
- **Service Processor:** Used for system configuration and the management console. It also provides notification and support capabilities to allow access to the system locally or remotely. The Service Processor automatically notifies the vendor's Customer Support Center whenever a component failure or environmental violation is detected.
- **Symmetrix Enginuity:** The operating environment for EMC Symmetrix. Enginuity manages and ensures the optimal flow and integrity of information through the various hardware components of the Symmetrix system. It manages all Symmetrix operations and system resources to optimize performance intelligently. Enginuity ensures system availability through advanced fault monitoring, detection, and correction capabilities and provides concurrent maintenance and serviceability features. It also offers a foundation for specific software features for disaster recovery, business continuance, and storage management.

4.4.3 Symmetrix VMAX Architecture

Each VMAX engine contains a portion of global memory and two directors capable of managing front-end, back-end, and remote connections simultaneously. The VMAX engine is connected to Virtual Matrix and allows all system resources, including CPU, memory, drives, and host ports, to be dynamically accessed and shared by any host. Additional VMAX engines can be added nondisruptively to efficiently scale system resources. The Virtual Matrix supports up to eight VMAX engines in a system, as shown in Figure 4-13.

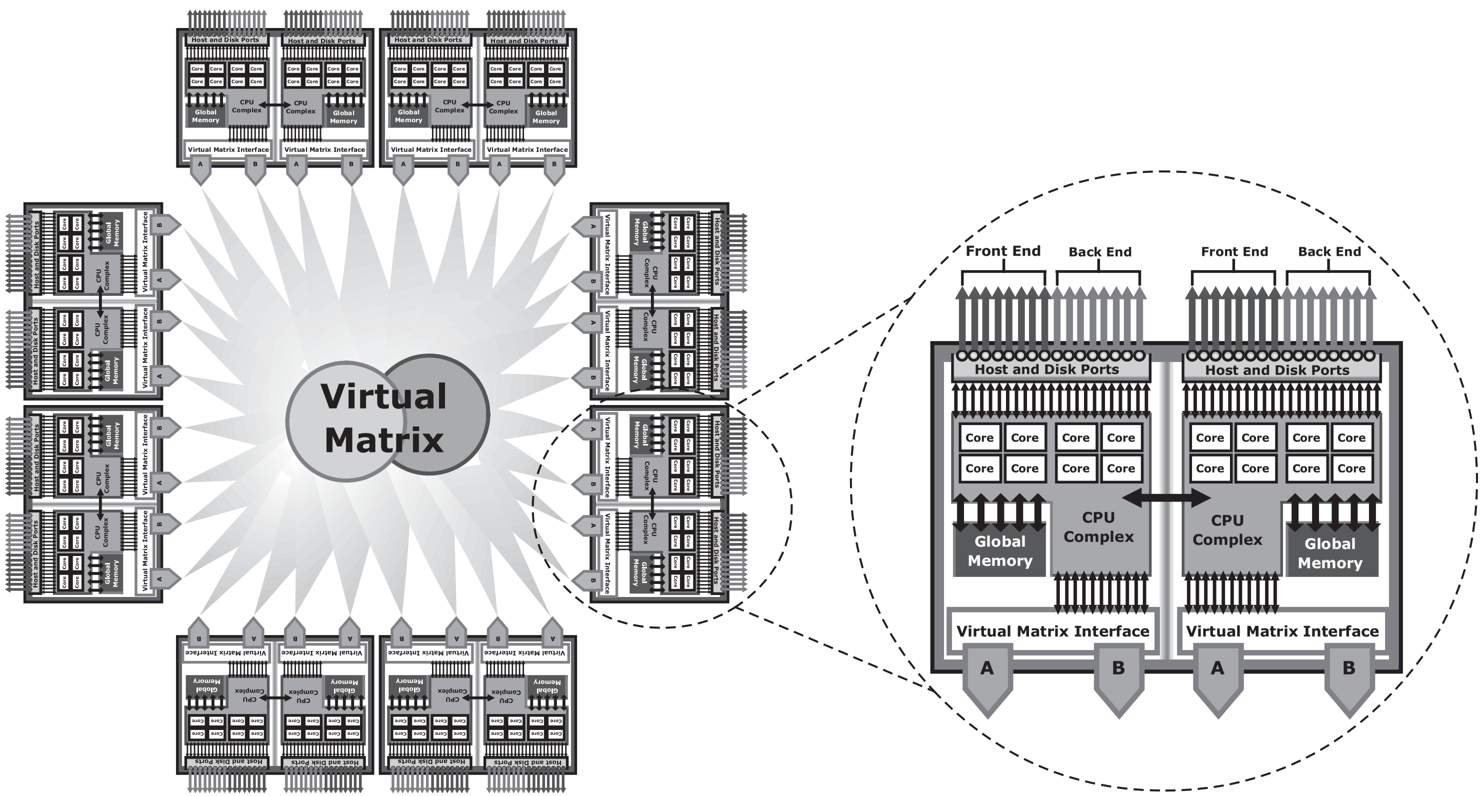


Figure 4-13: VMAX architecture

Summary

This chapter detailed the features and key components of modern intelligent storage systems. The different types of storage systems, high-end and midrange, and their characteristics were also explained. An intelligent storage system provides the following benefits to an organization:

- Increased storage capacity
- Improved I/O performance
- Easier storage management
- Improved data availability
- Improved scalability and flexibility
- Improved business continuity
- Improved security and access control

An intelligent storage system is an integral part of every data center. The large capacity and high performance supported by the intelligent storage system makes it necessary to share it among multiple hosts. Intelligent storage systems enable enterprises to share data easily and securely.

Storage networking is a flexible information-centric strategy that extends the reach of intelligent storage systems throughout an enterprise. It provides a common way to manage, share, and protect enterprise-information assets. Storage networking is detailed in the next part of this book.

EXERCISES

1. **Research Cache Coherency mechanisms, and explain how they address the environment with multiple shared caches.**
2. **Which type of application benefits the most by bypassing write cache? Justify your answer.**
3. **Research various cache parameters: cache page size, read versus write cache allocation, cache prefetch size, and write aside size.**
4. **An Oracle database uses a block size of 4 KB for its I/O operation. The application that uses this database primarily performs a sequential read operation. Suggest and explain the appropriate values for the following cache parameters: cache page size, cache allocation (read versus write), prefetch type, and write aside size.**
5. **Research and prepare a presentation on EMC VMAX architecture.**

Section

II

Storage Networking Technologies

In This Section

Chapter 5
Fibre Channel Storage Area Networks

KEY CONCEPTS

- Fibre Channel (FC) Architecture
- Fibre Channel Protocol Stack
- Ports in Fibre Channel SAN
- Fibre Channel Addressing
- World Wide Names
- Zoning
- Fibre Channel SAN Topologies
- Block-level Storage Virtualization
- Virtual SAN

Organizations are experiencing an explosive growth in information. This information needs to be stored, protected, optimized, and managed efficiently. Data center managers are burdened with the challenging task of providing low-cost, high-performance information management solutions. An effective information management solution must provide the following:

- **Just-in-time information to business users:** Information must be available to business users when they need it. 24 x 7 data availability is becoming one of the key requirements of today's storage infrastructure. The explosive growth in storage, proliferation of new servers and applications, and the spread of mission-critical data throughout enterprises are some of the challenges that need to be addressed to provide information availability in real time.
- **Integration of information infrastructure with business processes:** The storage infrastructure should be integrated with various business processes without compromising its security and integrity.
- **Flexible and resilient storage infrastructure:** The storage infrastructure must provide flexibility and resilience that aligns with changing business requirements. Storage should scale without compromising the performance

requirements of applications and, at the same time, the total cost of managing information must be low.

Direct-attached storage (DAS) is often referred to as a stovepiped storage environment. Hosts "own" the storage, and it is difficult to manage and share resources on these isolated storage devices. Efforts to organize this dispersed data led to the emergence of the *storage area network* (SAN). SAN is a high-speed, dedicated network of servers and shared storage devices. A SAN provides storage consolidation and facilitates centralized data management. It meets the storage demands efficiently with better economies of scale and also provides effective maintenance and protection of data. Virtualized SAN and block storage virtualization provide enhanced utilization and collaboration among dispersed storage resources. The implementation of virtualization in SAN provides improved productivity, resource utilization, and manageability.

Common SAN deployments are Fibre Channel (FC) SAN and IP SAN. Fibre Channel SAN uses Fibre Channel protocol for the transport of data, commands, and status information between servers (or hosts) and storage devices. IP SAN uses IP-based protocols for communication.

This chapter provides detailed insight into the FC technology on which an FC SAN is deployed. It also covers FC SAN components, topologies, and block storage virtualization.

5.1 Fibre Channel: Overview

The FC architecture forms the fundamental construct of the FC SAN infrastructure. *Fibre Channel* is a high-speed network technology that runs on high-speed optical fiber cables and serial copper cables. The FC technology was developed to meet the demand for increased speeds of data transfer between servers and mass storage systems. Although FC networking was introduced in 1988, the FC standardization process began when the American National Standards Institute (ANSI) chartered the Fibre Channel Working Group (FCWG). By 1994, the new high-speed computer interconnection standard was developed and the Fibre Channel Association (FCA) was founded with 70 charter member companies. Technical Committee T11, which is the committee within International Committee for Information Technology Standards (INCITS), is responsible for Fibre Channel interface standards.

High data transmission speed is an important feature of the FC networking technology. The initial implementation offered a throughput of 200 MB/s (equivalent to a raw bit rate of 1Gb/s), which was greater than the speeds of Ultra SCSI (20 MB/s), commonly used in DAS environments. In comparison with Ultra SCSI, FC is a significant leap in storage networking technology. The latest FC implementations of 16 GFC (Fibre Channel) offer a throughput of 3200 MB/s (raw bit rates of 16 Gb/s), whereas Ultra640 SCSI is available with a throughput of 640 MB/s. The FC architecture is highly scalable, and theoretically, a single FC network can accommodate approximately 15 million devices.

5.2 The SAN and Its Evolution

A SAN carries data between servers (or *hosts*) and storage devices through Fibre Channel network (see Figure 5-1). A SAN enables storage consolidation and enables storage to be shared across multiple servers. This improves the utilization of storage resources compared to direct-attached storage architecture and reduces the total amount of storage an organization needs to purchase and manage. With consolidation, storage management becomes centralized and less complex, which further reduces the cost of managing information. SAN also enables organizations to connect geographically dispersed servers and storage.

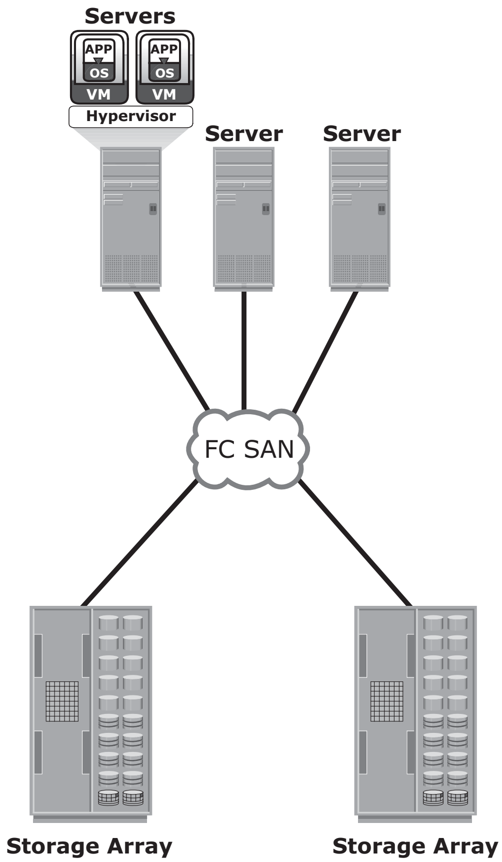


Figure 5-1: FC SAN implementation

In its earliest implementation, the FC SAN was a simple grouping of hosts and storage devices connected to a network using an FC hub as a connectivity device. This configuration of an FC SAN is known as a *Fibre Channel Arbitrated*

Loop (FC-AL). Use of hubs resulted in isolated FC-AL SAN islands because hubs provide limited connectivity and bandwidth.

The inherent limitations associated with hubs gave way to high-performance FC *switches*. Use of switches in SAN improved connectivity and performance and enabled FC SANs to be highly scalable. This enhanced data accessibility to applications across the enterprise. Now, FC-AL has been almost abandoned for FC SANs due to its limitations but still survives as a back-end connectivity option to disk drives. Figure 5-2 illustrates the FC SAN evolution from FC-AL to enterprise SANs.

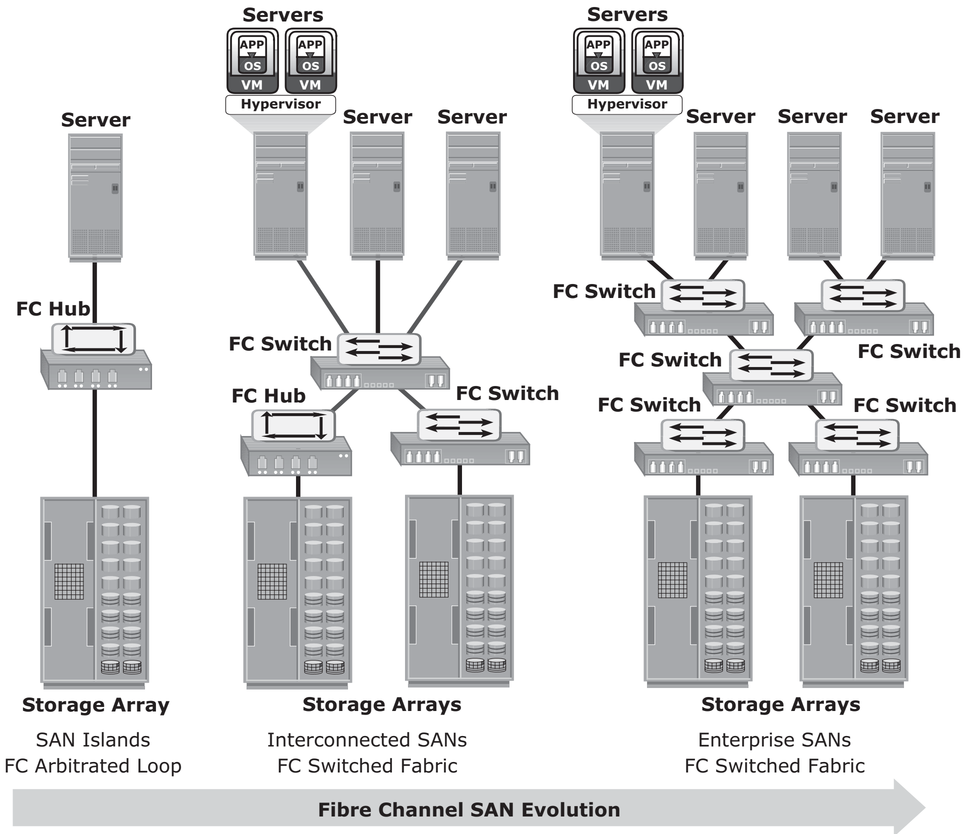


Figure 5-2: FC SAN evolution

5.3 Components of FC SAN

FC SAN is a network of servers and shared storage devices. Servers and storage are the end points or devices in the SAN (called *nodes*). FC SAN infrastructure consists of node ports, cables, connectors, and interconnecting devices (such as FC switches or hubs), along with SAN management software.

5.3.1 Node Ports

In a Fibre Channel network, the end devices, such as hosts, storage arrays, and tape libraries, are all referred to as *nodes*. Each node is a source or destination of information. Each node requires one or more ports to provide a physical interface for communicating with other nodes. These ports are integral components of host adapters, such as HBA, and storage front-end controllers or adapters. In an FC environment a port operates in full-duplex data transmission mode with a *transmit* (Tx) link and a *receive* (Rx) link (see Figure 5-3).

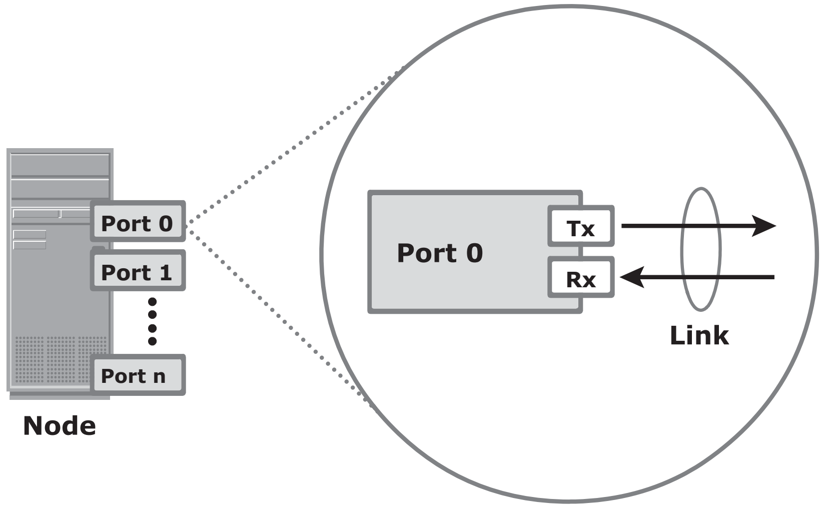


Figure 5-3: Nodes, ports, and links

5.3.2 Cables and Connectors

SAN implementations use optical fiber cabling. Copper can be used for shorter distances for back-end connectivity because it provides an acceptable signal-to-noise ratio for distances up to 30 meters. Optical fiber cables carry data in the form of light. There are two types of optical cables: multimode and single-mode. *Multimode fiber* (MMF) cable carries multiple beams of light projected at different angles simultaneously onto the core of the cable (see Figure 5-4 [a]). Based on the bandwidth, multimode fibers are classified as OM1 (62.5µm core), OM2 (50µm core), and laser-optimized OM3 (50µm core). In an MMF transmission, multiple light beams traveling inside the cable tend to disperse and collide. This collision weakens the signal strength after it travels a certain distance — a process known as *modal dispersion*. An MMF cable is typically used for short distances because of signal degradation (attenuation) due to modal dispersion.

Single-mode fiber (SMF) carries a single ray of light projected at the center of the core (see Figure 5-4 [b]). These cables are available in core diameters of 7 to 11 microns;

the most common size is 9 microns. In an SMF transmission, a single light beam travels in a straight line through the core of the fiber. The small core and the single light wave help to limit modal dispersion. Among all types of fiber cables, single-mode provides minimum signal attenuation over maximum distance (up to 10 km). A single-mode cable is used for long-distance cable runs, and distance usually depends on the power of the laser at the transmitter and sensitivity of the receiver.

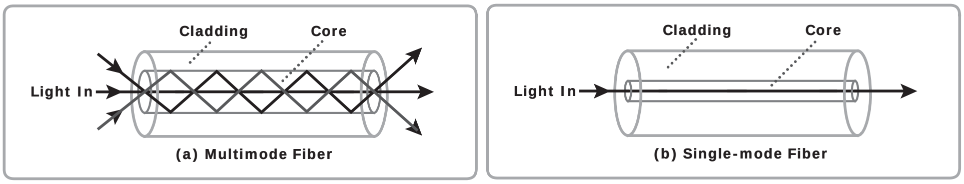


Figure 5-4: Multimode fiber and single-mode fiber

MMFs are generally used within data centers for shorter distance runs, whereas SMFs are used for longer distances.

A connector is attached at the end of a cable to enable swift connection and disconnection of the cable to and from a port. A *Standard connector* (SC) (see Figure 5-5 [a]) and a *Lucent connector* (LC) (see Figure 5-5 [b]) are two commonly used connectors for fiber optic cables. *Straight Tip* (ST) is another fiber-optic connector, which is often used with fiber patch panels (see Figure 5.5 [c]).

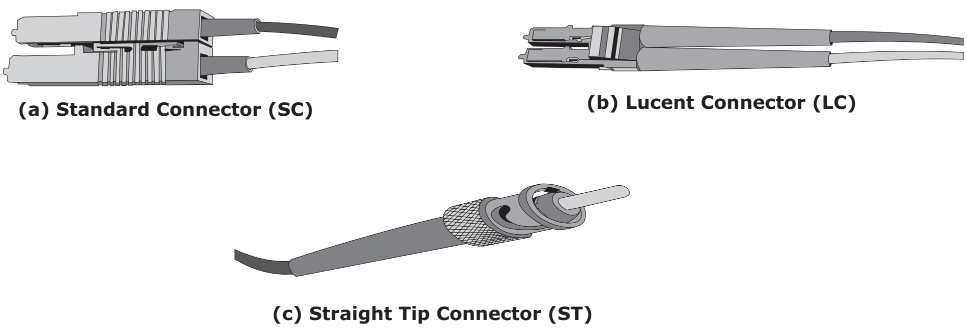


Figure 5-5: SC, LC, and ST connectors

5.3.3 Interconnect Devices

FC hubs, switches, and directors are the interconnect devices commonly used in FC SAN.

Hubs are used as communication devices in FC-AL implementations. Hubs physically connect nodes in a logical loop or a physical star topology. All the nodes must share the loop because data travels through all the connection points. Because of the availability of low-cost and high-performance switches, hubs are no longer used in FC SANs.

Switches are more intelligent than hubs and directly route data from one physical port to another. Therefore, nodes do not share the bandwidth. Instead, each node has a dedicated communication path.

Directors are high-end switches with a higher port count and better fault-tolerance capabilities.

Switches are available with a fixed port count or with modular design. In a modular switch, the port count is increased by installing additional port cards to open slots. The architecture of a director is always modular, and its port count is increased by inserting additional line cards or blades to the director's chassis. High-end switches and directors contain redundant components to provide high availability. Both switches and directors have management ports (Ethernet or serial) for connectivity to SAN management servers.

A port card or blade has multiple ports for connecting nodes and other FC switches. Typically, a Fibre Channel transceiver is installed at each port slot that houses the transmit (Tx) and receive (Rx) link. In a transceiver, the Tx and Rx links share common circuitry. Transceivers inside a port card are connected to an application specific integrated circuit, also called port ASIC. Blades in a director usually have more than one ASIC for higher throughput.

5.3.4 SAN Management Software

SAN management software manages the interfaces between hosts, interconnect devices, and storage arrays. The software provides a view of the SAN environment and enables management of various resources from one central console.

It provides key management functions, including mapping of storage devices, switches, and servers, monitoring and generating alerts for discovered devices, and *zoning* (discussed in section 5.9 "Zoning" later in this chapter).

FC SWITCH VERSUS FC HUB

Scalability and performance are the primary differences between switches and hubs. Addressing in a switched fabric supports more than 15 million nodes within the fabric, whereas the FC-AL implemented in hubs supports only a maximum of 126 nodes.

Fabric switches provide full bandwidth between multiple pairs of ports in a fabric, resulting in a scalable architecture that supports multiple simultaneous communications.

Hubs support only one communication at a time. They provide a low-cost connectivity expansion solution. Switches, conversely, can be used to build dynamic, high-performance fabrics through which multiple communications can take place simultaneously. Switches are more expensive than hubs.

5.4 FC Connectivity

The FC architecture supports three basic interconnectivity options: point-to-point, arbitrated loop, and Fibre Channel switched fabric.

5.4.1 Point-to-Point

Point-to-point is the simplest FC configuration — two devices are connected directly to each other, as shown in Figure 5-6. This configuration provides a dedicated connection for data transmission between nodes. However, the point-to-point configuration offers limited connectivity, because only two devices can communicate with each other at a given time. Moreover, it cannot be scaled to accommodate a large number of nodes. Standard DAS uses point-to-point connectivity.

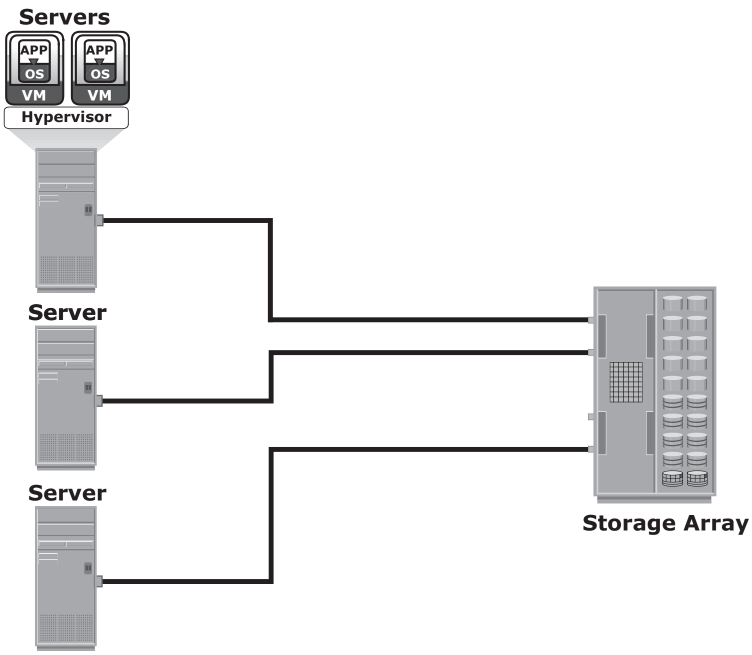


Figure 5-6: Point-to-point connectivity

5.4.2 Fibre Channel Arbitrated Loop

In the FC-AL configuration, devices are attached to a shared loop. FC-AL has the characteristics of a token ring topology and a physical star topology. In FC-AL, each device contends with other devices to perform I/O operations. Devices on the loop must "arbitrate" to gain control of the loop. At any given time, only one device can perform I/O operations on the loop (see Figure 5-7).

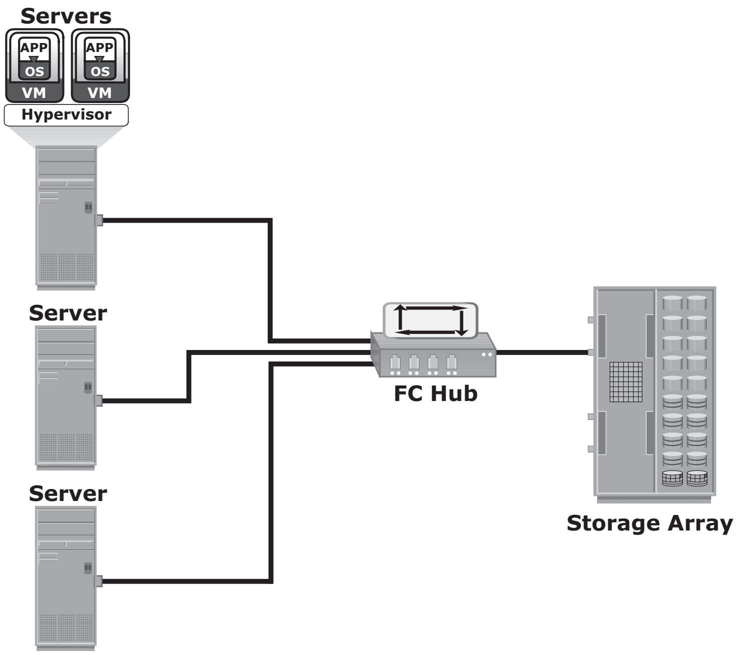


Figure 5-7: Fibre Channel Arbitrated Loop

As a loop configuration, FC-AL can be implemented without any interconnecting devices by directly connecting one device to another two devices in a ring through cables.

However, FC-AL implementations may also use hubs whereby the arbitrated loop is physically connected in a star topology.

The FC-AL configuration has the following limitations in terms of scalability:

- FC-AL shares the loop and only one device can perform I/O operations at a time. Because each device in a loop must wait for its turn to process an I/O request, the overall performance in FC-AL environments is low.
- FC-AL uses only 8-bits of 24-bit Fibre Channel addressing (the remaining 16-bits are masked) and enables the assignment of 127 valid addresses to the ports. Hence, it can support up to 127 devices on a loop. One address is reserved for optionally connecting the loop to an FC switch port. Therefore, up to 126 nodes can be connected to the loop.
- Adding or removing a device results in loop re-initialization, which can cause a momentary pause in loop traffic.

5.4.3 Fibre Channel Switched Fabric

Unlike a loop configuration, a Fibre Channel switched fabric (FC-SW) network provides dedicated data path and scalability. The addition or removal of a device in a switched fabric is minimally disruptive; it does not affect the ongoing traffic between other devices.

FC-SW is also referred to as *fabric connect*. A fabric is a logical space in which all nodes communicate with one another in a network. This virtual space can be created with a switch or a network of switches. Each switch in a fabric contains a unique domain identifier, which is part of the fabric's addressing scheme. In FC-SW, nodes do not share a loop; instead, data is transferred through a dedicated path between the nodes. Each port in a fabric has a unique 24-bit Fibre Channel address for communication. Figure 5-8 shows an example of the FC-SW fabric.

In a switched fabric, the link between any two switches is called an *Interswitch link* (ISL). ISLs enable switches to be connected together to form a single, larger fabric. ISLs are used to transfer host-to-storage data and fabric management traffic from one switch to another. By using ISLs, a switched fabric can be expanded to connect a large number of nodes.

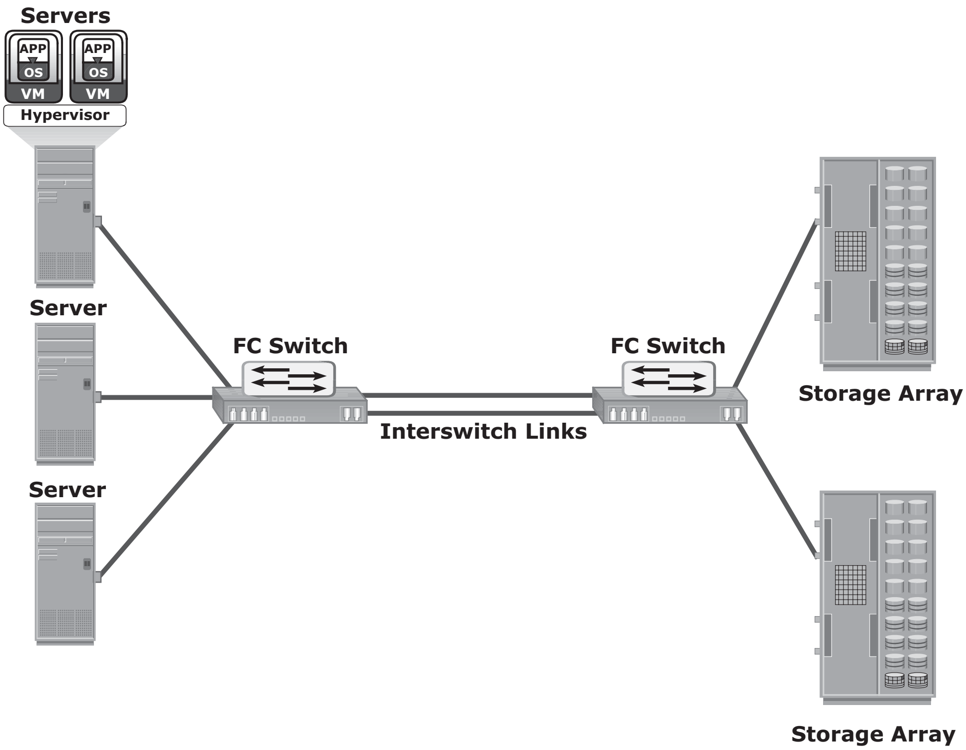


Figure 5-8: Fibre Channel switched fabric

A fabric can be described by the number of tiers it contains. The number of tiers in a fabric is based on the number of switches traversed between two points that are farthest from each other. This number is based on the infrastructure constructed by the fabric instead of how the storage and server are connected across the switches.

When the number of tiers in a fabric increases, the distance that the fabric management traffic must travel to reach each switch also increases. This increase in the distance also increases the time taken to propagate and complete a fabric reconfiguration event, such as the addition of a new switch or a zone set propagation event. Figure 5-9 illustrates two-tier and three-tier fabric architecture.

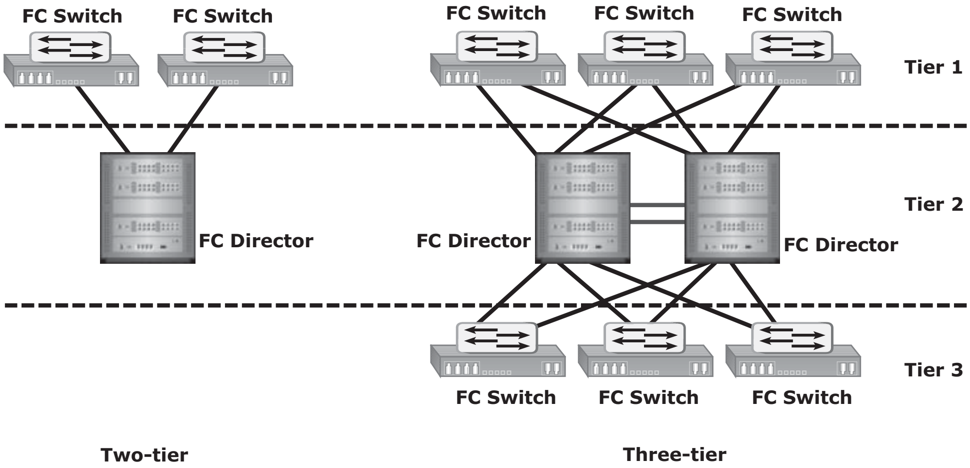


Figure 5-9: Tiered structure of Fibre Channel switched fabric

FC-SW Transmission

FC-SW uses switches that can switch data traffic between nodes directly through switch ports. Frames are routed between source and destination by the fabric.

As shown in Figure 5-10, if node B wants to communicate with node D, the nodes should individually login first and then transmit data via the FC-SW. This link is considered a dedicated connection between the initiator and the target.

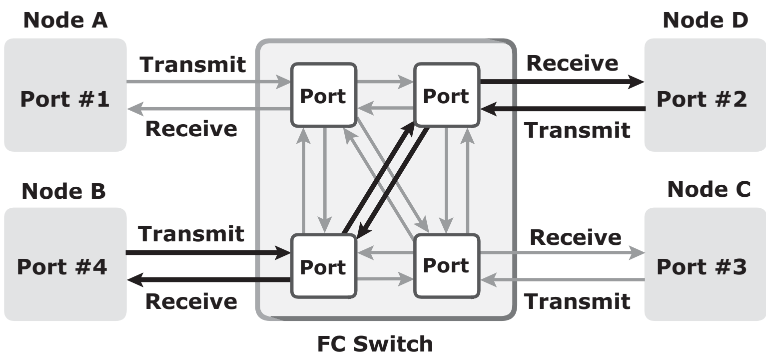


Figure 5-10: Data transmission in Fibre Channel switched fabric

5.5 Switched Fabric Ports

Ports in a switched fabric can be one of the following types:

- **N_Port:** An end point in the fabric. This port is also known as the *node port*. Typically, it is a host port (HBA) or a storage array port connected to a switch in a switched fabric.
- **E_Port:** A port that forms the connection between two FC switches. This port is also known as the *expansion port*. The E_Port on an FC switch connects to the E_Port of another FC switch in the fabric through ISLs.
- **F_Port:** A port on a switch that connects an N_Port. It is also known as a *fabric port*.
- **G_Port:** A generic port on a switch that can operate as an E_Port or an F_Port and determines its functionality automatically during initialization.

Figure 5-11 shows various FC ports located in a switched fabric.

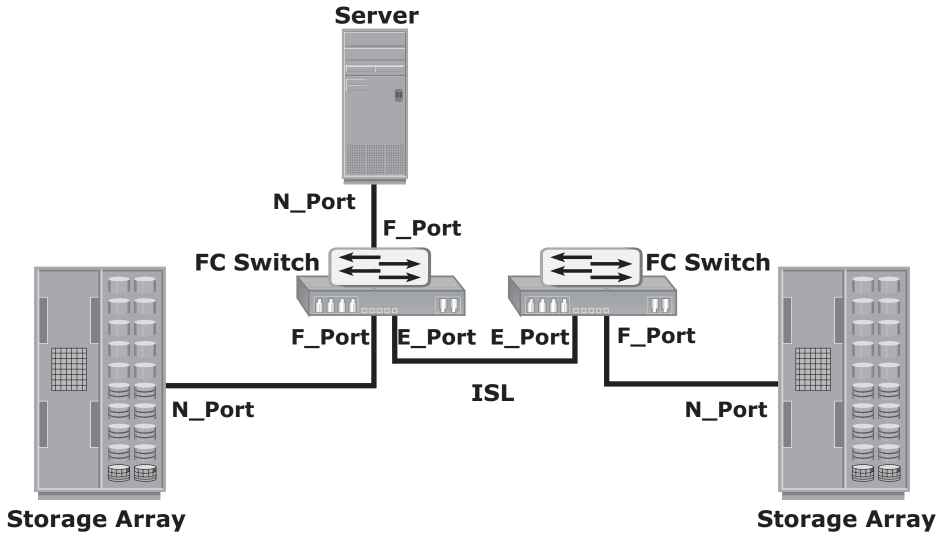


Figure 5-11: Switched fabric ports

5.6 Fibre Channel Architecture

Traditionally, host computer operating systems have communicated with peripheral devices over channel connections, such as ESCON and SCSI. Channel technologies provide high levels of performance with low protocol overheads. Such performance is achievable due to the static nature of channels and the high level of hardware and software integration provided by the channel technologies.

However, these technologies suffer from inherent limitations in terms of the number of devices that can be connected and the distance between these devices.

In contrast to channel technology, network technologies are more flexible and provide greater distance capabilities. Network connectivity provides greater scalability and uses shared bandwidth for communication. This flexibility results in greater protocol overhead and reduced performance.

The FC architecture represents true channel/network integration and captures some of the benefits of both channel and network technology. FC SAN uses the *Fibre Channel Protocol* (FCP) that provides both channel speed for data transfer with low protocol overhead and scalability of network technology.

FCP forms the fundamental construct of the FC SAN infrastructure. Fibre Channel provides a serial data transfer interface that operates over copper wire and optical fiber. FCP is the implementation of serial SCSI over an FC network. In FCP architecture, all external and remote storage devices attached to the SAN appear as local devices to the host operating system. The key advantages of FCP are as follows:

- Sustained transmission bandwidth over long distances.
- Support for a larger number of addressable devices over a network. Theoretically, FC can support more than 15 million device addresses on a network.
- Support speeds up to 16 Gbps (16 GFC).

5.6.1 Fibre Channel Protocol Stack

It is easier to understand a communication protocol by viewing it as a structure of independent layers. FCP defines the communication protocol in five layers: FC-0 through FC-4 (except FC-3 layer, which is not implemented). In a layered communication model, the peer layers on each node talk to each other through defined protocols. Figure 5-12 illustrates the Fibre Channel protocol stack.

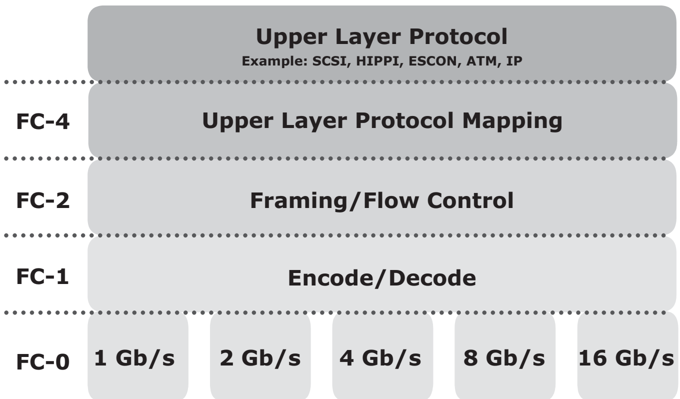


Figure 5-12: Fibre Channel protocol stack

FC-4 Layer

FC-4 is the uppermost layer in the FCP stack. This layer defines the application interfaces and the way *Upper Layer Protocols* (ULPs) are mapped to the lower FC layers. The FC standard defines several protocols that can operate on the FC-4 layer (see Figure 5-12). Some of the protocols include SCSI, High Performance Parallel Interface (HIPPI) Framing Protocol, Enterprise Storage Connectivity (ESCON), Asynchronous Transfer Mode (ATM), and IP.

FC-2 Layer

The FC-2 layer provides Fibre Channel addressing, structure, and organization of data (frames, sequences, and exchanges). It also defines fabric services, classes of service, flow control, and routing.

FC-1 Layer

The FC-1 layer defines how data is encoded prior to transmission and decoded upon receipt. At the transmitter node, an 8-bit character is encoded into a 10-bit transmissions character. This character is then transmitted to the receiver node. At the receiver node, the 10-bit character is passed to the FC-1 layer, which decodes the 10-bit character into the original 8-bit character. FC links with speeds of 10 Gbps and above use 64-bit to 66-bit encoding algorithms. The FC-1 layer also defines the transmission words, such as FC frame delimiters, which identify the start and end of a frame and primitive signals that indicate events at a transmitting port. In addition to these, the FC-1 layer performs link initialization and error recovery.

FC-0 Layer

FC-0 is the lowest layer in the FCP stack. This layer defines the physical interface, media, and transmission of bits. The FC-0 specification includes cables, connectors, and optical and electrical parameters for a variety of data rates. The FC transmission can use both electrical and optical media.

Mainframe SANs use *Fibre Connectivity* (FICON) for a low-latency, high-bandwidth connection to the storage controller. FICON was designed as a replacement for *Enterprise System Connection* (ESCON) to support mainframe-attached storage systems.

5.6.2 Fibre Channel Addressing

An FC address is dynamically assigned when a node port logs on to the fabric. The FC address has a distinct format, as shown in Figure 5-13. The addressing mechanism provided here corresponds to the fabric with the switch as an interconnecting device.

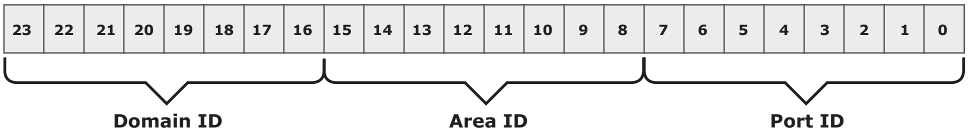


Figure 5-13: 24-bit FC address of N_Port

The first field of the FC address contains the domain ID of the switch. A *domain ID* is a unique number provided to each switch in the fabric. Although this is an 8-bit field, there are only 239 available addresses for domain ID because some addresses are deemed special and reserved for fabric management services. For example, FFFFFC is reserved for the name server, and FFFFFE is reserved for the fabric login service. The *area ID* is used to identify a group of switch ports used for connecting nodes. An example of a group of ports with a common area ID is a port card on the switch. The last field, the *port ID*, identifies the port within the group.

Therefore, the maximum possible number of node ports in a switched fabric is calculated as:

239 domains × 256 areas × 256 ports = 15,663,104

N_PORT ID VIRTUALIZATION (NPIV)

***NPIV* is a Fibre Channel configuration that enables multiple N_Port IDs to share a single physical N_Port. A typical use of NPIV would be for SAN storage provisioning to virtual machines in a virtualized server environment. With NPIV, several virtual machines on a host may share a common physical N_Port in the host, with each virtual machine using its own N_Port_ID for that physical node port. For this to work, the FC switch must be NPIV-enabled.**

5.6.3 World Wide Names

Each device in the FC environment is assigned a 64-bit unique identifier called the *World Wide Name* (WWN). The Fibre Channel environment uses two types of WWNs: *World Wide Node Name* (WWNN) and *World Wide Port Name* (WWPN). Unlike an FC address, which is assigned dynamically, a WWN is a static name for each node on an FC network. WWNs are similar to the Media Access Control (MAC) addresses used in IP networking. WWNs are *burned* into the hardware or assigned through software. Several configuration definitions in a SAN use WWN for identifying storage devices and HBAs. The name server in an FC environment keeps the association of WWNs to the dynamically created FC addresses for nodes. Figure 5-14 illustrates the WWN structure examples for an array and an HBA.

World Wide Name - Array															
5	0	0	6	0	1	6	0	0	0	6	0	0	1	B	2
0101	0000	0000	0110	0000	0001	0110	0000	0000	0000	0110	0000	0000	0001	1011	0010
Format Type	Company ID 24 bits						Port	Model Seed 32 bits							

World Wide Name - HBA															
1	0	0	0	0	0	0	0	c	9	2	0	d	c	4	0
Format Type	Reserved 12 bits			Company ID 24 bits						Company Specific 24 bits					

Figure 5-14: World Wide Names

5.6.4 FC Frame

An FC frame (Figure 5-15) consists of five parts: *start of frame* (SOF), *frame header*, *data field*, *cyclic redundancy check* (CRC), and *end of frame* (EOF).

SOF 4 Bytes	Frame Header 24 Bytes	Data Field 0 – 2112 Bytes	CRC 4 Bytes	EOF 4 Bytes

R_CTL	Destination ID	
CS_CTL	Source ID	
TYPE	F_CTL	
SEQ_ID	DF_CTL	Sequence Count
OX_ID		RX_ID
Offset		

Figure 5-15: FC frame

The SOF and EOF act as delimiters. In addition to this role, the SOF also indicates whether the frame is the first frame in a sequence of frames.

The frame header is 24 bytes long and contains addressing information for the frame. It includes the following information: Source ID (S_ID), Destination ID (D_ID), Sequence ID (SEQ_ID), Sequence Count (SEQ_CNT), Originating Exchange ID (OX_ID), and Responder Exchange ID (RX_ID), in addition to some control fields.

The S_ID and D_ID are FC addresses for the source port and the destination port, respectively. The SEQ_ID and OX_ID identify the frame as a component of a specific sequence and exchange, respectively.

The frame header also defines the following fields:

- **Routing Control (R_CTL):** This field denotes whether the frame is a link control frame or a data frame. Link control frames are frames that do not carry any user data. These frames are used for setup and messaging. In contrast, data frames carry the user data.
- **Class Specific Control (CS_CTL):** This field specifies link speeds for class 1 and class 4 data transmission. (Class of service is discussed in section 5.6.7 "Classes of Service" later in the chapter.)
- **TYPE:** This field describes the upper layer protocol (ULP) to be carried on the frame if it is a data frame. However, if it is a link control frame, this field is used to signal an event such as "fabric busy." For example, if the TYPE is 08, and the frame is a data frame, it means that the SCSI will be carried on an FC.
- **Data Field Control (DF_CTL):** A 1-byte field that indicates the existence of any optional headers at the beginning of the data payload. It is a mechanism to extend header information into the payload.
- **Frame Control (F_CTL):** A 3-byte field that contains control information related to frame content. For example, one of the bits in this field indicates whether this is the first sequence of the exchange.

The data field in an FC frame contains the data payload, up to 2,112 bytes of actual data with 36 bytes of fixed overhead.

The CRC checksum facilitates error detection for the content of the frame. This checksum verifies data integrity by checking whether the content of the frames are received correctly. The CRC checksum is calculated by the sender before encoding at the FC-1 layer. Similarly, it is calculated by the receiver after decoding at the FC-1 layer.

5.6.5. Structure and Organization of FC Data

In an FC network, data transport is analogous to a conversation between two people, whereby a frame represents a word, a sequence represents a sentence, and an exchange represents a conversation.

- **Exchange:** An exchange operation enables two node ports to identify and manage a set of information units. Each upper layer protocol has its protocol-specific information that must be sent to another port to perform certain operations. This protocol-specific information is called an information unit. The structure of these information units is defined in the FC-4 layer. This unit maps to a sequence. An exchange is composed of one or more sequences.
- **Sequence:** A sequence refers to a contiguous set of frames that are sent from one port to another. A sequence corresponds to an information unit, as defined by the ULP.
- **Frame:** A frame is the fundamental unit of data transfer at Layer 2. Each frame can contain up to 2,112 bytes of payload.

5.6.6 Flow Control

Flow control defines the pace of the flow of data frames during data transmission. FC technology uses two flow-control mechanisms: buffer-to-buffer credit (BB_Credit) and end-to-end credit (EE_Credit).

BB_Credit

FC uses the *BB_Credit* mechanism for flow control. BB_Credit controls the maximum number of frames that can be present over the link at any given point in time. In a switched fabric, BB_Credit management may take place between any two FC ports. The transmitting port maintains a count of free receiver buffers and continues to send frames if the count is greater than 0. The BB_Credit mechanism uses *Receiver Ready* (R_RDY) primitive that indicates a buffer has been freed on the port that transmitted the R_RDY.

EE_Credit

The function of end-to-end credit, known as *EE_Credit*, is similar to that of BB_Credit. When an initiator and a target establish themselves as nodes communicating with each other, they exchange the EE_Credit parameters (part of Port login). The EE_Credit mechanism provides the flow control for class 1 and class 2 traffic only.

5.6.7 Classes of Service

The FC standards define different classes of service to meet the requirements of a wide range of applications. Table 5-1 shows three classes of services and their features.

Table 5-1: FC Class of Services

	CLASS 1	CLASS 2	CLASS 3
Communication type	Dedicated connection	Nondedicated connection	Nondedicated connection
Flow control	End-to-end credit	End-to-end credit B-to-B credit	B-to-B credit
Frame delivery	In order delivery	Order not guaranteed	Order not guaranteed
Frame acknowledgment	Acknowledged	Acknowledged	Not acknowledged
Multiplexing	No	Yes	Yes
Bandwidth utilization	Poor	Moderate	High

Another class of service is *class F*, which is used for fabric management. Class F is similar to Class 2 and provides notification of nondelivery of frames.

5.7 Fabric Services

All FC switches, regardless of the manufacturer, provide a common set of services as defined in the Fibre Channel standards. These services are available at certain predefined addresses. Some of these services are Fabric Login Server, Fabric Controller, Name Server, and Management Server (see Figure 5-16).

The *Fabric Login Server* is located at the predefined address of FFFFFE and is used during the initial part of the node's fabric login process.

The *Name Server* (formally known as *Distributed Name Server*) is located at the predefined address FFFFFC and is responsible for name registration and management of node ports. Each switch exchanges its Name Server information with other switches in the fabric to maintain a synchronized, distributed name service.

Each switch has a *Fabric Controller* located at the predefined address FFFFFD. The Fabric Controller provides services to both node ports and other switches. The Fabric Controller is responsible for managing and distributing Registered State Change Notifications (RSCNs) to the node ports registered with the

Fabric Controller. If there is a change in the fabric, RSCNs are sent out by a switch to the attached node ports. The Fabric Controller also generates Switch Registered State Change Notifications (SW-RSCNs) to every other domain (switch) in the fabric. These RSCNs keep the name server up-to-date on all switches in the fabric.

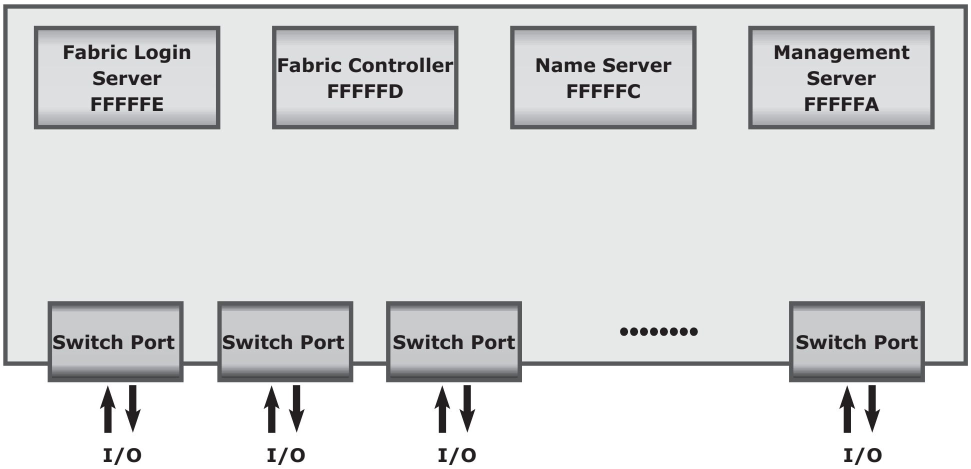


Figure 5-16: Fabric services provided by FC switches

FFFFFA is the Fibre Channel address for the *Management Server*. The Management Server is distributed to every switch within the fabric. The Management Server enables the FC SAN management software to retrieve information and administer the fabric.

5.8 Switched Fabric Login Types

Fabric services define three login types:

- ***Fabric login*** **(FLOGI):** Performed between an N_Port and an F_Port. To log on to the fabric, a node sends a FLOGI frame with the WWNN and WWPN parameters to the login service at the predefined FC address FFFFFE (Fabric Login Server). In turn, the switch accepts the login and returns an Accept (ACC) frame with the assigned FC address for the node. Immediately after the FLOGI, the N_Port registers itself with the local Name Server on the switch, indicating its WWNN, WWPN, port type, class of service, assigned FC address and so on. After the N_Port has logged in, it can query the name server database for information about all other logged in ports.

- *Port login* **(PLOGI):** Performed between two N_Ports to establish a session. The initiator N_Port sends a PLOGI request frame to the target N_Port, which accepts it. The target N_Port returns an ACC to the initiator N_Port. Next, the N_Ports exchange service parameters relevant to the session.
- *Process login* **(PRLI):** Also performed between two N_Ports. This login relates to the FC-4 ULPs, such as SCSI. If the ULP is SCSI, N_Ports exchange SCSI-related service parameters.

5.9 Zoning

Zoning is an FC switch function that enables node ports within the fabric to be logically segmented into groups and to communicate with each other within the group (see Figure 5-17).

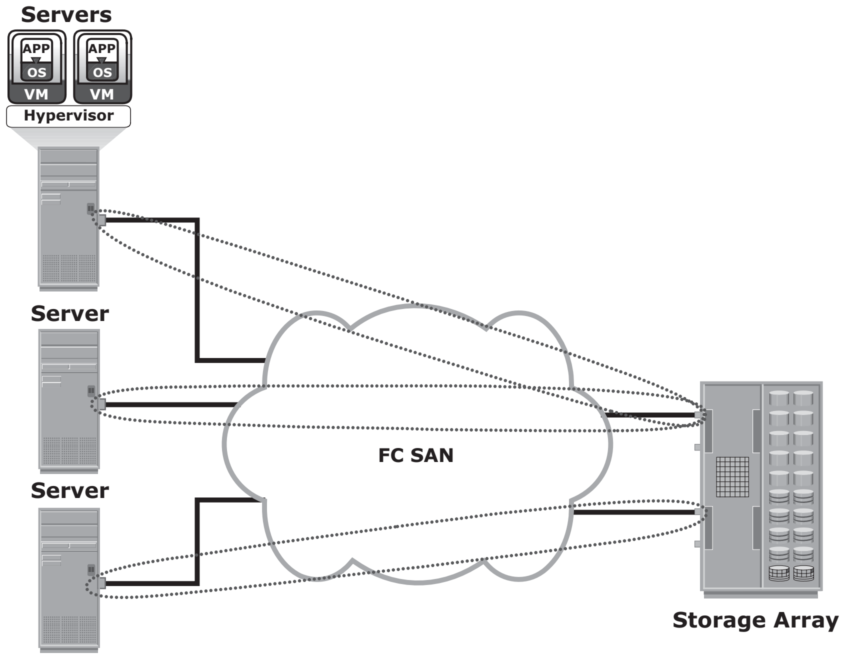


Figure 5-17: Zoning

Whenever a change takes place in the name server database, the fabric controller sends a Registered State Change Notification (RSCN) to all the nodes impacted by the change. If zoning is not configured, the fabric controller sends an RSCN to all the nodes in the fabric. Involving the nodes that are not impacted by the change results in increased fabric-management traffic. For a large fabric, the amount of FC traffic generated due to this process can be significant and might impact the host-to-storage data traffic. Zoning helps to limit the number of RSCNs in a fabric. In the presence of zoning, a fabric sends the RSCN to only those nodes in a zone where the change has occurred.

Zone members, zones, and zone sets form the hierarchy defined in the zoning process (see Figure 5-18). A *zone set* is composed of a group of zones that can be activated or deactivated as a single entity in a fabric. Multiple zone sets may be defined in a fabric, but only one zone set can be active at a time. *Members* are nodes within the SAN that can be included in a zone. Switch ports, HBA ports, and storage device ports can be members of a zone. A port or node can be a member of multiple zones. Nodes distributed across multiple switches in a switched fabric may also be grouped into the same zone. Zone sets are also referred to as *zone configurations*.

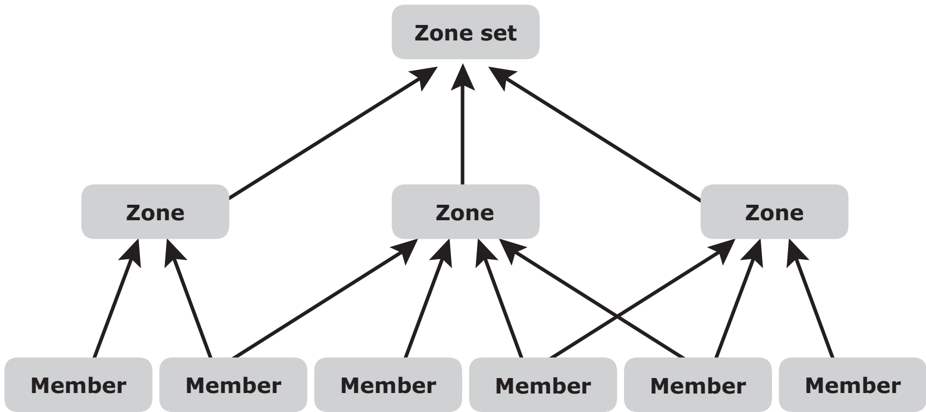


Figure 5-18: Members, zones, and zone sets

Zoning provides control by allowing only the members in the same zone to establish communication with each other.

5.9.1 Types of Zoning

Zoning can be categorized into three types:

- **Port zoning:** Uses the physical address of switch ports to define zones. In port zoning, access to node is determined by the physical switch port to which a node is connected. The zone members are the port identifier (switch domain ID and port number) to which HBA and its targets (storage devices) are connected. If a node is moved to another switch port in the fabric, then zoning must be modified to allow the node, in its new port, to participate in its original zone. However, if an HBA or storage device port fails, an administrator just has to replace the failed device without changing the zoning configuration.

- **WWN zoning:** Uses World Wide Names to define zones. The zone members are the unique WWN addresses of the HBA and its targets (storage devices). A major advantage of WWN zoning is its flexibility. WWN zoning allows nodes to be moved to another switch port in the fabric and maintain connectivity to its zone partners without having to modify the zone configuration. This is possible because the WWN is static to the node port.
- **Mixed zoning:** Combines the qualities of both WWN zoning and port zoning. Using mixed zoning enables a specific node port to be tied to the WWN of another node.

Figure 5-19 shows the three types of zoning on an FC network.

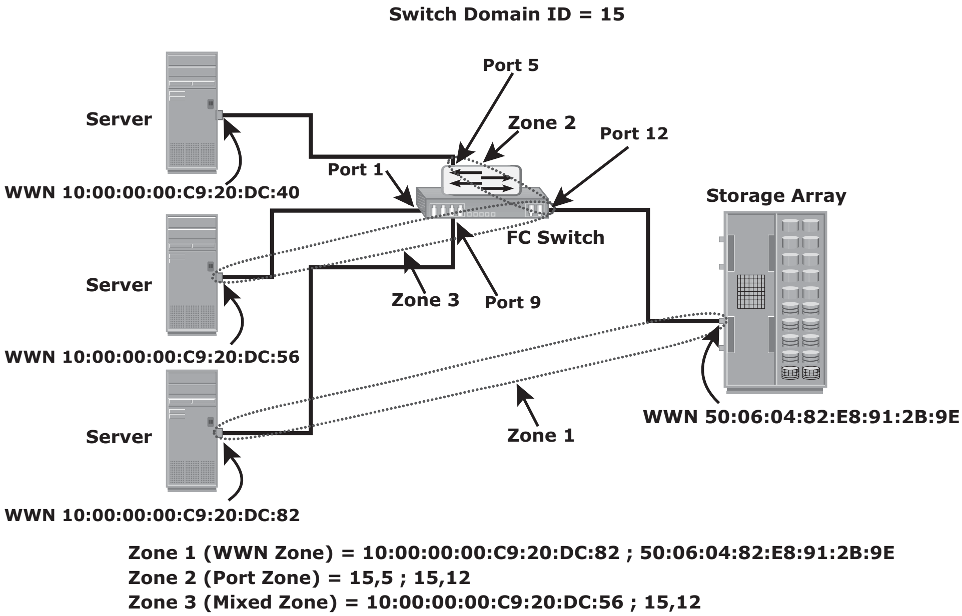


Figure 5-19: Types of zoning

Zoning is used with LUN masking to control server access to storage. However, these are two different activities. Zoning takes place at the fabric level and LUN masking is performed at the array level.

SINGLE HBA ZONING

Single HBA zoning is considered as an industry best practice to configure a zone set. A single HBA zone consists of one HBA port and one or more storage device ports. Single HBA zoning eliminates unnecessary host-to-host interaction and minimizes RSCNs. Single HBA zoning in a large fabric leads to configuring a large number of zones and more administrative actions. However, this practice improves the FC SAN performance and reduces the time to troubleshoot FC SAN-related problems.

5.10 FC SAN Topologies

Fabric design follows standard topologies to connect devices. Core-edge fabric is one of the popular topologies for fabric designs. Variations of core-edge fabric and mesh topologies are most commonly deployed in FC SAN implementations.

5.10.1 Mesh Topology

A mesh topology may be one of the two types: full mesh or partial mesh. In a *full mesh*, every switch is connected to every other switch in the topology. A full mesh topology may be appropriate when the number of switches involved is small. A typical deployment would involve up to four switches or directors, with each of them servicing highly localized host-to-storage traffic. In a full mesh topology, a maximum of one ISL or hop is required for host-to-storage traffic. However, with the increase in the number of switches, the number of switch ports used for ISL also increases. This reduces the available switch ports for node connectivity.

In a *partial mesh topology*, several hops or ISLs may be required for the traffic to reach its destination. Partial mesh offers more scalability than full mesh topology. However, without proper placement of host and storage devices, traffic management in a partial mesh fabric might be complicated and ISLs could become overloaded due to excessive traffic aggregation. Figure 5-20 depicts both partial mesh and full mesh topologies.

A SINGLE-SWITCH TOPOLOGY

A single-switch fabric consists of only a single switch or single director. This topology is becoming popular, especially in large data centers, due to their inherent simplicity. Larger port count and modular and scalable architecture of switches and directors allow SAN design to start small and grow as needed by adding port cards/blades in the switch rather than adding new switches.

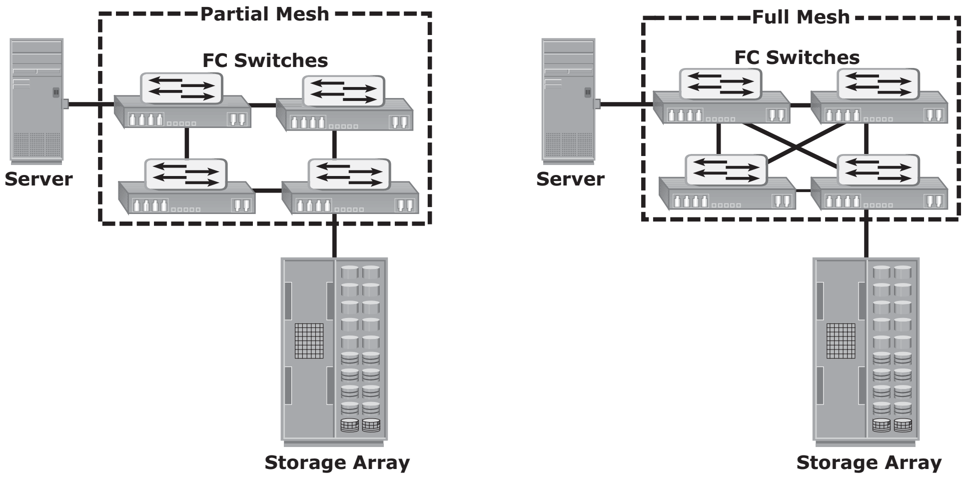


Figure 5-20: Partial mesh and full mesh topologies

5.10.2 Core-Edge Fabric

The *core-edge fabric* topology has two types of switch tiers. The *edge tier* is usually composed of switches and offers an inexpensive approach to adding more hosts in a fabric. Each switch at the edge tier is attached to a switch at the core tier through ISLs.

The *core tier* is usually composed of enterprise directors that ensure high fabric availability. In addition, typically all traffic must either traverse this tier or terminate at this tier. In this configuration, all storage devices are connected to the core tier, enabling host-to-storage traffic to traverse only one ISL. Hosts that require high performance may be connected directly to the core tier and consequently avoid ISL delays.

In *core-edge topology*, the edge-tier switches are not connected to each other. The core-edge fabric topology increases connectivity within the SAN while conserving the overall port utilization. If fabric expansion is required, additional edge switches are connected to the core. The core of the fabric is also extended by adding more switches or directors at the core tier. Based on the number of core-tier switches, this topology has different variations, such as, *single-core topology* (see Figure 5-21) and *dual-core topology* (see Figure 5-22). To transform a single-core topology to dual-core, new ISLs are created to connect each edge switch to the new core switch in the fabric.

Benefits and Limitations of Core-Edge Fabric

The core-edge fabric provides maximum one-hop storage access to all storage devices in the system. Because traffic travels in a deterministic pattern (from the edge to the core and vice versa), a core-edge provides easier calculation of the ISL load and traffic patterns. In this topology, because each tier's switch port is used for either storage or hosts, it's easy to identify which network resources are approaching their capacity, making it easier to develop a set of rules for scaling and apportioning.

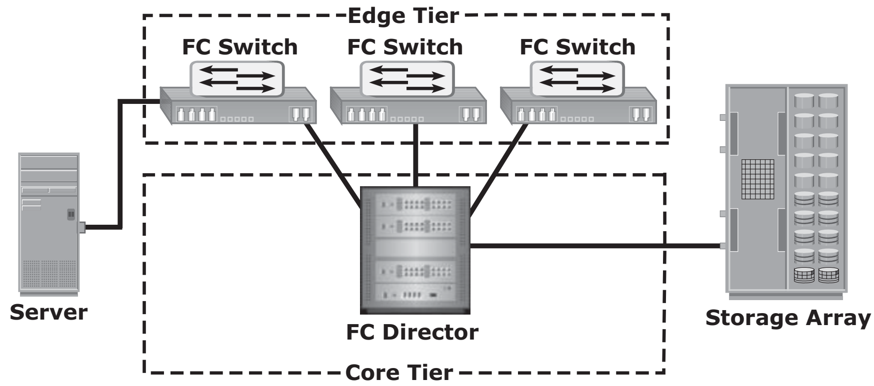


Figure 5-21: Single-core topology

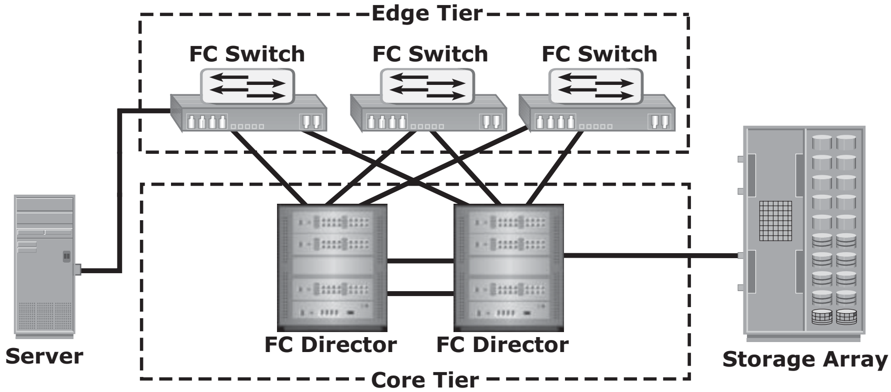


Figure 5-22: Dual-core topology

Core-edge fabrics are scaled to larger environments by adding more core switches and linking them, or adding more edge switches. This method enables extending the existing simple core-edge model or expanding the fabric into a compound or complex core-edge model.

However, the core-edge fabric might lead to some performance-related problems because scaling a core-edge topology involves increasing the number of hop counts in the fabric. *Hop count* represents the total number of ISLs traversed by a packet between its source and destination. A common best practice is to keep the number of host-to-storage hops unchanged, at one hop, in a core-edge. Generally, a large hop count means a high data transmission delay between the source and destination.

As the number of cores increases, it is prohibitive to continue to maintain ISLs from each core to each edge switch. When this happens, the fabric design is changed to a compound or complex core-edge design (see Figure 5-23).

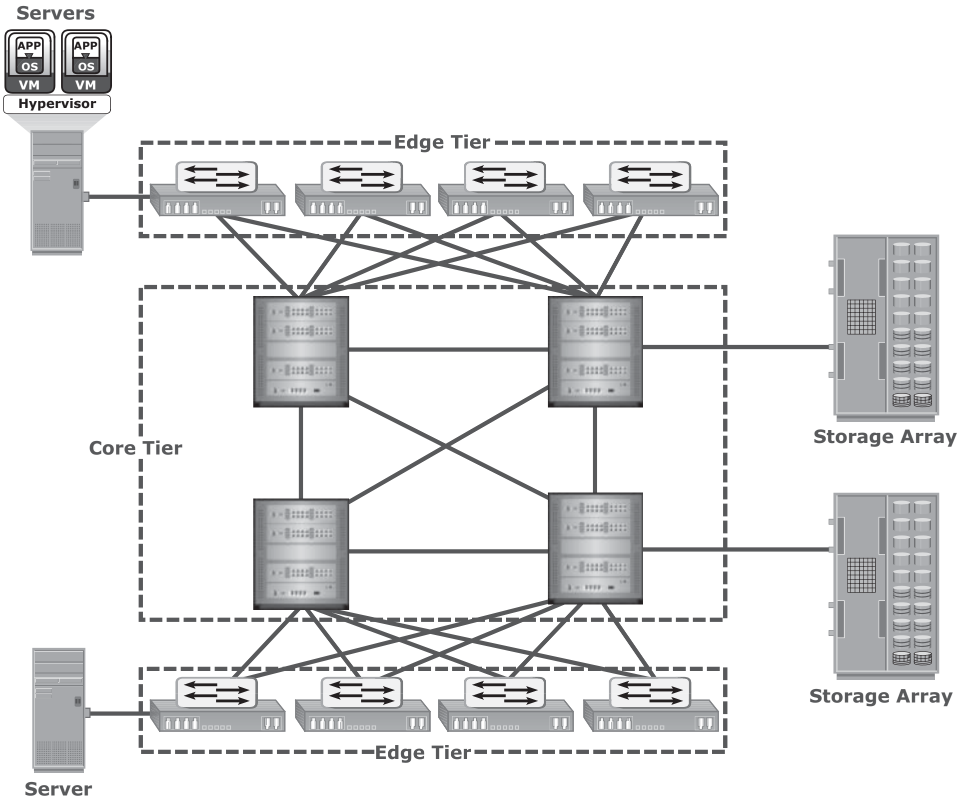


Figure 5-23: Compound core-edge topology

FAN-OUT AND FAN-IN

Fan-out **enables multiple server ports to communicate to a single storage port. A four-server connection to a single-storage port results in a fan-out ratio of 4. The fan-out ratio of a storage port is dependent on the capabilities of the storage system. The key parameter that governs the fan-out ratio of a storage port is the front-end processing capability of the storage system. Typically, the product vendor specifies the fan-out ratio of a storage system.**

Fan-in **refers to the number of storage ports that a single server port uses. Similar to fan-out, the restriction on fan-in is based on the capability of the host-bus adapter.**

5.11 Virtualization in SAN

This section details two network-based virtualization techniques in a SAN environment: block-level storage virtualization and virtual SAN (VSAN).

5.11.1 Block-level Storage Virtualization

Block-level storage virtualization aggregates block storage devices (LUNs) and enables provisioning of virtual storage volumes, independent of the underlying physical storage. A virtualization layer, which exists at the SAN, abstracts the identity of physical storage devices and creates a storage pool from heterogeneous storage devices. Virtual volumes are created from the storage pool and assigned to the hosts. Instead of being directed to the LUNs on the individual storage arrays, the hosts are directed to the virtual volumes provided by the virtualization layer. For hosts and storage arrays, the virtualization layer appears as the target and initiator devices, respectively. The virtualization layer maps the virtual volumes to the LUNs on the individual arrays. The hosts remain unaware of the mapping operation and access the virtual volumes as if they were accessing the physical storage attached to them. Typically, the virtualization layer is managed via a dedicated virtualization appliance to which the hosts and the storage arrays are connected.

Figure 5-24 illustrates a virtualized environment. It shows two physical servers, each of which has one virtual volume assigned. These virtual volumes are used by the servers. These virtual volumes are mapped to the LUNs in the storage arrays. When an I/O is sent to a virtual volume, it is redirected through the virtualization layer at the storage network to the mapped LUNs. Depending on the capabilities of the virtualization appliance, the architecture may allow for more complex mapping between array LUNs and virtual volumes.

Block-level storage virtualization enables extending the storage volumes online to meet application growth requirements. It consolidates heterogeneous storage arrays and enables transparent volume access.

Block-level storage virtualization also provides the advantage of nondisruptive data migration. In a traditional SAN environment, LUN migration from one array to another is an offline event because the hosts needed to be updated to reflect the new array configuration. In other instances, host CPU cycles were required to migrate data from one array to the other, especially in a multivendor environment. With a block-level virtualization solution in place, the virtualization layer handles the back-end migration of data, which enables LUNs to remain online and accessible while data is migrating. No physical changes are required because the host still points to the same virtual targets on the virtualization layer. However, the mappings information on the virtualization layer should be changed. These changes can be executed dynamically and are transparent to the end user.

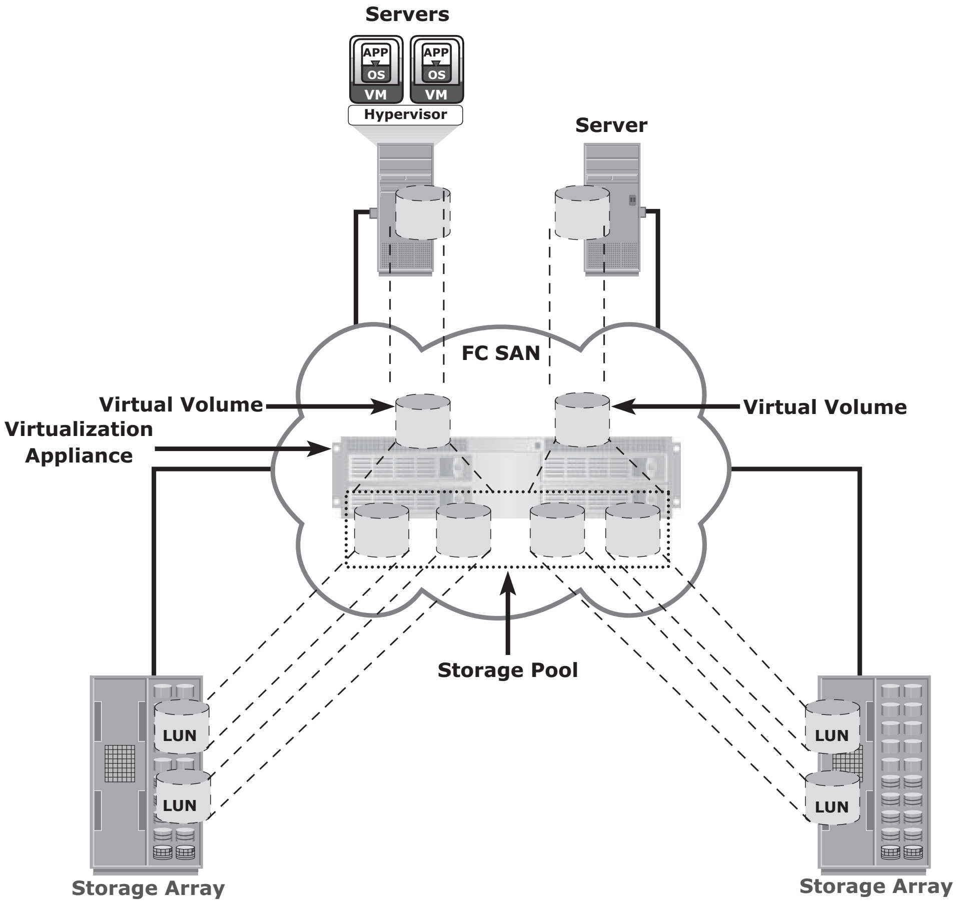


Figure 5-24: Block-level storage virtualization

Previously, block-level storage virtualization provided nondisruptive data migration only within a data center. The new generation of block-level storage virtualization enables nondisruptive data migration both within and between data centers. It provides the capability to connect the virtualization layers at multiple data centers. The connected virtualization layers are managed centrally and work as a single virtualization layer stretched across data centers (see Figure 5-25). This enables the federation of block-storage resources both within and across data centers. The virtual volumes are created from the federated storage resources.

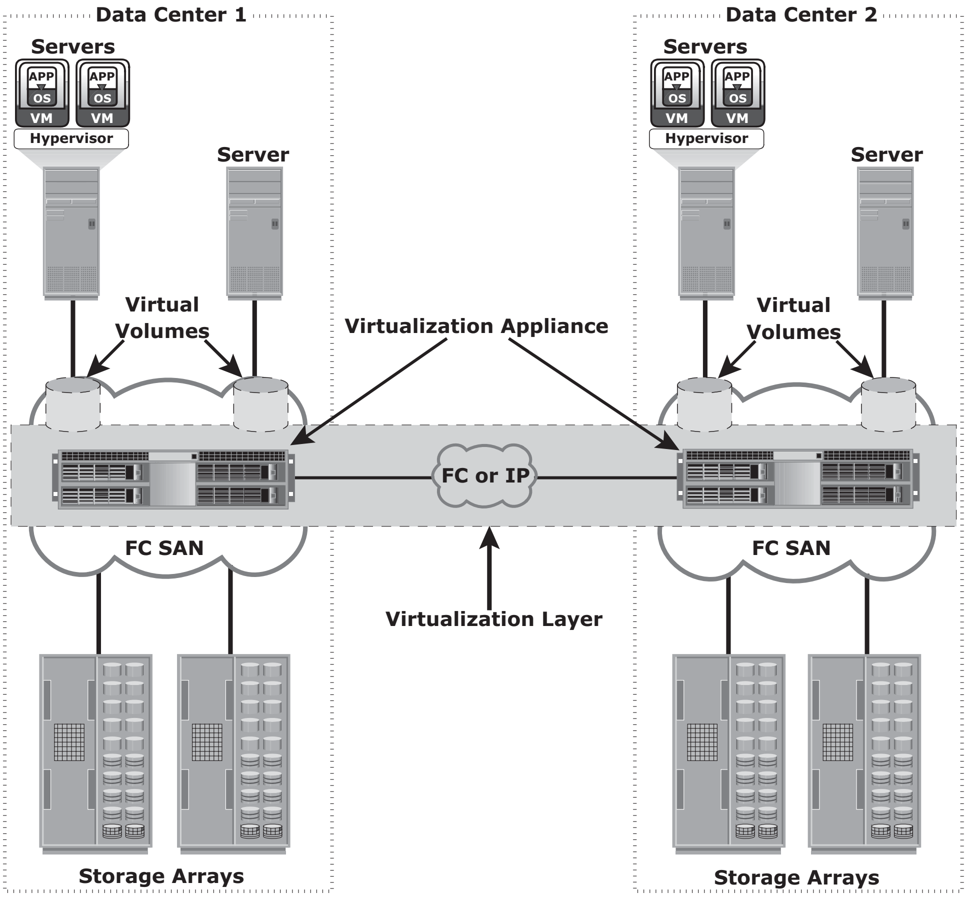


Figure 5-25: Federation of block storage across data centers

5.11.2 Virtual SAN (VSAN)

Virtual SAN (also called *virtual fabric*) is a logical fabric on an FC SAN, which enables communication among a group of nodes regardless of their physical location in the fabric. In a VSAN, a group of hosts or storage ports communicate with each other using a virtual topology defined on the physical SAN. Multiple VSANs may be created on a single physical SAN. Each VSAN acts as an independent fabric with its own set of fabric services, such as name server, and zoning. Fabric-related configurations in one VSAN do not affect the traffic in another.

VSANs improve SAN security, scalability, availability, and manageability. VSANs provide enhanced security by isolating the sensitive data in a VSAN and by restricting access to the resources located within that VSAN. The same Fibre Channel address can be assigned to nodes in different VSANs, thus increasing the fabric scalability. Events causing traffic disruptions in one VSAN are contained within that VSAN and are not propagated to other VSANs. VSANs facilitate an easy, flexible, and less expensive way to manage networks. Configuring VSANs is easier and quicker compared to building separate physical FC SANs for various node groups. To regroup nodes, an administrator simply changes the VSAN configurations without moving nodes and recabling. VSAN is further discussed in Chapter 14.

5.12 Concepts in Practice: EMC Connectrix and EMC VPLEX

The EMC Connectrix family represents the industry's most extensive selection of networked storage connectivity products. Connectrix integrates high-speed Fibre Channel connectivity, highly resilient switching technology, options for intelligent IP storage networking, and I/O consolidation with products that support Fibre Channel over Ethernet.

EMC VPLEX is the next-generation solution for block-level virtualization and data mobility within, across, and between data centers. EMC VPLEX provides storage federation by aggregating storage arrays that can be located either in a single data center or multiple data centers. VPLEX is also used as the data mobility solution for environments like cloud computing.

For the latest information on Connectrix connectivity products and VPLEX, visit `www.emc.com`.

5.12.1 EMC Connectrix

EMC offers the following connectivity products under the Connectrix brand (see Figure 5-26):

- Enterprise directors
- Departmental switches
- Multi-purpose switches

Enterprise directors are ideal for large enterprise connectivity. They offer high port density and high component redundancy. Enterprise directors are deployed in high-availability or large-scale environments. Connectrix directors offer several hundred ports per domain. Departmental switches are best suited for workgroup, mid-tier environments. Multi-purpose switches support various protocols such as iSCSI, FCIP, FCoE, FICON, in addition to FC protocol. In addition to FC ports, Connectrix switches and directors have Ethernet ports and serial ports for communication and switch management functions. The Connectrix management software enables configuration, monitoring, and management of Connectrix switches.

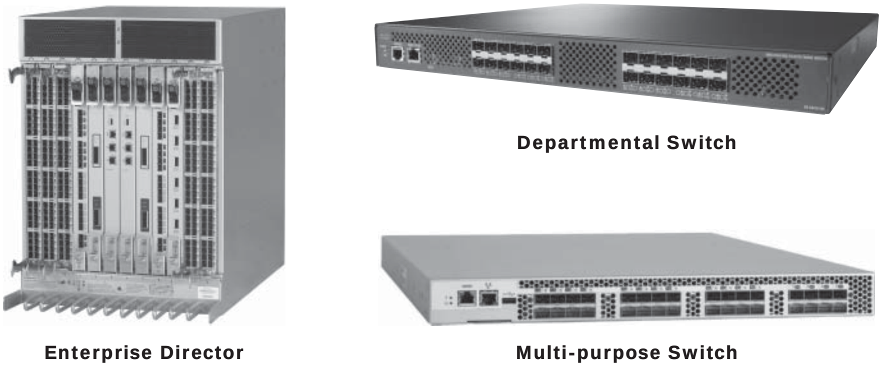


Figure 5-26: EMC Connectrix

Connectrix Switches

B-series and MDS-series make up the Connectrix family of switches offered by EMC. These switches are designed to meet workgroup, department-level, and enterprise-level requirements. They are designed with a nonblocking architecture and can operate in heterogeneous environments. Nonblocking architecture refers to the capability of a switch to handle independent packets simultaneously because the switch has sufficient internal resources to handle maximum transfer rates from all ports. The features of these switches that ensure their high availability are their nondisruptive software and port upgrade, and redundant and hot-swappable components. These switches can be managed through CLI, HTTP, and standalone GUI applications.

Connectrix Directors

EMC offers the high-end Connectrix family of directors. Their modular architectural design offers high scalability by providing over 500 ports. They are suitable for server and storage consolidation across enterprises. These directors have redundant components for high availability and provide multiprotocol connectivity for both mainframe and open system environments. Connectrix directors offer high speeds (up to 16 Gb/s) and support ISL aggregation. Similar to switches, directors can also be managed through CLI or with other GUI tools.

Connectrix Multi-purpose Switches

Multi-purpose switches provide support for multiple protocols, such as FC, FCIP, iSCSI, FCoE, and FICON. They perform protocol translation and route frames between two dissimilar networks, such as FC and IP. These multiprotocol capabilities offer many benefits, including long-distance SAN extension, greater resource sharing, and simplified management. Connectrix multi-purpose switches include FCoE switches, FCIP routers, iSCSI gateways, and so on.

Connectrix Management Tools

There are several ways to monitor and manage FC switches in a fabric. Individual switch management is accomplished through the CLI or browser-based tools.

Command-line utilities such as Telnet and Secure Shell (SSH) are used to log on to the switch over IP and issue CLI commands. The primary purpose of the CLI is to automate the management of a large number of switches or directors with the use of scripts. The browser-based tools provide GUIs. These tools also display the topology map.

Fabric-wide management and monitoring is accomplished by using vendor-specific tools and Simple Network Management Protocol (SNMP)-based, third-party software.

EMC ControlCenter SAN Manager provides a single interface for managing a Storage Area Network. With SAN Manager, an administrator can discover, monitor, manage, and configure complex heterogeneous SAN environments. It streamlines and centralizes SAN management operations across multivendor storage networks and storage devices. It enables storage administrators to manage SAN zones and LUN masking consistently across multivendor SAN arrays and switches. EMC ControlCenter SAN Manager also supports virtual environments, including VMware, and virtual SANs.

EMC ProSphere is a newly launched tool with additional features specifically for the cloud computing environment. A future release of EMC ProSphere will include all the functionalities of EMC ControlCenter.

5.12.2 EMC VPLEX

EMC VPLEX provides a virtual storage infrastructure that enables federation of heterogeneous storage resources both within and across datacenters. The VPLEX appliance resides between the servers and heterogeneous storage devices. It forms a pool of distributed block storage resources and enables creating virtual storage volumes from the pool. These virtual volumes are then allocated to the servers. The virtual-to-physical-storage mapping remains hidden to the servers.

VPLEX provides nondisruptive data mobility among physical storage devices to balance the application workload and to enable both local and remote data access. The mapping of virtual volumes to physical volumes can be changed dynamically by the administrator. This allows for a virtual volume to be moved across storage arrays while still in production.

VPLEX uses a unique clustering architecture and distributed cache coherency that enable multiple hosts located across two locations to access a single copy of data. This eliminates the operational overhead and time required to copy and distribute data across locations VPLEX also provides the capability to mirror data of a virtual volume both within and across locations. This enables hosts at different data centers to access cache-coherent copies of the same virtual volume. Practical applications of this capability include mobility, load-balancing, and high availability across data centers.

To avoid application downtime due to outage at a data center, the workload can be moved quickly to another data center. Applications continue accessing the same virtual volume and remain uninterrupted by the data mobility.

VPLEX Family of Products

The VPLEX family consists of three products: VPLEX Local, VPLEX Metro, and VPLEX Geo.

EMC VPLEX Local delivers local federation, which provides simplified management and nondisruptive data mobility across heterogeneous arrays within a data center. EMC VPLEX Metro delivers distributed federation, which provides data access and mobility between two VPEX clusters within synchronous distances that support round-trip latency up to 5 ms. EMC VPLEX Geo delivers data access and mobility between two VPLEX clusters within asynchronous distances (that support round-trip latency up to 50 ms).

Summary

The FC SAN has enabled the consolidation of storage and benefited organizations by lowering the cost of storage infrastructure. FC SAN reduces overall operational cost and downtime. Virtualization of storage and storage networks further minimizes resource management complexity and cost. The adoption of FC SANs has increased with the decline of hardware prices and has enhanced to the maturity of storage network standards.

This chapter detailed the components of an FC SAN, its topologies, and the FC technology that forms its backbone. FC meets today's demands for reliable, and high-performance applications. The chapter also covered virtualization in a SAN environment.

The interoperability between FC switches from different vendors has enhanced significantly compared to early SAN deployments. The standards published by a dedicated study group within T11 on FC SAN routing, and the new product offerings from vendors, are now revolutionizing the way FC SANs are deployed and operated.

Although FC SANs have eliminated islands of storage, their implementation requires additional equipment and infrastructure in an enterprise. The emergence of the iSCSI and FCIP technologies, detailed in Chapter 6, has pushed the convergence of FC SAN with IP technology, providing a cost-effective method to leverage existing IP based infrastructure for storage networking.

EXERCISES

1. **What is zoning? Discuss a scenario:**
   a. **Where WWN zoning is preferred over port zoning.**
   b. **Where port zoning is preferred over WWN zoning.**
2. **Describe the process of assigning an FC address to a node when logging on to the network for the first time.**
3. **Seventeen switches, with 16 ports each, are connected in a full mesh topology. How many ports are available for host and storage connectivity?**
4. **Discuss the roles of the name server and fabric controller in an FC-switched fabric.**
5. **How does flow control work in an FC network?**
6. **Explain storage migration using block-level storage virtualization. Compare this migration with traditional migration methods.**
7. **How do VSANs improve the manageability of an FC SAN?**

Chapter 6
IP SAN and FCoE

KEY CONCEPTS

- iSCSI Protocol
- Native and Bridged iSCSI
- FCIP Protocol
- FCoE Protocol

Traditional SAN enables the transfer of block I/O over Fibre Channel and provides high performance and scalability. These advantages of FC SAN come with the additional cost of buying FC components, such as FC HBA and switches. Organizations typically have an existing Internet Protocol (IP)-based infrastructure, which could be leveraged for storage networking. Advancements in technology have enabled IP to be used for transporting block I/O over the IP network. This technology of transporting block I/Os over an IP is referred to as IP SAN. IP is a mature technology, and using IP as a storage networking option provides several advantages. When block I/O is run over IP, the existing network infrastructure can be leveraged, which is more economical than investing in a new SAN infrastructure. In addition, many robust and mature security options are now available for IP networks. Many long-distance, disaster recovery (DR) solutions are already leveraging IP-based networks. With IP SAN, organizations can extend the geographical reach of their storage infrastructure.

Two primary protocols that leverage IP as the transport mechanism are *Internet SCSI* (iSCSI) and *Fibre Channel over IP* (FCIP). iSCSI is encapsulation of SCSI I/O over IP. FCIP is a protocol in which an FCIP entity such as an FCIP gateway is used to tunnel FC fabrics through an IP network. In FCIP, FC frames are encapsulated onto the IP payload. An FCIP implementation is capable of merging interconnected fabrics into a single fabric. Frequently, only a small subset of nodes at either end require connectivity across fabrics. Thus, the majority of FCIP implementations today use switch-specific features such as IVR (Inter-VSAN Routing) or FCRS

(Fibre Channel Routing Services) to create a tunnel. In this manner, traffic may be routed between specific nodes without actually merging the fabrics.

This chapter describes both iSCSI and FCIP protocols, components, and topologies in detail. This chapter also covers an emerging protocol, *Fibre Channel over Ethernet* (FCoE). FCoE converges Ethernet and FC traffic over a single physical link. Therefore, it eliminates the complexity of managing two separate networks in the data center.

6.1 iSCSI

iSCSI is an IP based protocol that establishes and manages connections between host and storage over IP, as shown in Figure 6-1. iSCSI encapsulates SCSI commands and data into an IP packet and transports them using TCP/IP. iSCSI is widely adopted for connecting servers to storage because it is relatively inexpensive and easy to implement, especially in environments in which an FC SAN does not exist.

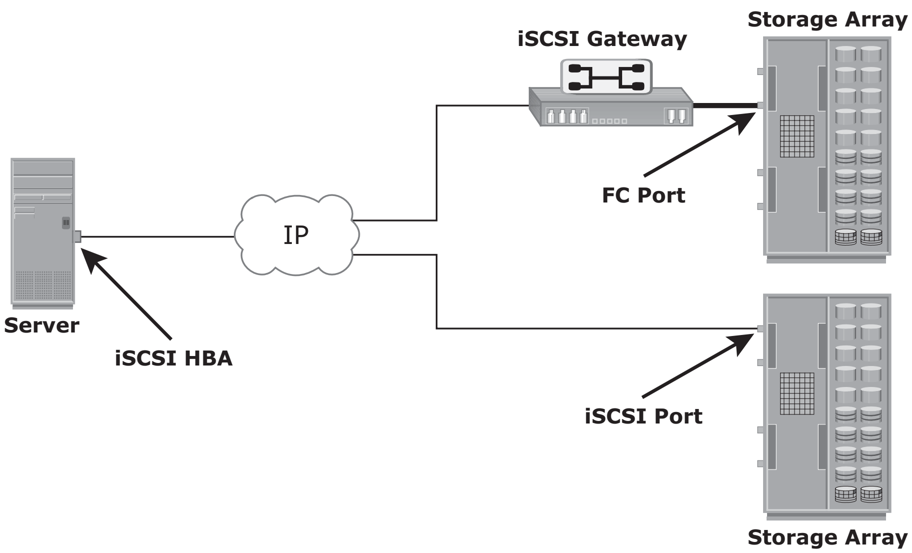


Figure 6-1: iSCSI implementation

6.1.1 Components of iSCSI

An initiator (host), target (storage or iSCSI gateway), and an IP-based network are the key iSCSI components. If an iSCSI-capable storage array is deployed, then a host with the iSCSI initiator can directly communicate with the storage array over an IP network. However, in an implementation that uses an existing FC array for iSCSI communication, an iSCSI gateway is used. These devices perform the translation of IP packets to FC frames and vice versa, thereby bridging the connectivity between the IP and FC environments.

6.1.2 iSCSI Host Connectivity

A standard NIC with software iSCSI initiator, a TCP offload engine (TOE) NIC with software iSCSI initiator, and an iSCSI HBA are the three iSCSI host connectivity options. The function of the iSCSI initiator is to route the SCSI commands over an IP network.

A standard NIC with a software iSCSI initiator is the simplest and least expensive connectivity option. It is easy to implement because most servers come with at least one, and in many cases two, embedded NICs. It requires only a software initiator for iSCSI functionality. Because NICs provide standard IP function, encapsulation of SCSI into IP packets and decapsulation are carried out by the host CPU. This places additional overhead on the host CPU. If a standard NIC is used in heavy I/O load situations, the host CPU might become a bottleneck. TOE NIC helps alleviate this burden. A TOE NIC offloads TCP management functions from the host and leaves only the iSCSI functionality to the host processor. The host passes the iSCSI information to the TOE card, and the TOE card sends the information to the destination using TCP/IP. Although this solution improves performance, the iSCSI functionality is still handled by a software initiator that requires host CPU cycles.

An iSCSI HBA is capable of providing performance benefits because it offloads the entire iSCSI and TCP/IP processing from the host processor. The use of an iSCSI HBA is also the simplest way to boot hosts from a SAN environment via iSCSI. If there is no iSCSI HBA, modifications must be made to the basic operating system to boot a host from the storage devices because the NIC needs to obtain an IP address before the operating system loads. The functionality of an iSCSI HBA is similar to the functionality of an FC HBA.

6.1.3 iSCSI Topologies

Two topologies of iSCSI implementations are native and bridged. *Native topology* does not have FC components. The initiators may be either directly attached to targets or connected through the IP network. *Bridged topology* enables the coexistence of FC with IP by providing iSCSI-to-FC bridging functionality. For example, the initiators can exist in an IP environment while the storage remains in an FC environment.

Native iSCSI Connectivity

FC components are not required for iSCSI connectivity if an iSCSI-enabled array is deployed. In Figure 6-2 (a), the array has one or more iSCSI ports configured with an IP address and is connected to a standard Ethernet switch.

After an initiator is logged on to the network, it can access the available LUNs on the storage array. A single array port can service multiple hosts or initiators as long as the array port can handle the amount of storage traffic that the hosts generate.

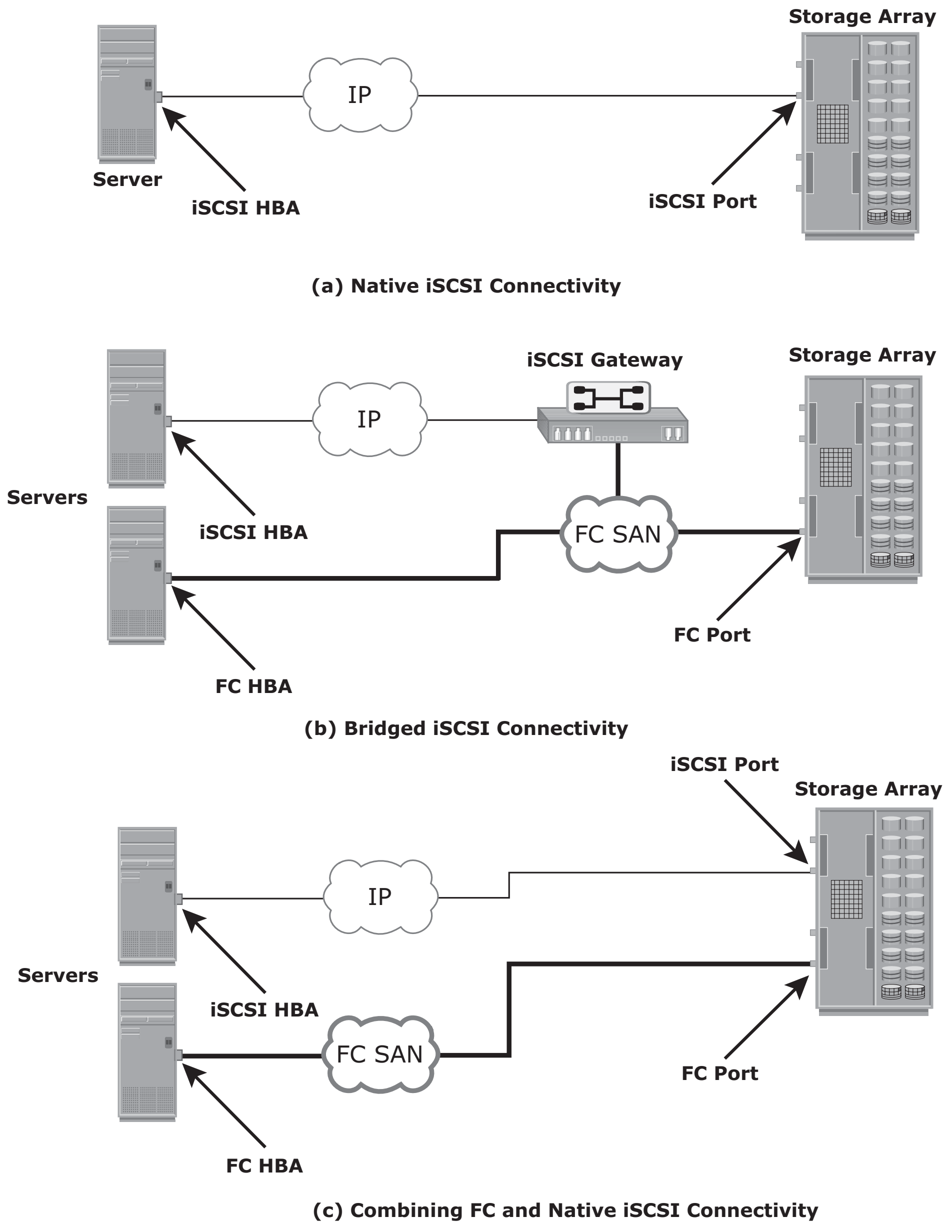


Figure 6-2: iSCSI Topologies

Bridged iSCSI Connectivity

A bridged iSCSI implementation includes FC components in its configuration. Figure 6-2 (b) illustrates iSCSI host connectivity to an FC storage array.

In this case, the array does not have any iSCSI ports. Therefore, an external device, called a gateway or a multiprotocol router, must be used to facilitate the communication between the iSCSI host and FC storage. The gateway converts IP packets to FC frames and vice versa. The bridge devices contain both FC and Ethernet ports to facilitate the communication between the FC and IP environments.

In a bridged iSCSI implementation, the iSCSI initiator is configured with the gateway's IP address as its target destination. On the other side, the gateway is configured as an FC initiator to the storage array.

Combining FC and Native iSCSI Connectivity

The most common topology is a combination of FC and native iSCSI. Typically, a storage array comes with both FC and iSCSI ports that enable iSCSI and FC connectivity in the same environment, as shown in Figure 6-2 (c).

6.1.4 iSCSI Protocol Stack

Figure 6-3 displays a model of the iSCSI protocol layers and depicts the encapsulation order of the SCSI commands for their delivery through a physical carrier.

SCSI is the command protocol that works at the application layer of the Open System Interconnection (OSI) model. The initiators and targets use SCSI commands and responses to talk to each other. The SCSI command descriptor blocks, data, and status messages are encapsulated into TCP/IP and transmitted across the network between the initiators and targets.

iSCSI is the session-layer protocol that initiates a reliable session between devices that recognize SCSI commands and TCP/IP. The iSCSI session-layer interface is responsible for handling login, authentication, target discovery, and session management. TCP is used with iSCSI at the transport layer to provide reliable transmission.

TCP controls message flow, windowing, error recovery, and retransmission. It relies upon the network layer of the OSI model to provide global addressing and connectivity. The Layer 2 protocols at the data link layer of this model enable node-to-node communication through a physical network.

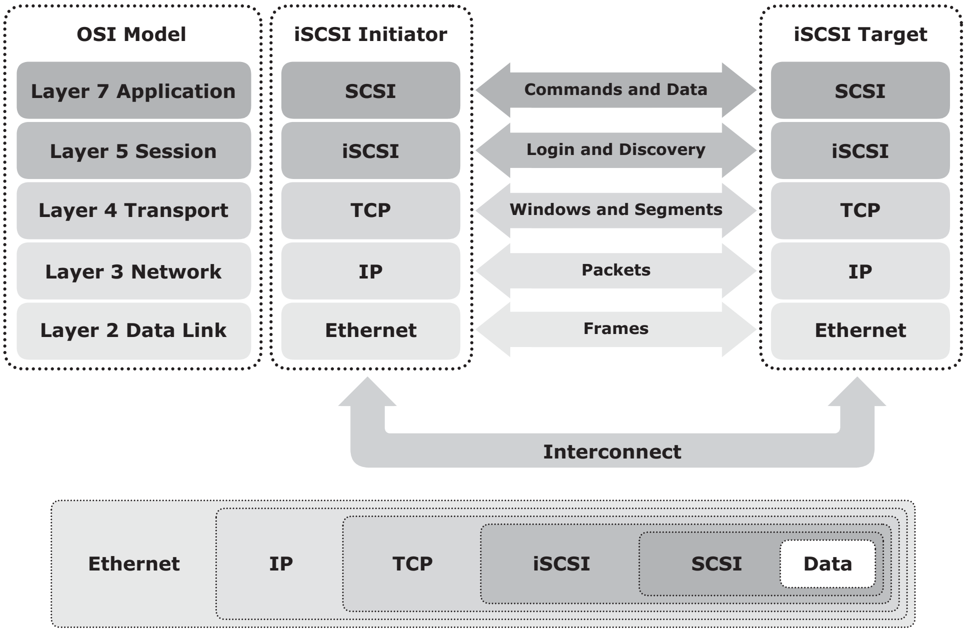


Figure 6-3: iSCSI protocol stack

6.1.5 iSCSI PDU

A *protocol data unit* (PDU) is the basic "information unit" in the iSCSI environment. The iSCSI initiators and targets communicate with each other using iSCSI PDUs. This communication includes establishing iSCSI connections and iSCSI sessions, performing iSCSI discovery, sending SCSI commands and data, and receiving SCSI status. All iSCSI PDUs contain one or more header segments followed by zero or more data segments. The PDU is then encapsulated into an IP packet to facilitate the transport.

A PDU includes the components shown in Figure 6-4. The IP header provides packet-routing information to move the packet across a network. The TCP header contains the information required to guarantee the packet delivery to the target. The iSCSI header (basic header segment) describes how to extract SCSI commands and data for the target. iSCSI adds an optional CRC, known as the *digest*, to ensure datagram integrity. This is in addition to TCP checksum and Ethernet CRC. The header and the data digests are optionally used in the PDU to validate integrity and data placement.

As shown in Figure 6-5, each iSCSI PDU does not correspond in a 1:1 relationship with an IP packet. Depending on its size, an iSCSI PDU can span an IP packet or even coexist with another PDU in the same packet.

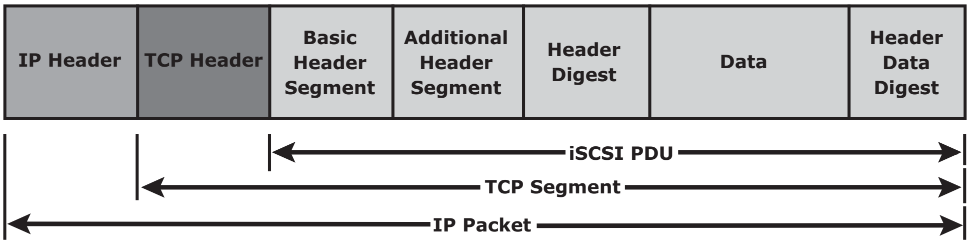


Figure 6-4: iSCSI PDU encapsulated in an IP packet

> **A message transmitted on a network is divided into a number of packets. If necessary, each packet can be sent by a different route across the network. Packets can arrive in a different order than the order in which they were sent. IP only delivers them; it is up to TCP to organize them in the right sequence. The target extracts the SCSI commands and data on the basis of the information in the iSCSI header.**

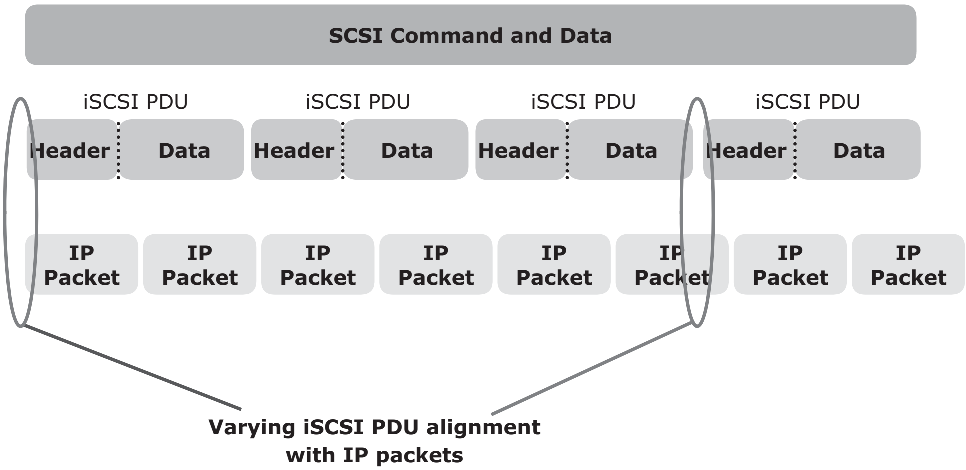


Figure 6-5: Alignment of iSCSI PDUs with IP packets

To achieve the 1:1 relationship between the IP packet and the iSCSI PDU, the maximum transmission unit (MTU) size of the IP packet is modified. This eliminates fragmentation of the IP packet, which improves the transmission efficiency.

6.1.6 iSCSI Discovery

An initiator must discover the location of its targets on the network and the names of the targets available to it before it can establish a session. This discovery can take place in two ways: *SendTargets discovery* or *internet Storage Name Service* (iSNS).

In *SendTargets discovery*, the initiator is manually configured with the target's network portal to establish a discovery session. The initiator issues the `SendTargets` command, and the target network portal responds with the names and addresses of the targets available to the host.

iSNS (see Figure 6-6) enables automatic discovery of iSCSI devices on an IP network. The initiators and targets can be configured to automatically register themselves with the iSNS server. Whenever an initiator wants to know the targets that it can access, it can query the iSNS server for a list of available targets.

The discovery can also take place by using service location protocol (SLP). However, this is less commonly used than `SendTargets` discovery and iSNS.

6.1.7 iSCSI Names

A unique worldwide iSCSI identifier, known as an *iSCSI name*, is used to identify the initiators and targets within an iSCSI network to facilitate communication. The unique identifier can be a combination of the names of the department, application, or manufacturer, serial number, asset number, or any tag that can be used to recognize and manage the devices. Following are two types of iSCSI names commonly used:

- **iSCSI Qualified Name (IQN):** An organization must own a registered domain name to generate iSCSI Qualified Names. This domain name does not need to be active or resolve to an address. It just needs to be reserved to prevent other organizations from using the same domain name to generate iSCSI names. A date is included in the name to avoid potential conflicts caused by the transfer of domain names. An example of an IQN is `iqn.2008-02.com.example:`*`optional_string`*.

  The *optional_string* provides a serial number, an asset number, or any other device identifiers. An iSCSI Qualified Name enables storage administrators to assign meaningful names to iSCSI devices, and therefore, manage those devices more easily.

- **Extended Unique Identifier (EUI):** An EUI is a globally unique identifier based on the IEEE EUI-64 naming standard. An EUI is composed of the eui prefix followed by a 16-character hexadecimal name, such as `eui.0300732A32598D26`.

In either format, the allowed special characters are dots, dashes, and blank spaces.

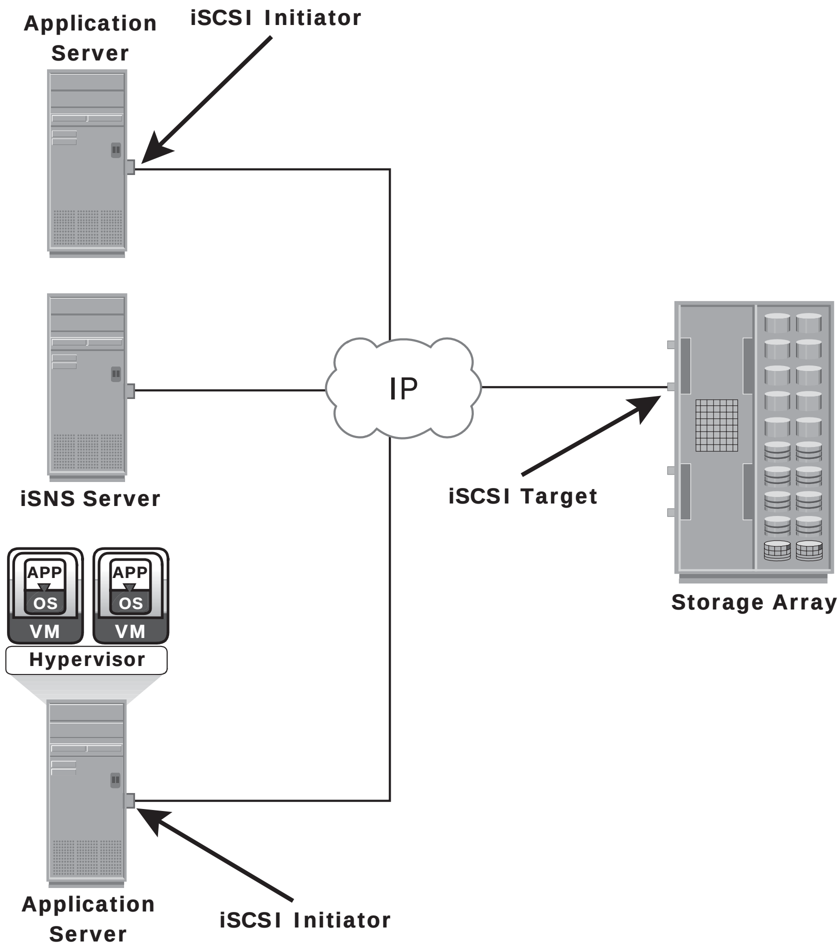


Figure 6-6: Discovery using iSNS

NETWORK ADDRESS AUTHORITY

***Network Address Authority* (NAA) is an additional iSCSI node name type to enable a worldwide naming format as defined by the InterNational Committee for Information Technology Standards (INCITS) T11. This format enables the SCSI storage devices that contain both iSCSI ports and SAS ports to use the same NAA-based SCSI device name. This format is defined by RFC 3980, "T11 Network Address Authority (NAA) Naming Format for iSCSI Node Names."**

6.1.8 iSCSI Session

An iSCSI session is established between an initiator and a target, as shown in Figure 6-7. A session is identified by a session ID (SSID), which includes part of an initiator ID and a target ID. The session can be intended for one of the following:

- The discovery of the available targets by the initiators and the location of a specific target on a network
- The normal operation of iSCSI (transferring data between initiators and targets)

There might be one or more TCP connections within each session. Each TCP connection within the session has a unique connection ID (CID).

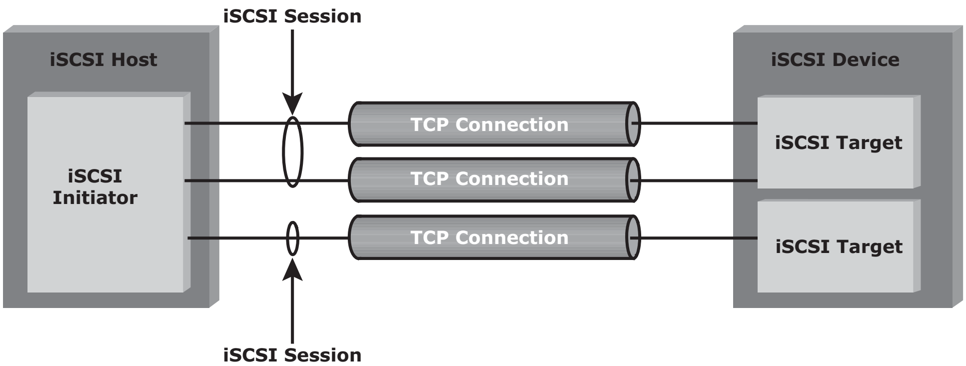


Figure 6-7: iSCSI session

An iSCSI session is established via the iSCSI login process. The login process is started when the initiator establishes a TCP connection with the required target either via the well-known port 3260 or a specified target port. During the login phase, the initiator and the target authenticate each other and negotiate on various parameters.

After the login phase is successfully completed, the iSCSI session enters the full-feature phase for normal SCSI transactions. In this phase, the initiator may send SCSI commands and data to the various LUNs on the target by encapsulating them in iSCSI PDUs that travel over the established TCP connection.

The final phase of the iSCSI session is the connection termination phase, which is referred to as the logout procedure. The initiator is responsible for commencing the logout procedure; however, the target may also prompt termination by sending an iSCSI message, indicating the occurrence of an internal error condition. After the logout request is sent from the initiator and accepted by the target, no further request and response can be sent on that connection.

6.1.9 iSCSI Command Sequencing

The iSCSI communication between the initiators and targets is based on the request-response command sequences. A command sequence may generate multiple PDUs. A *command sequence number* (CmdSN) within an iSCSI session is used for numbering all initiator-to-target command PDUs belonging to the session. This number ensures that every command is delivered in the same order in which it is transmitted, regardless of the TCP connection that carries the command in the session.

Command sequencing begins with the first login command, and the CmdSN is incremented by one for each subsequent command. The iSCSI target layer is responsible for delivering the commands to the SCSI layer in the order of their CmdSN. This ensures the correct order of data and commands at a target even when there are multiple TCP connections between an initiator and the target that use portal groups.

Similar to command numbering, a *status sequence number* (StatSN) is used to sequentially number status responses, as shown in Figure 6-8. These unique numbers are established at the level of the TCP connection.

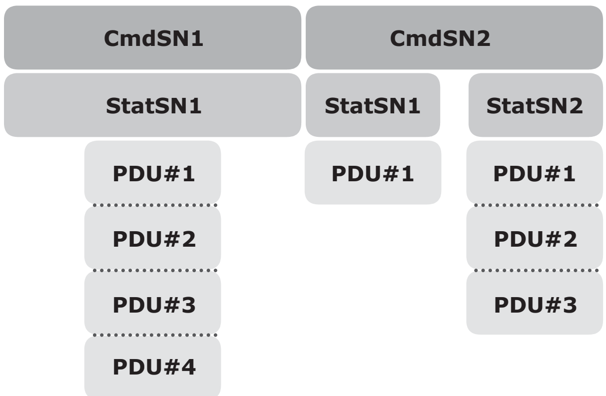


Figure 6-8: Command and status sequence number

A target sends *request-to-transfer* (R2T) PDUs to the initiator when it is ready to accept data. A *data sequence number* (DataSN) is used to ensure in-order delivery of data within the same command. The DataSN and R2TSN are used to sequence data PDUs and R2Ts, respectively. Each of these sequence numbers is stored locally as an unsigned 32-bit integer counter defined by iSCSI. These numbers are communicated between the initiator and target in the appropriate iSCSI PDU fields during command, status, and data exchanges.

For read operations, the DataSN begins at zero and is incremented by one for each subsequent data PDU in that command sequence. For a write operation, the first unsolicited data PDU or the first data PDU in response to an R2T begins with a DataSN of zero and increments by one for each subsequent data PDU. R2TSN is set to zero at the initiation of the command and incremented by one for each subsequent R2T sent by the target for that command.

6.2 FCIP

FC SAN provides a high-performance infrastructure for localized data movement. Organizations are now looking for ways to transport data over a long distance between their disparate SANs at multiple geographic locations. One of the best ways to achieve this goal is to interconnect geographically dispersed SANs through reliable, high-speed links. This approach involves transporting the FC block data over the IP infrastructure. FCIP is a tunneling protocol that enables distributed FC SAN islands to be interconnected over the existing IP-based networks.

The FCIP standard has rapidly gained acceptance as a manageable, cost-effective way to blend the best of the two worlds: FC SAN and the proven, widely deployed IP infrastructure. As a result, organizations now have a better way to store, protect and move their data by leveraging investments in their existing IP infrastructure. FCIP is extensively used in disaster recovery implementations in which data is duplicated to the storage located at a remote site.

FCIP might require high network bandwidth when replicating or backing up data. FCIP does not handle data traffic throttling or flow control; these are controlled by the communicating FC switches and devices within the fabric.

6.2.1 FCIP Protocol Stack

The FCIP protocol stack is shown in Figure: 6-9. Applications generate SCSI commands and data, which are processed by various layers of the protocol stack.

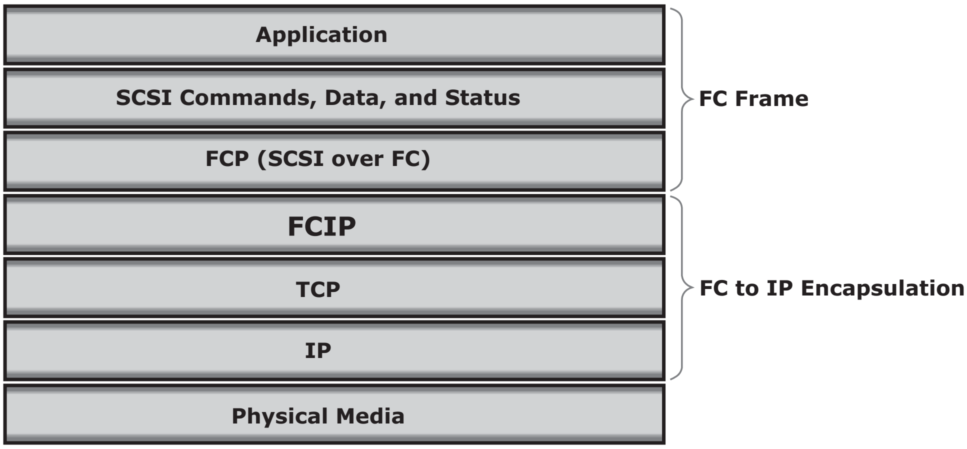


Figure 6-9: FCIP protocol stack

The upper layer protocol SCSI includes the SCSI driver program that executes the read-and-write commands. Below the SCSI layer is the Fibre Channel Protocol (FCP) layer, which is simply a Fibre Channel frame whose payload is SCSI. The FCP layer rides on top of the Fibre Channel transport layer. This enables the FC frames to run natively within a SAN fabric environment. In addition, the FC frames can be encapsulated into the IP packet and sent to a remote SAN over the IP. The FCIP layer encapsulates the Fibre Channel frames onto the IP payload and passes them to the TCP layer (see Figure 6-10). TCP and IP are used for transporting the encapsulated information across Ethernet, wireless, or other media that support the TCP/IP traffic.

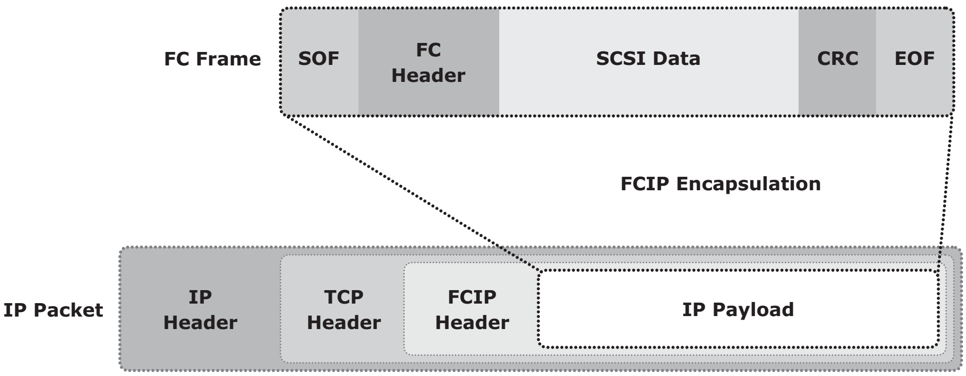


Figure 6-10: FCIP encapsulation

Encapsulation of FC frame into an IP packet could cause the IP packet to be fragmented when the data link cannot support the maximum transmission unit

(MTU) size of an IP packet. When an IP packet is fragmented, the required parts of the header must be copied by all fragments. When a TCP packet is segmented, normal TCP operations are responsible for receiving and re-sequencing the data prior to passing it on to the FC processing portion of the device.

6.2.2 FCIP Topology

In an FCIP environment, an FCIP gateway is connected to each fabric via a standard FC connection (see Figure 6-11). The FCIP gateway at one end of the IP network encapsulates the FC frames into IP packets. The gateway at the other end removes the IP wrapper and sends the FC data to the layer 2 fabric. The fabric treats these gateways as layer 2 fabric switches. An IP address is assigned to the port on the gateway, which is connected to an IP network. After the IP connectivity is established, the nodes in the two independent fabrics can communicate with each other.

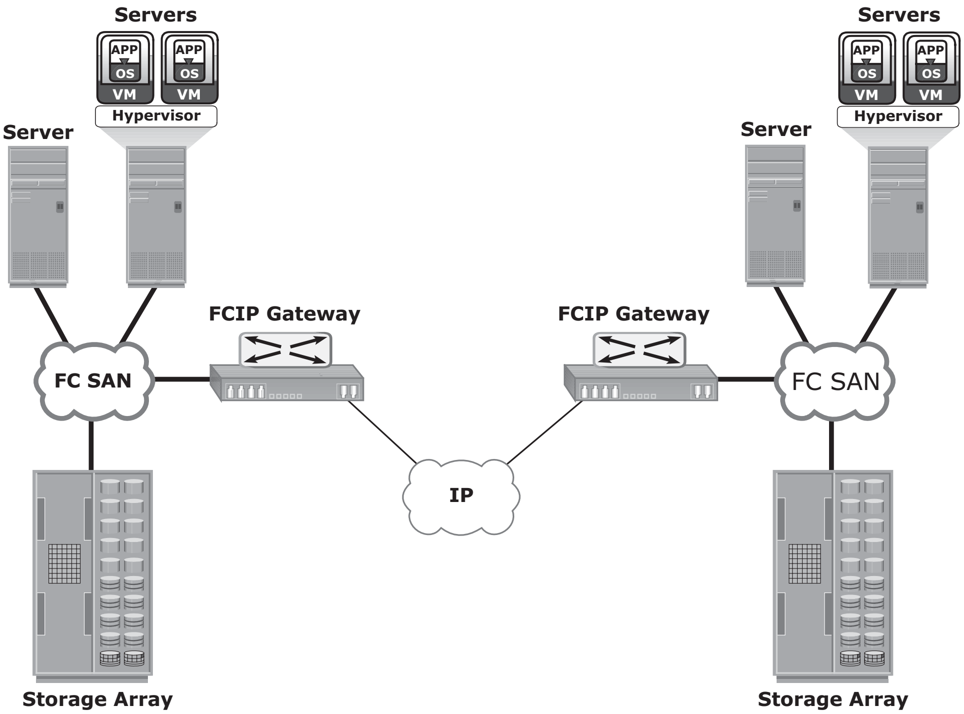


Figure 6-11: FCIP topology

6.2.3 FCIP Performance and Security

Performance, reliability, and security should always be taken into consideration when implementing storage solutions. The implementation of FCIP is also subject to the same considerations.

From the perspective of performance, configuring multiple paths between FCIP gateways eliminates single points of failure and provides increased bandwidth. In a scenario of extended distance, the IP network might be a bottleneck if sufficient bandwidth is not available. In addition, because FCIP creates a unified fabric, disruption in the underlying IP network can cause instabilities in the SAN environment. These instabilities include a segmented fabric, excessive RSCNs, and host timeouts.

The vendors of FC switches have recognized some of the drawbacks related to FCIP and have implemented features to enhance stability, such as the capability to segregate the FCIP traffic into a separate virtual fabric.

Security is also a consideration in an FCIP solution because the data is transmitted over public IP channels. Various security options are available to protect the data based on the router's support. IPSec is one such security measure that can be implemented in the FCIP environment.

6.3 FCoE

Data centers typically have multiple networks to handle various types of I/O traffic — for example, an Ethernet network for TCP/IP communication and an FC network for FC communication. TCP/IP is typically used for client-server communication, data backup, infrastructure management communication, and so on. FC is typically used for moving block-level data between storage and servers. To support multiple networks, servers in a data center are equipped with multiple redundant physical network interfaces — for example, multiple Ethernet and FC cards/adapters. In addition, to enable the communication, different types of networking switches and physical cabling infrastructure are implemented in data centers. The need for two different kinds of physical network infrastructure increases the overall cost and complexity of data center operation.

Fibre Channel over Ethernet (FCoE) protocol provides consolidation of LAN and SAN traffic over a single physical interface infrastructure. FCoE helps organizations address the challenges of having multiple discrete network infrastructures. FCoE uses the Converged Enhanced Ethernet (CEE) link (10 Gigabit Ethernet) to send FC frames over Ethernet.

6.3.1 I/O Consolidation Using FCoE

The key benefit of FCoE is I/O consolidation. Figure 6-12 represents the infrastructure before FCoE deployment. Here, the storage resources are accessed using HBAs, and the IP network resources are accessed using NICs by the servers. Typically, in a data center, a server is configured with 2 to 4 NIC cards and redundant HBA cards. If the data center has hundreds of servers, it would require a large number of adapters, cables, and switches. This leads to a complex environment, which is difficult to manage and scale. The cost of power, cooling, and floor space further adds to the challenge.

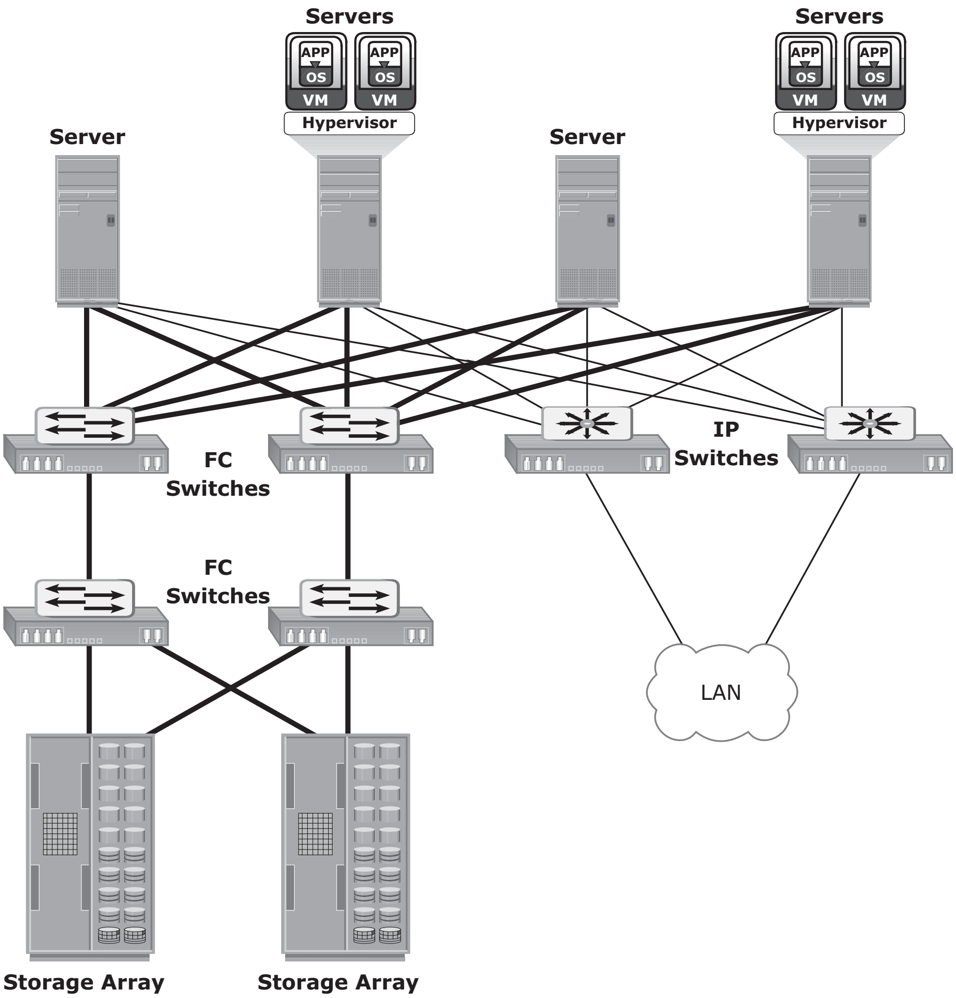


Figure 6-12: Infrastructure before using FCoE

Figure 6-13 shows the I/O consolidation with FCoE using FCoE switches and Converged Network Adapters (CNAs). A CNA (discussed in the section "Converged Network Adapter") replaces both HBAs and NICs in the server and consolidates both the IP and FC traffic. This reduces the requirement of multiple network adapters at the server to connect to different networks. Overall, this reduces the requirement of adapters, cables, and switches. This also considerably reduces the cost and management overhead.

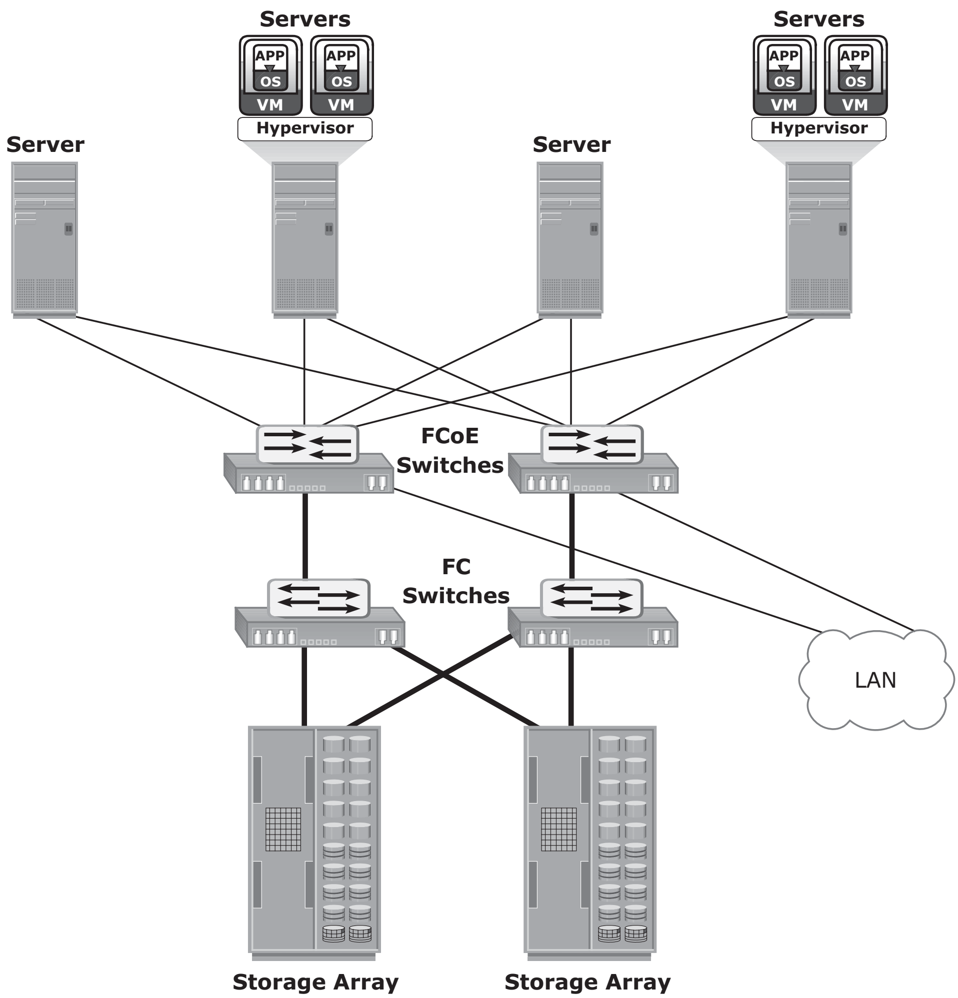


Figure 6-13: Infrastructure after using FCoE

6.3.2 Components of an FCoE Network

This section describes the key physical components required to implement FCoE in a data center. The key FCoE components are:

- Converged Network Adapter (CNA)
- Cables
- FCoE switches

Converged Network Adapter

A CNA provides the functionality of both a standard NIC and an FC HBA in a single adapter and consolidates both types of traffic. CNA eliminates the need to deploy separate adapters and cables for FC and Ethernet communications, thereby reducing the required number of server slots and switch ports. CNA offloads the FCoE protocol processing task from the server, thereby freeing the server CPU resources for application processing. As shown in Figure 6-14, a CNA contains separate modules for 10 Gigabit Ethernet, Fibre Channel, and FCoE Application Specific Integrated Circuits (ASICs). The FCoE ASIC encapsulates FC frames into Ethernet frames. One end of this ASIC is connected to 10GbE and FC ASICs for server connectivity, while the other end provides a 10GbE interface to connect to an FCoE switch.

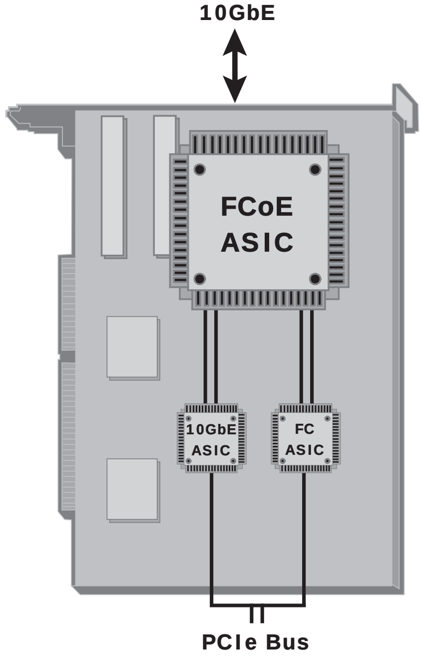


Figure 6-14: Converged Network Adapter

Cables

Currently two options are available for FCoE cabling: Copper based Twinax and standard fiber optical cables. A Twinax cable is composed of two pairs of copper cables covered with a shielded casing. The Twinax cable can transmit data at the speed of 10 Gbps over shorter distances up to 10 meters. Twinax cables require less power and are less expensive than fiber optic cables. The Small Form Factor Pluggable Plus (SFP+) connector is the primary connector used for FCoE links and can be used with both optical and copper cables.

A typical strategy for FCoE deployment is the *top of rack* implementation. Here, a pair of redundant FCoE switches is installed at the top of each rack of servers. Both FC and IP connectivity to each server is accomplished using inexpensive Twinax cabling from the server to the top of rack FCoE switches. This short distance is well supported with Twinax. Connectivity from the top of rack switches to existing backbone LAN and SAN infrastructures, that is connections across racks, is typically done with optical links, which can support the longer cable runs that may be required.

FCoE Switches

An FCoE switch has both Ethernet switch and Fibre Channel switch functionalities. The FCoE switch has a Fibre Channel Forwarder (FCF), Ethernet Bridge, and set of Ethernet ports and optional FC ports, as shown in Figure 6-15. The function of the FCF is to encapsulate the FC frames, received from the FC port, into the FCoE frames and also to de-encapsulate the FCoE frames, received from the Ethernet Bridge, to the FC frames.

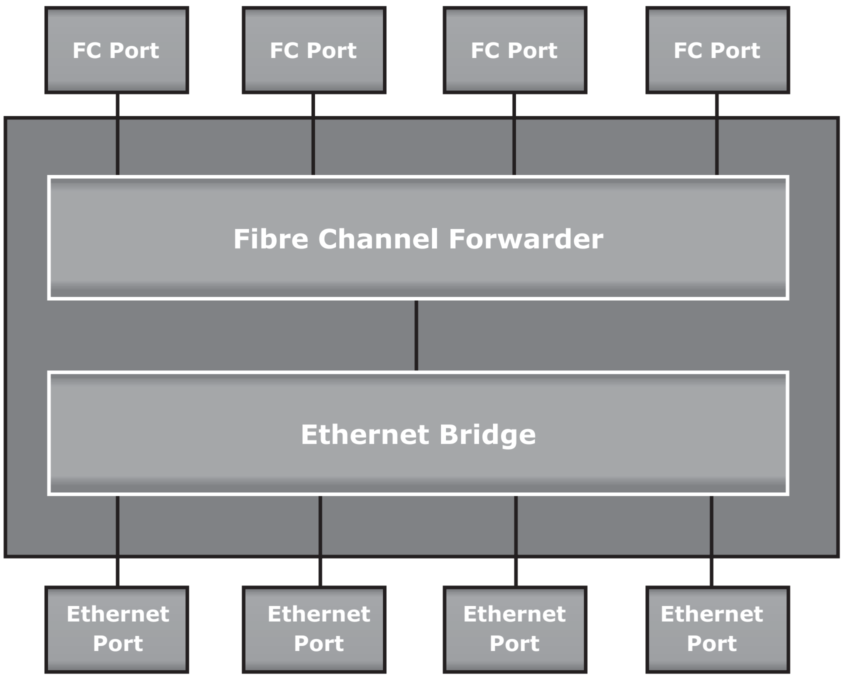


Figure 6-15: FCoE switch generic architecture

Upon receiving the incoming traffic, the FCoE switch inspects the Ethertype (used to indicate which protocol is encapsulated in the payload of an Ethernet frame) of the incoming frames and uses that to determine the destination. If the Ethertype of the frame is FCoE, the switch recognizes that the frame contains an FC payload and forwards it to the FCF. From there, the FC is extracted from the FCoE frame and transmitted to FC SAN over the FC ports. If the Ethertype is not FCoE, the switch handles the traffic as usual Ethernet traffic and forwards it over the Ethernet ports.

6.3.3 FCoE Frame Structure

An FCoE frame is an Ethernet frame that contains an FCoE Protocol Data Unit. Figure 6-16 shows the FCoE frame structure. The first 48-bits in the frame are used to specify the destination MAC address, and the next 48-bits specify the source MAC address. The 32-bit IEEE 802.1Q tag supports the creation of multiple virtual networks (VLANs) across a single physical infrastructure. FCoE has its own Ethertype, as designated by the next 16 bits, followed by the 4-bit version field. The next 100-bits are reserved and are followed by the 8-bit Start of Frame and then the actual FC frame. The 8-bit End of Frame delimiter is followed by 24 reserved bits. The frame ends with the final 32-bits dedicated to the Frame Check Sequence (FCS) function that provides error detection for the Ethernet frame.

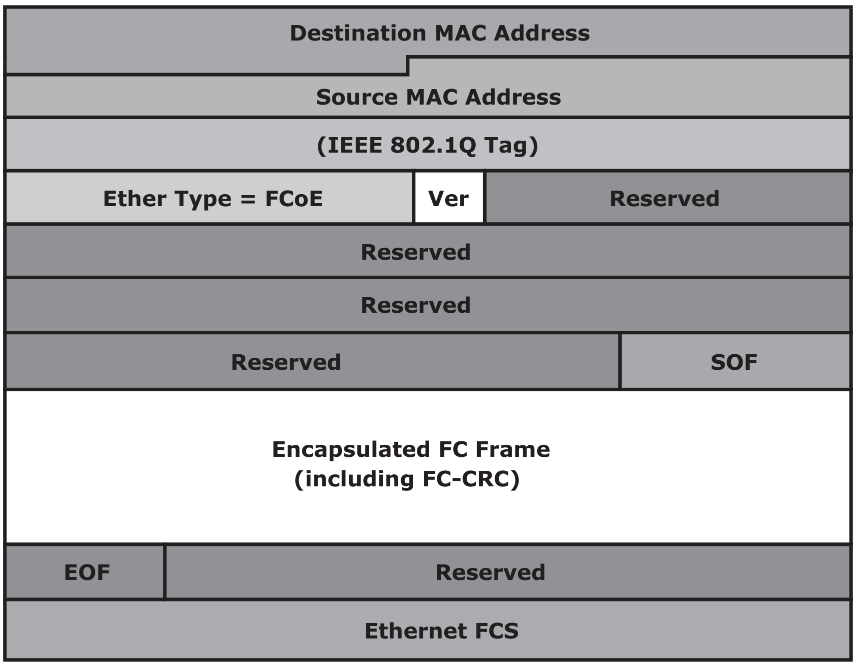


Figure 6-16: FCoE frame structure

The encapsulated Fibre Channel frame consists of the original 24-byte FC header and the data being transported (including the Fibre Channel CRC). The FC frame structure is maintained such that when a traditional FC SAN is connected to an FCoE capable switch, the FC frame is de-encapsulated from the FCoE frame and transported to FC SAN seamlessly. This capability enables FCoE to integrate with the existing FC SANs without the need for a gateway.

Frame size is also an important factor in FCoE. A typical Fibre Channel data frame has a 2,112-byte payload, a 24-byte header, and an FCS. A standard Ethernet frame has a default payload capacity of 1,500 bytes. To maintain good performance, FCoE must use jumbo frames to prevent a Fibre Channel frame from being split into two Ethernet frames. The next chapter discusses jumbo frames in detail. FCoE requires Converged Enhanced Ethernet, which provides lossless Ethernet and jumbo frame support.

FCoE Frame Mapping

The encapsulation of the Fibre Channel frame occurs through the mapping of the FC frames onto Ethernet, as shown in Figure 6-17. Fibre Channel and traditional networks have stacks of layers where each layer in the stack represents a set of functionalities. The FC stack consists of five layers: FC-0 through FC-4. Ethernet is typically considered as a set of protocols that operates at the physical and data link layers in the seven layer OSI stack. The FCoE protocol specification replaces the FC-0 and FC-1 layers of the FC stack with Ethernet. This provides the capability to carry the FC-2 to the FC-4 layer over the Ethernet layer.

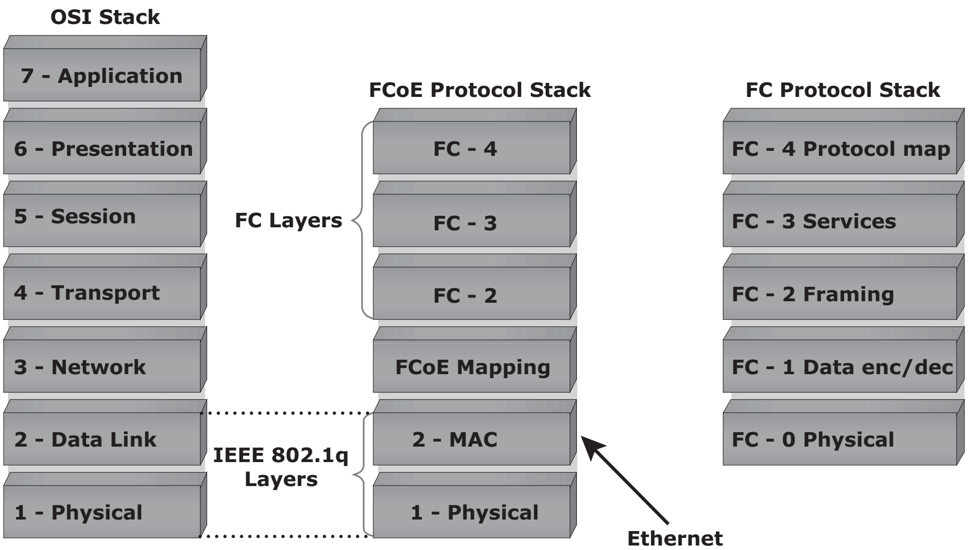
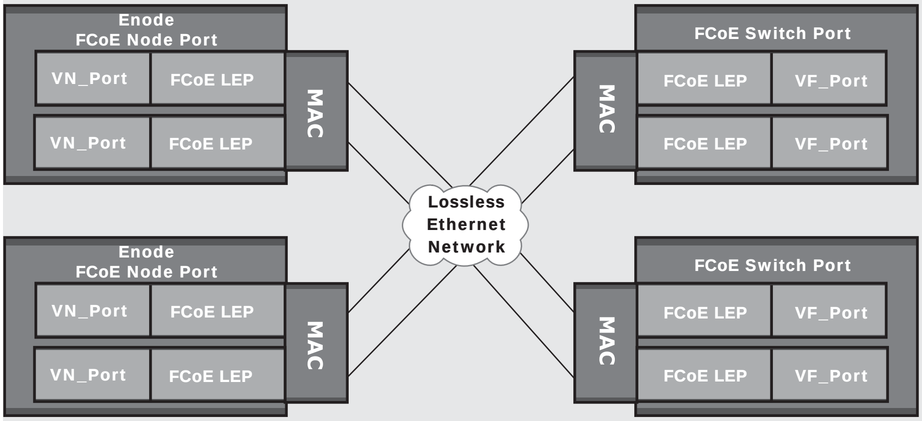


Figure 6-17: FCoE frame mapping

FCoE PORTS

To transport FC frames, the FCoE ports need to emulate the behavior of FC ports and become virtual FC ports. FCoE uses similar terminology as FC to define various ports in the network. FCoE has the following ports (See the figure following this list):

- **VN_Port (Virtual N_Port): Port in an Enhanced Ethernet node (or Enode). Enodes are end points, such as a server, with CNA.**
- **VF_Port (Virtual F_Port): Virtual Fabric Port in an FCoE Switch**
- **VE_Port (Virtual E_Port): Virtual Extension Port in an FCoE Switch for ISLs**



FCoE Link End Points (LEP) are located between the MAC and the virtual ports. LEPs are responsible for FC frame encapsulation/de-capsulation and for transmitting and receiving the encapsulated frames through a virtual port.

6.3.4 FCoE Enabling Technologies

Conventional Ethernet is lossy in nature, which means that frames might be dropped or lost during transmission. *Converged Enhanced Ethernet* (CEE), or lossless Ethernet, provides a new specification to the existing Ethernet standard that eliminates the lossy nature of Ethernet. This makes 10 Gb Ethernet a viable storage networking option, similar to FC. Lossless Ethernet requires certain functionalities. These functionalities are defined and maintained by the data center bridging (DCB) task group, which is a part of the IEEE 802.1 working group, and they are:

- Priority-based flow control
- Enhanced transmission selection

- Congestion Notification
- Data center bridging exchange protocol

Priority-Based Flow Control (PFC)

Traditional FC manages congestion through the use of a link-level, credit-based flow control that guarantees no loss of frames. Typical Ethernet, coupled with TCP/IP, uses a packet drop flow control mechanism. The packet drop flow control is not lossless. This challenge is eliminated by using an IEEE 802.3x Ethernet PAUSE control frame to create a lossless Ethernet. A receiver can send a PAUSE request to a sender when the receiver's buffer is filling up. Upon receiving a PAUSE frame, the sender stops transmitting frames, which guarantees no loss of frames. The downside of using the Ethernet PAUSE frame is that it operates on the entire link, which might be carrying multiple traffic flows.

PFC provides a link level flow control mechanism. PFC creates eight separate virtual links on a single physical link and allows any of these links to be paused and restarted independently. PFC enables the pause mechanism based on user priorities or classes of service. Enabling the pause based on priority allows creating lossless links for traffic, such as FCoE traffic. This PAUSE mechanism is typically implemented for FCoE while regular TCP/IP traffic continues to drop frames. Figure 6-18 illustrates how a physical Ethernet link is divided into eight virtual links and allows a PAUSE for a single virtual link without affecting the traffic for the others.

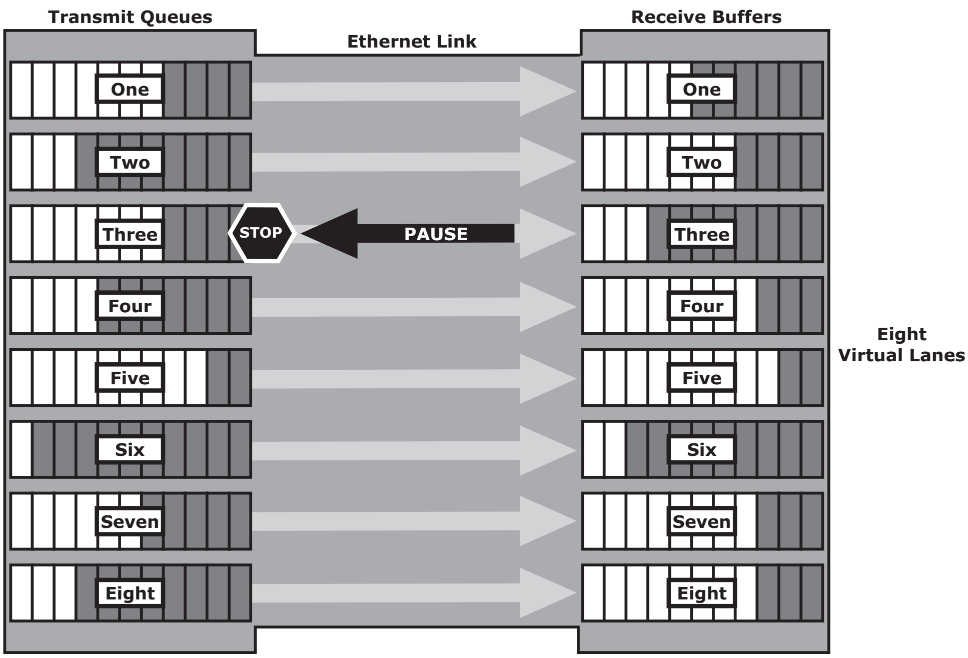


Figure 6-18: Priority-based flow control

Enhanced Transmission Selection (ETS)

Enhanced transmission selection provides a common management framework for the assignment of bandwidth to different traffic classes, such as LAN, SAN, and Inter Process Communication (IPC). When a particular class of traffic does not use its allocated bandwidth, ETS enables other traffic classes to use the available bandwidth.

Congestion Notification (CN)

Congestion notification provides end-to-end congestion management for protocols, such as FCoE, that do not have built-in congestion control mechanisms. Link level congestion notification provides a mechanism for detecting congestion and notifying the source to move the traffic flow away from the congested links. Link level congestion notification enables a switch to send a signal to other ports that need to stop or slow down their transmissions. The process of congestion notification and its management is shown in Figure 6-19, which represents the communication between the nodes A (sender) and B (receiver). If congestion at the receiving end occurs, the algorithm running on the switch generates a congestion notification message to the sending node (Node A). In response to the CN message, the sending end limits the rate of data transfer.

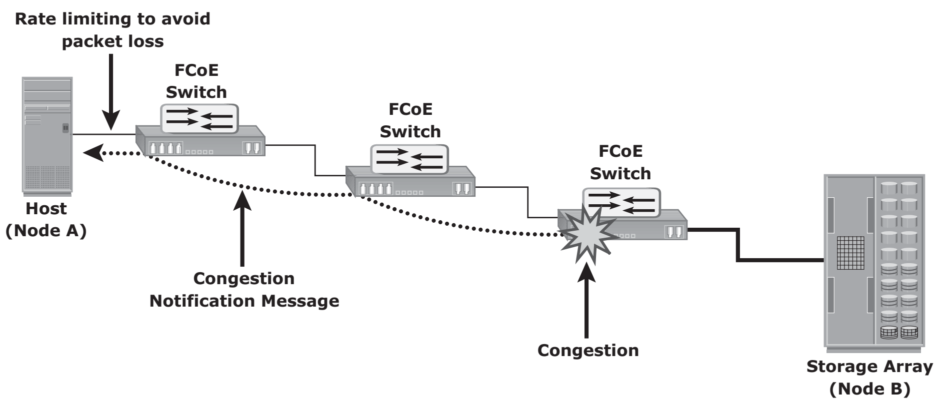


Figure 6-19: Congestion Notification

Data Center Bridging Exchange Protocol (DCBX)

DCBX protocol is a discovery and capability exchange protocol, which helps Converged Enhanced Ethernet devices to convey and configure their features with the other CEE devices in the network. DCBX is used to negotiate capabilities between the switches and the adapters, and it allows the switch to distribute the configuration values to all the attached adapters. This helps to ensure consistent configuration across the entire network.

Summary

IP SAN has enabled IT organizations to adopt storage networking infrastructure at reasonable costs. Storage networks can now be geographically distributed with the help of the IP SAN technology, which enhances storage utilization across enterprises. FCIP has emerged as a solution for implementing viable business continuity across data centers.

Because IP SANs are based on standard IP protocols, the concepts, security mechanisms, and management tools are familiar to network administrators. This has enabled the rapid adoption of IP SAN in organizations. This chapter detailed the two IP SAN technologies: iSCSI and FCIP. This chapter also detailed the emerging FCoE technology that enables transportation of both the LAN and SAN traffic on a single physical network infrastructure.

SAN offers a high-performance storage networking solution; however, SAN does not enable sharing of data among multiple hosts. Organizations might require sharing of data or files among multiple heterogeneous clients for collaboration purposes.

The next chapter details network-attached storage (NAS), a solution that provides a file-sharing environment to heterogeneous clients. Because NAS is dedicated for file sharing, it provides better performance than traditional file servers.

EXERCISES

1. How does iSCSI handle the process of authentication? Research the available options.
2. Compared to a standard IP packet, what percentage of reduction can be realized in protocol overhead in an iSCSI, configured to use jumbo frames with an MTU value of 9,000 bytes?
3. Why should an MTU value of at least 2,500 bytes be configured in a bridged iSCSI environment?
4. Why does the lossy nature of standard Ethernet make it unsuitable for a layered FCoE implementation? How does Converged Enhanced Ethernet (CEE) address this problem?
5. Compare various data center protocols that use Ethernet as the physical medium for transporting storage traffic.

Chapter 7
Network-Attached Storage

KEY CONCEPTS

- NAS Devices
- Network File Sharing
- Unified, Gateway, and Scale-Out NAS
- NAS Connectivity and Protocols
- NAS Performance
- MTU and Jumbo Frames
- TCP Window and Link Aggregation
- File-Level Virtualization

File sharing, as the name implies, enables users to share files with other users. Traditional methods of file sharing involve copying files to portable media such as floppy diskette, CD, DVD, or USB drives and delivering them to other users with whom it is being shared. However, this approach is not suitable in an enterprise environment in which a large number of users at different locations need access to common files.

Network-based file sharing provides the flexibility to share files over long distances among a large number of users. File servers use client-server technology to enable file sharing over a network. To address the tremendous growth of file data in enterprise environments, organizations have been deploying large numbers of file servers. These servers are either connected to direct-attached storage (DAS) or storage area network (SAN)-attached storage. This has resulted in the proliferation of islands of over-utilized and under-utilized file servers and storage. In addition, such environments have poor scalability, higher management cost, and greater complexity. *Network-attached storage* (NAS) emerged as a solution to these challenges.

NAS is a dedicated, high-performance file sharing and storage device. NAS enables its clients to share files over an IP network. NAS provides the advantages of server consolidation by eliminating the need for multiple file servers. It also consolidates the storage used by the clients onto a single system, making it easier to manage the storage. NAS uses network and file-sharing protocols to provide access to the file data. These protocols include TCP/IP for data transfer, and Common Internet File System (CIFS) and Network File System (NFS) for network file service. NAS enables both UNIX and Microsoft Windows users to share the same data seamlessly.

A NAS device uses its own operating system and integrated hardware and software components to meet specific file-service needs. Its operating system is optimized for file I/O and, therefore, performs file I/O better than a general-purpose server. As a result, a NAS device can serve more clients than general-purpose servers and provide the benefit of server consolidation.

A network-based file sharing environment is composed of multiple file servers or NAS devices. It might be required to move the files from one device to another due to reasons such as cost or performance. File-level virtualization, implemented in the file sharing environment, provides a simple, nondisruptive file-mobility solution. It enables the movement of files across NAS devices, even if the files are being accessed.

This chapter describes the components of NAS, different types of NAS implementations, and the file-sharing protocols used in NAS implementations. The chapter also explains factors that affect NAS performance, and file-level virtualization.

7.1 General-Purpose Servers versus NAS Devices

A NAS device is optimized for file-serving functions such as storing, retrieving, and accessing files for applications and clients. As shown in Figure 7-1, a general-purpose server can be used to host any application because it runs a general-purpose operating system. Unlike a general-purpose server, a NAS device is dedicated to file-serving. It has specialized operating system dedicated to file serving by using industry-standard protocols. Some NAS vendors support features, such as native clustering for high availability.

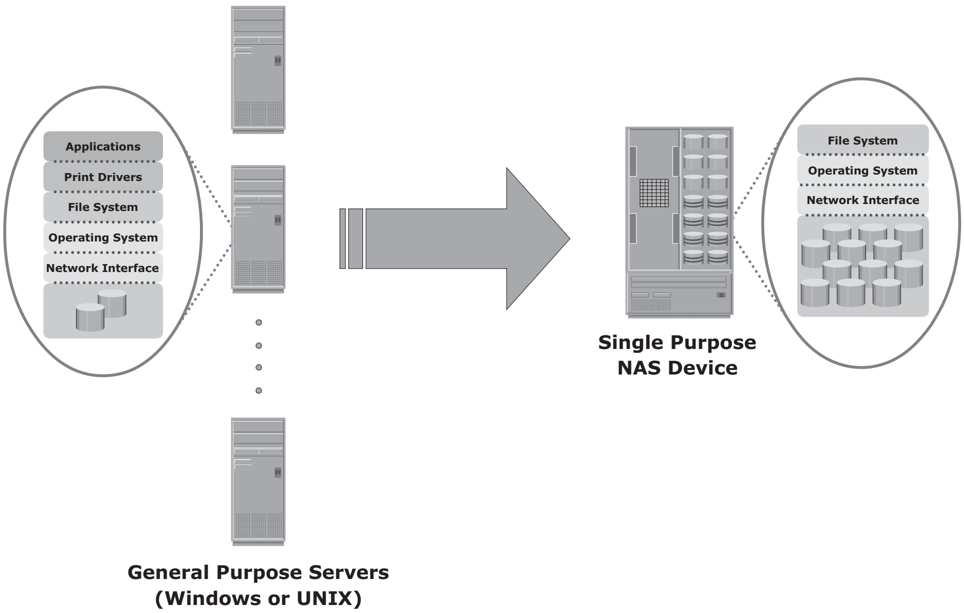


Figure 7-1: General purpose server versus NAS device

7.2 Benefits of NAS

NAS offers the following benefits:

- **Comprehensive access to information:** Enables efficient file sharing and supports many-to-one and one-to-many configurations. The many-to-one configuration enables a NAS device to serve many clients simultaneously. The one-to-many configuration enables one client to connect with many NAS devices simultaneously.
- **Improved efficiency:** NAS delivers better performance compared to a general-purpose file server because NAS uses an operating system specialized for file serving.
- **Improved flexibility:** Compatible with clients on both UNIX and Windows platforms using industry-standard protocols. NAS is flexible and can serve requests from different types of clients from the same source.
- **Centralized storage:** Centralizes data storage to minimize data duplication on client workstations, and ensure greater data protection
- **Simplified management:** Provides a centralized console that makes it possible to manage file systems efficiently

- **Scalability:** Scales well with different utilization profiles and types of business applications because of the high-performance and low-latency design
- **High availability:** Offers efficient replication and recovery options, enabling high data availability. NAS uses redundant components that provide maximum connectivity options. A NAS device supports clustering technology for failover.
- **Security:** Ensures security, user authentication, and file locking with industry-standard security schemas
- **Low cost:** NAS uses commonly available and inexpensive Ethernet components.
- **Ease of deployment:** Configuration at the client is minimal, because the clients have required NAS connection software built in.

7.3 File Systems and Network File Sharing

A *file system* is a structured way to store and organize data files. Many file systems maintain a file access table to simplify the process of searching and accessing files.

7.3.1 Accessing a File System

A file system must be mounted before it can be used. In most cases, the operating system mounts a local file system during the boot process. The mount process creates a link between the file system on the NAS and the operating system on the client. When mounting a file system, the operating system organizes files and directories in a tree-like structure and grants the privilege to the user to access this structure. The tree is rooted at a mount point. The mount point is named using operating system conventions. Users and applications can traverse the entire tree from the root to the leaf nodes as file system permissions allow. Files are located at leaf nodes, and directories and subdirectories are located at intermediate roots. The access to the file system terminates when the file system is unmounted. Figure 7-2 shows an example of a UNIX directory structure.

7.3.2 Network File Sharing

Network file sharing refers to storing and accessing files over a network. In a file-sharing environment, the user who creates a file (the creator or owner of a file) determines the type of access (such as read, write, execute, append, and delete) to be given to other users and controls changes to the file. When multiple users try to access a shared file at the same time, a locking scheme is required to maintain data integrity and, at the same time, make this sharing possible.

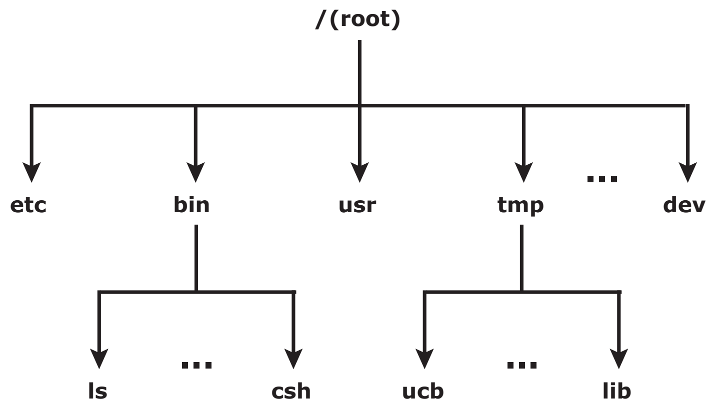


Figure 7-2: UNIX directory structure

Some examples of file-sharing methods are file transfer protocol (FTP), Distributed File System (DFS), client-server models that use file-sharing protocols such as NFS and CIFS, and the peer-to-peer (P2P) model

FTP is a client-server protocol that enables data transfer over a network. An FTP server and an FTP client communicate with each other using TCP as the transport protocol. FTP, as defined by the standard, is not a secure method of data transfer because it uses unencrypted data transfer over a network. FTP over Secure Shell (SSH) adds security to the original FTP specification. When FTP is used over SSH, it is referred to as Secure FTP (SFTP).

A *distributed file system* (DFS) is a file system that is distributed across several hosts. A DFS can provide hosts with direct access to the entire file system, while ensuring efficient management and data security. Standard client-server file-sharing protocols, such as NFS and CIFS, enable the owner of a file to set the required type of access, such as read-only or read-write, for a particular user or group of users. Using this protocol, the clients mount remote file systems that are available on dedicated file servers.

A name service, such as Domain Name System (DNS), and directory services such as Microsoft Active Directory, and Network Information Services (NIS), helps users identify and access a unique resource over the network. A *name service protocol* such as the Lightweight Directory Access Protocol (LDAP) creates a namespace, which holds the unique name of every network resource and helps recognize resources on the network.

A *peer-to-peer* (P2P) file sharing model uses a peer-to-peer network. P2P enables client machines to directly share files with each other over a network. Clients use a file sharing software that searches for other peer clients. This differs from the client-server model that uses file servers to store files for sharing.

7.4 Components of NAS

A NAS device has two key components: NAS head and storage (see Figure 7-3). In some NAS implementations, the storage could be external to the NAS device and shared with other hosts. The NAS head includes the following components:

- CPU and memory
- One or more network interface cards (NICs), which provide connectivity to the client network. Examples of network protocols supported by NIC include Gigabit Ethernet, Fast Ethernet, ATM, and Fiber Distributed Data Interface (FDDI).
- An optimized operating system for managing the NAS functionality. It translates file-level requests into block-storage requests and further converts the data supplied at the block level to file data.
- NFS, CIFS, and other protocols for file sharing
- Industry-standard storage protocols and ports to connect and manage physical disk resources

The NAS environment includes clients accessing a NAS device over an IP network using file-sharing protocols.

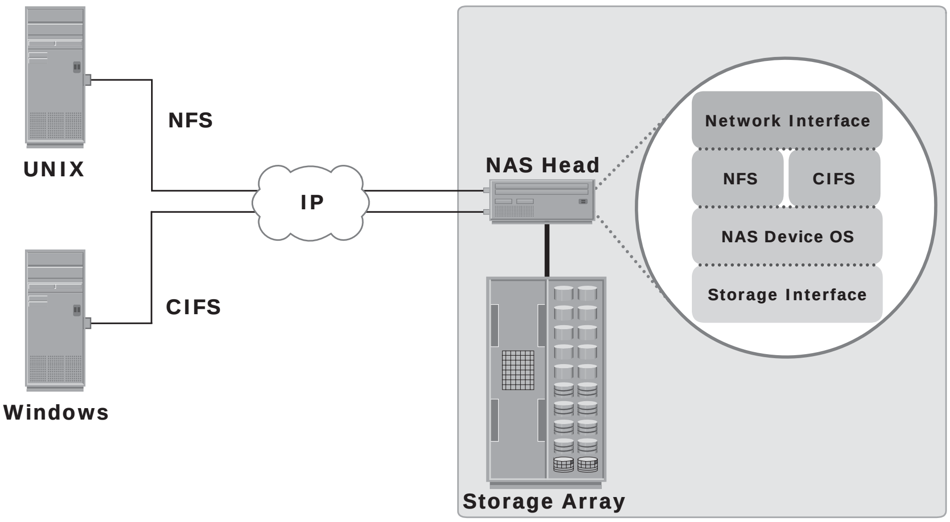


Figure 7-3: Components of NAS

7.5 NAS I/O Operation

NAS provides file-level data access to its clients. File I/O is a high-level request that specifies the file to be accessed. For example, a client may request a file by specifying its name, location, or other attributes. The NAS operating system keeps track of the location of files on the disk volume and converts client file I/O into block-level I/O to retrieve data. The process of handling I/Os in a NAS environment is as follows:

1. The requestor (client) packages an I/O request into TCP/IP and forwards it through the network stack. The NAS device receives this request from the network.
2. The NAS device converts the I/O request into an appropriate physical storage request, which is a block-level I/O, and then performs the operation on the physical storage.
3. When the NAS device receives data from the storage, it processes and repackages the data into an appropriate file protocol response.
4. The NAS device packages this response into TCP/IP again and forwards it to the client through the network.

Figure 7-4 illustrates this process.

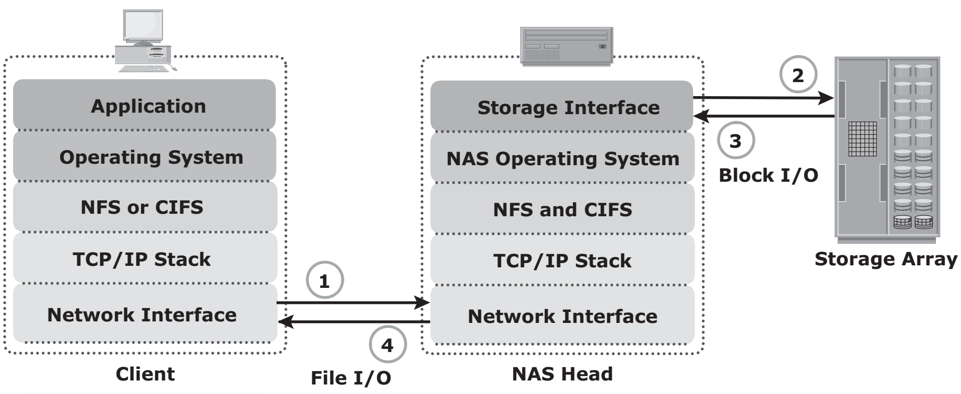


Figure 7-4: NAS I/O operation

7.6 NAS Implementations

Three common NAS implementations are unified, gateway, and scale-out. The *unified* NAS consolidates NAS-based and SAN-based data access within a unified storage platform and provides a unified management interface for managing both the environments.

In a *gateway* implementation, the NAS device uses external storage to store and retrieve data, and unlike unified storage, there are separate administrative tasks for the NAS device and storage.

The *scale-out* NAS implementation pools multiple nodes together in a cluster. A node may consist of either the NAS head or storage or both. The cluster performs the NAS operation as a single entity.

7.6.1 Unified NAS

Unified NAS performs file serving and storing of file data, along with providing access to block-level data. It supports both CIFS and NFS protocols for file access and iSCSI and FC protocols for block level access. Due to consolidation of NAS-based and SAN-based access on a single storage platform, unified NAS reduces an organization's infrastructure and management costs.

A unified NAS contains one or more NAS heads and storage in a single system. NAS heads are connected to the storage controllers (SCs), which provide access to the storage. These storage controllers also provide connectivity to iSCSI and FC hosts. The storage may consist of different drive types, such as SAS, ATA, FC, and flash drives, to meet different workload requirements.

7.6.2 Unified NAS Connectivity

Each NAS head in a unified NAS has front-end Ethernet ports, which connect to the IP network. The front-end ports provide connectivity to the clients and service the file I/O requests. Each NAS head has back-end ports, to provide connectivity to the storage controllers.

iSCSI and FC ports on a storage controller enable hosts to access the storage directly or through a storage network at the block level. Figure 7-5 illustrates an example of unified NAS connectivity.

7.6.3 Gateway NAS

A gateway NAS device consists of one or more NAS heads and uses external and independently managed storage. Similar to unified NAS, the storage is shared with other applications that use block-level I/O. Management functions in this type of solution are more complex than those in a unified NAS environment because there are separate administrative tasks for the NAS head and the storage. A gateway solution can use the FC infrastructure, such as switches and directors for accessing SAN-attached storage arrays or direct-attached storage arrays.

The gateway NAS is more scalable compared to unified NAS because NAS heads and storage arrays can be independently scaled up when required.

For example, NAS heads can be added to scale up the NAS device performance. When the storage limit is reached, it can scale up, adding capacity on the SAN, independent of NAS heads. Similar to a unified NAS, a gateway NAS also enables high utilization of storage capacity by sharing it with the SAN environment.

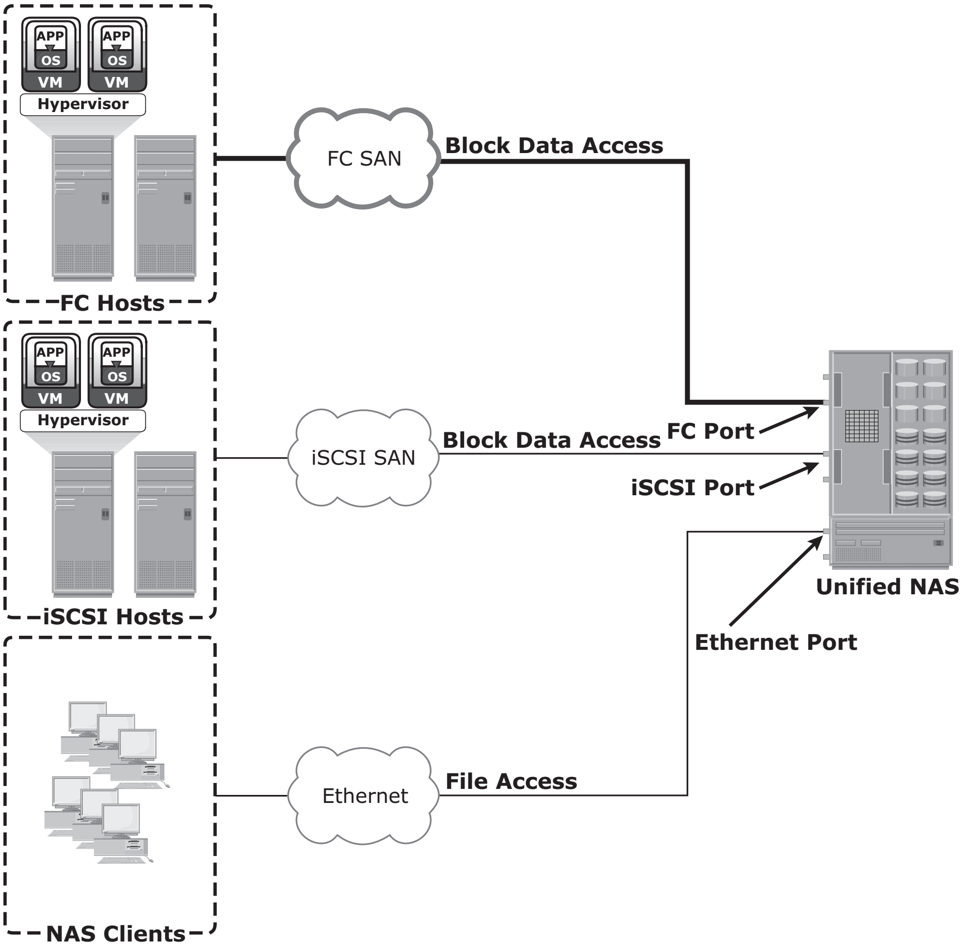


Figure 7-5: Unified NAS connectivity

7.6.4 Gateway NAS Connectivity

In a gateway solution, the front-end connectivity is similar to that in a unified storage solution. Communication between the NAS gateway and the storage system in a gateway solution is achieved through a traditional FC SAN. To deploy a gateway NAS solution, factors, such as multiple paths for data, redundant fabrics, and load distribution, must be considered. Figure 7-6 illustrates an example of gateway NAS connectivity.

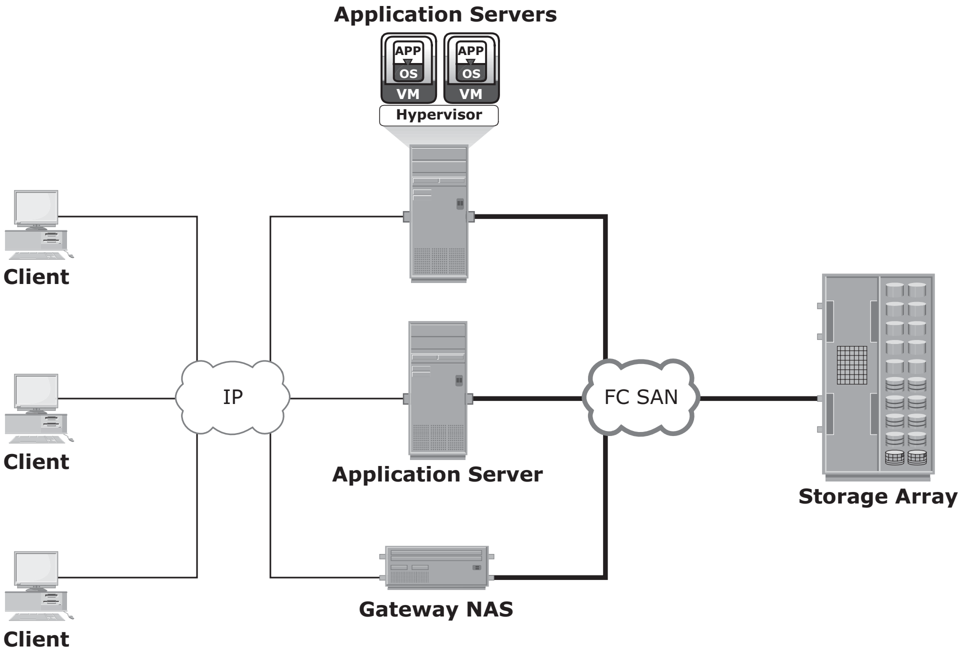


Figure 7-6: Gateway NAS connectivity

Implementation of both unified and gateway solutions requires analysis of the SAN environment. This analysis is required to determine the feasibility of combining the NAS workload with the SAN workload. Analyze the SAN to determine whether the workload is primarily read or write, and if it is random or sequential. Also determine the predominant I/O size in use. Typically, NAS workloads are random with small I/O sizes. Introducing sequential workload with random workloads can be disruptive to the sequential workload. Therefore, it is recommended to separate the NAS and SAN disks. Also, determine whether the NAS workload performs adequately with the configured cache in the storage system.

7.6.5 Scale-Out NAS

Both unified and gateway NAS implementations provide the capability to scale-up their resources based on data growth and rise in performance requirements. Scaling up these NAS devices involves adding CPUs, memory, and storage to the NAS device. Scalability is limited by the capacity of the NAS device to house and use additional NAS heads and storage.

Scale-out NAS enables grouping multiple nodes together to construct a clustered NAS system. A scale-out NAS provides the capability to scale its resources by simply adding nodes to a clustered NAS architecture. The cluster works as a single NAS device and is managed centrally. Nodes can be added to the cluster, when more performance or more capacity is needed, without causing any downtime. Scale-out NAS provides the flexibility to use many nodes of moderate performance and availability characteristics to produce a total system that has better aggregate performance and availability. It also provides ease of use, low cost, and theoretically unlimited scalability.

Scale-out NAS creates a single file system that runs on all nodes in the cluster. All information is shared among nodes, so the entire file system is accessible by clients connecting to any node in the cluster. Scale-out NAS stripes data across all nodes in a cluster along with mirror or parity protection. As data is sent from clients to the cluster, the data is divided and allocated to different nodes in parallel. When a client sends a request to read a file, the scale-out NAS retrieves the appropriate blocks from multiple nodes, recombines the blocks into a file, and presents the file to the client. As nodes are added, the file system grows dynamically and data is evenly distributed to every node. Each node added to the cluster increases the aggregate storage, memory, CPU, and network capacity. Hence, cluster performance also increases.

Scale-out NAS is suitable to solve the "Big Data" challenges that enterprises and customers face today. It provides the capability to manage and store large, high-growth data in a single place with the flexibility to meet a broad range of performance requirements.

7.6.6 Scale-Out NAS Connectivity

Scale-out NAS clusters use separate internal and external networks for back-end and front-end connectivity, respectively. An internal network provides connections for intracluster communication, and an external network connection enables clients to access and share file data. Each node in the cluster connects to the internal network. The internal network offers high throughput and low latency and uses high-speed networking technology, such as InfiniBand or Gigabit Ethernet. To enable clients to access a node, the node must be connected to the external Ethernet network. Redundant internal or external networks may be used for high availability. Figure 7-7 illustrates an example of scale-out NAS connectivity.

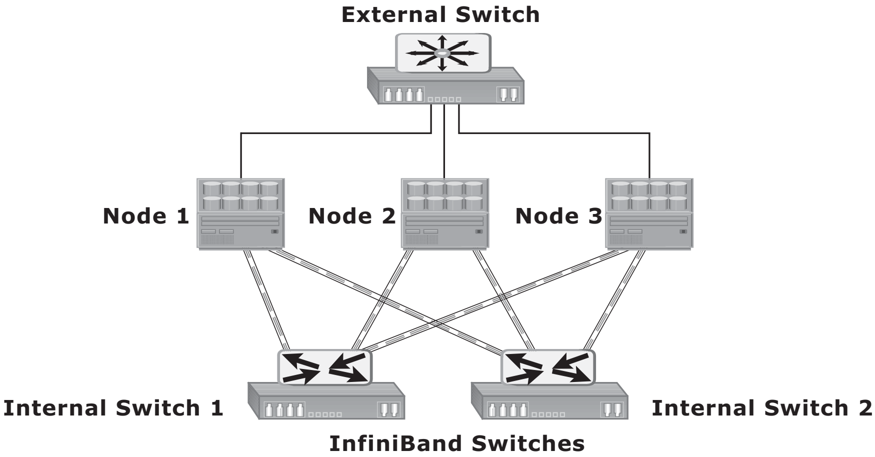


Figure 7-7: Scale-out NAS with dual internal and single external networks

INFINIBAND

InfiniBand is a networking technology that provides a low-latency, high-bandwidth communication link between hosts and peripherals. It provides serial connection and is often used for inter-server communications in high-performance computing environments. InfiniBand enables remote direct memory access (RDMA) that enables a device (host or peripheral) to access data directly from the memory of a remote device. InfiniBand also enables a single physical link to carry multiple channels of data simultaneously using a multiplexing technique. The InfiniBand networking infrastructure consists of host channel adapters (HCAs), target channel adapters (TCAs), and InfiniBand switches. HCAs are located within hosts. HCAs provide the mechanism to connect CPUs and memory of the hosts to the InfiniBand network. Similarly, TCAs enable storage and other peripheral devices to connect to the InfiniBand network. InfiniBand switches provide connectivity among HCAs and TCAs.

7.7 NAS File-Sharing Protocols

Most NAS devices support multiple file-service protocols to handle file I/O requests to a remote file system. As discussed earlier, NFS and CIFS are the common protocols for file sharing. NAS devices enable users to share file data across different operating environments and provide a means for users to migrate transparently from one operating system to another.

7.7.1 NFS

NFS is a client-server protocol for file sharing that is commonly used on UNIX systems. NFS was originally based on the connectionless *User Datagram Protocol* (UDP). It uses a machine-independent model to represent user data. It also uses Remote Procedure Call (RPC) as a method of inter-process communication between two computers. The NFS protocol provides a set of RPCs to access a remote file system for the following operations:

- Searching files and directories
- Opening, reading, writing to, and closing a file
- Changing file attributes
- Modifying file links and directories

NFS creates a connection between the client and the remote system to transfer data. NFS (NFSv3 and earlier) is a *stateless protocol*, which means that it does not maintain any kind of table to store information about open files and associated pointers. Therefore, each call provides a full set of arguments to access files on the server. These arguments include a file handle reference to the file, a particular position to read or write, and the versions of NFS.

Currently, three versions of NFS are in use:

- **NFS version 2 (NFSv2):** Uses UDP to provide a stateless network connection between a client and a server. Features, such as locking, are handled outside the protocol.
- **NFS version 3 (NFSv3):** The most commonly used version, which uses UDP or TCP, and is based on the stateless protocol design. It includes some new features, such as a 64-bit file size, asynchronous writes, and additional file attributes to reduce refetching.
- **NFS version 4 (NFSv4):** Uses TCP and is based on a stateful protocol design. It offers enhanced security. The latest NFS version 4.1 is the enhancement of NFSv4 and includes some new features, such as session model, parallel NFS (pNFS), and data retention.

PNFS AND MPFS

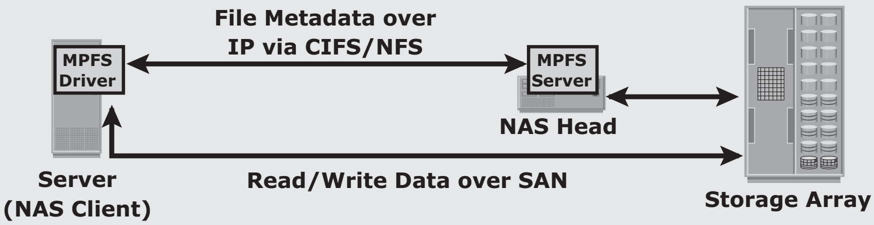
pNFS, as part of NFSv4.1, separates the file system protocol processing into two parts: metadata processing and data processing. The metadata includes information about a file system object, such as its name, location within the namespace, owner, access control list (ACL), and other attributes. The pNFS server, also called a metadata server,

(Continued)

PNFS AND MPFS *(continued)*

does the metadata processing and is kept out of the data path. pNFS clients send the metadata information to the pNFS server. The pNFS clients access storage devices directly using multiple parallel data paths. The pNFS client uses a storage network protocol, such as iSCSI or FC, to perform I/O to storage devices. The pNFS clients get information about the storage devices from the metadata server. Because the pNFS server is relieved of data processing and pNFS clients can access the storage devices directly using parallel paths, the pNFS mechanism significantly improves the pNFS client performance.

The EMC-patented Multi-Path File System (MPFS) protocol works similar to pNFS. The MPFS driver software, installed at the NAS clients, sends the file's metadata to the NAS device (MPFS server) via the IP network. The MPFS driver obtains information about the location of the data from the NAS device over the IP network. After knowing the data location, the MPFS driver communicates directly to the storage devices and enables the NAS clients to access the data over SAN. The following Figure shows the MPFS architecture that provides different paths for transferring a file's metadata and data.



7.7.2 CIFS

CIFS is a client-server application protocol that enables client programs to make requests for files and services on remote computers over TCP/IP. It is a public, or open, variation of Server Message Block (SMB) protocol.

The CIFS protocol enables remote clients to gain access to files on a server. CIFS enables file sharing with other clients by using special locks. Filenames in CIFS are encoded using unicode characters. CIFS provides the following features to ensure data integrity:

- It uses file and record locking to prevent users from overwriting the work of another user on a file or a record.
- It supports fault tolerance and can automatically restore connections and reopen files that were open prior to an interruption. The fault tolerance features of CIFS depend on whether an application is written to take advantage of these features. Moreover, CIFS is a stateful protocol because the CIFS server maintains connection information regarding every connected

client. If a network failure or CIFS server failure occurs, the client receives a disconnection notification. User disruption is minimized if the application has the embedded intelligence to restore the connection. However, if the embedded intelligence is missing, the user must take steps to reestablish the CIFS connection.

Users refer to remote file systems with an easy-to-use file-naming scheme: `\\server\share` or `\\servername.domain.suffix\share`.

The file naming scheme in an NFS environment is:
`Server:/export` or `Server.domain.suffix:/export`.

7.8 Factors Affecting NAS Performance

NAS uses IP network; therefore, bandwidth and latency issues associated with IP affect NAS performance. Network congestion is one of the most significant sources of latency (Figure 7-8) in a NAS environment. Other factors that affect NAS performance at different levels follow:

1. **Number of hops:** A large number of hops can increase latency because IP processing is required at each hop, adding to the delay caused at the router.
2. **Authentication with a directory service such as Active Directory or NIS:** The authentication service must be available on the network with enough resources to accommodate the authentication load. Otherwise, a large number of authentication requests can increase latency.
3. **Retransmission:** Link errors and buffer overflows can result in retransmission. This causes packets that have not reached the specified destination to be re-sent. Care must be taken to match both speed and duplex settings on the network devices and the NAS heads. Improper configuration might result in errors and retransmission, adding to latency.
4. **Overutilized routers and switches:** The amount of time that an overutilized device in a network takes to respond is always more than the response time of an optimally utilized or underutilized device. Network administrators can view utilization statistics to determine the optimum utilization of switches and routers in a network. Additional devices should be added if the current devices are overutilized.

5. **File system lookup and metadata requests:** NAS clients access files on NAS devices. The processing required to reach the appropriate file or directory can cause delays. Sometimes a delay is caused by deep directory structures and can be resolved by flattening the directory structure. Poor file system layout and an overutilized disk system can also degrade performance.
6. **Over utilized NAS devices:** Clients accessing multiple files can cause high utilization levels on a NAS device, which can be determined by viewing utilization statistics. High memory, CPU, or disk subsystem utilization levels can be caused by a poor file system structure or insufficient resources in a storage subsystem.
7. **Over utilized clients:** The client accessing CIFS or NFS data might also be over utilized. An overutilized client requires a longer time to process the requests and responses. Specific performance-monitoring tools are available for various operating systems to help determine the utilization of client resources.

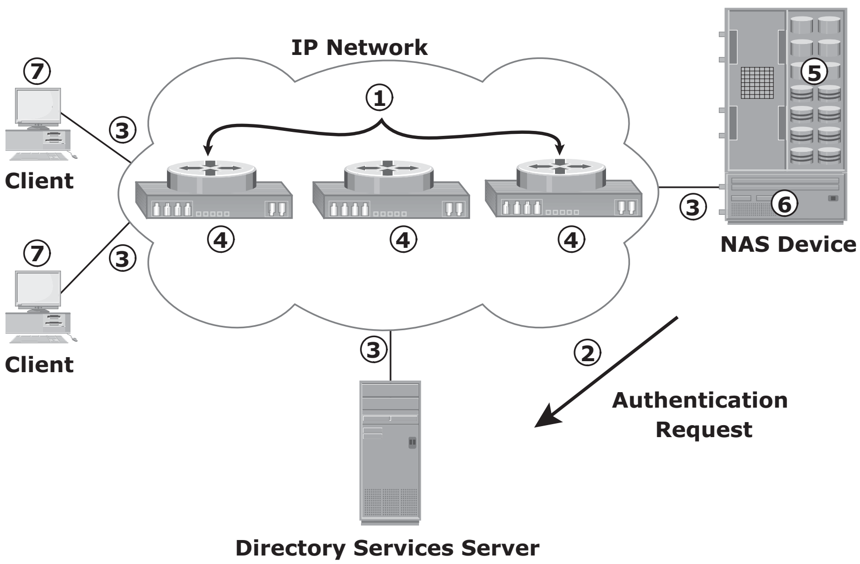


Figure 7-8: Causes of latency

Configuring *virtual LANs* (VLANs), setting proper Maximum Transmission Unit (MTU) and TCP window sizes, and link aggregation can improve NAS performance. Link aggregation and redundant network configurations also ensure high availability.

A VLAN is a logical segment of a switched network or logical grouping of end devices connected to different physical networks. An end device could be a client or a NAS device. The segmentation or grouping can be done based on business functions, project teams, or applications. VLAN is a Layer 2 (data link layer) construct and works similar to a physical LAN. A network switch can be logically divided among multiple VLANs, enabling better utilization of the switch and reducing the overall cost of deploying a network infrastructure.

The broadcast traffic on one VLAN is not transmitted outside that VLAN, which substantially reduces the broadcast overhead, makes bandwidth available for applications, and reduces the network's vulnerability to broadcast storms.

VLANs also provide enhanced security by restricting user access, flagging network intrusions, and controlling the size and composition of the broadcast domain. The *MTU* setting determines the size of the largest packet that can be transmitted without data fragmentation. *Path maximum transmission unit discovery* is the process of discovering the maximum size of a packet that can be sent across a network without fragmentation. The default MTU setting for an Ethernet interface card is 1,500 bytes. A feature called *jumbo frames* sends, receives, or transports Ethernet frames with an MTU of more than 1,500 bytes. The most common deployments of jumbo frames have an MTU of 9,000 bytes. However not all vendors use the same MTU size for jumbo frames. Servers send and receive larger frames more efficiently than smaller ones in heavy network traffic conditions. Jumbo frames ensure increased efficiency because it takes fewer, larger frames to transfer the same amount of data. Larger packets also reduce the amount of raw network bandwidth being consumed for the same amount of payload. Larger frames also help to smooth sudden I/O bursts.

The *TCP window size* is the maximum amount of data that can be sent at any time for a connection. For example, if a pair of hosts is talking over a TCP connection that has a TCP window size of 64 KB, the sender can send only 64 KB of data and must then wait for an acknowledgment from the receiver. If the receiver acknowledges that all the data has been received, then the sender is free to send another 64 KB of data. If the sender receives an acknowledgment from the receiver that only the first 32 KB of data has been received, which can happen only if another 32 KB of data is in transit or was lost, the sender can send only another 32 KB of data because the transmission cannot have more than 64 KB of unacknowledged data outstanding.

In theory, the TCP window size should be set to the product of the available bandwidth of the network and the round-trip time of data sent over the network.

For example, if a network has a bandwidth of 100 Mbps and the round-trip time is 5 milliseconds, the TCP window should be as follows:

$$100 \text{ Mb/s} \times .005 \text{ seconds} = 524{,}288 \text{ bits or } 65{,}536 \text{ bytes}$$

The size of the TCP window field that controls the flow of data is between 2 bytes and 65,535 bytes.

Link aggregation is the process of combining two or more network interfaces into a logical network interface, enabling higher throughput, load sharing or load balancing, transparent path failover, and scalability. Due to link aggregation, multiple active Ethernet connections to the same switch appear as one link. If a connection or a port in the aggregation is lost, then all the network traffic on that link is redistributed across the remaining active connections.

7.9 File-Level Virtualization

File-level virtualization eliminates the dependencies between the data accessed at the file level and the location where the files are physically stored. Implementation of file-level virtualization is common in NAS or file-server environments. It provides non-disruptive file mobility to optimize storage utilization.

Before virtualization, each host knows exactly where its file resources are located. This environment leads to underutilized storage resources and capacity problems because files are bound to a specific NAS device or file server. It may be required to move the files from one server to another because of performance reasons or when the file server fills up. Moving files across the environment is not easy and may make files inaccessible during file movement. Moreover, hosts and applications need to be reconfigured to access the file at the new location. This makes it difficult for storage administrators to improve storage efficiency while maintaining the required service level.

File-level virtualization simplifies file mobility. It provides user or application independence from the location where the files are stored. File-level virtualization creates a logical pool of storage, enabling users to use a logical path, rather than a physical path, to access files. File-level virtualization facilitates the movement of files across the online file servers or NAS devices. This means that while the files are being moved, clients can access their files nondisruptively. Clients can also read their files from the old location and write them back to the new location without realizing that the physical location has changed. A global namespace is used to map the logical path of a file to the physical path names.

Figure 7-9 illustrates a file-serving environment before and after the implementation of file-level virtualization.

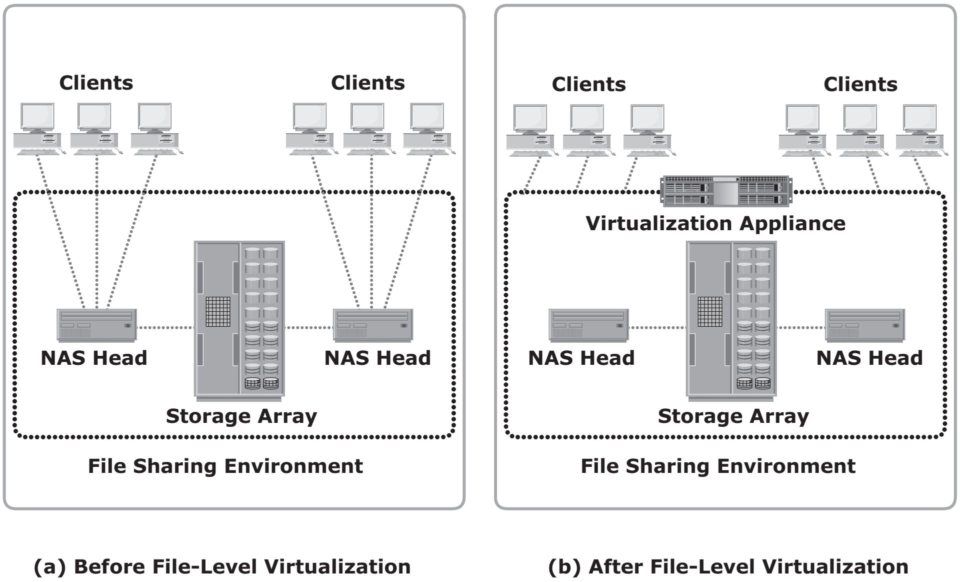


Figure 7-9: File-serving environment before and after file-level virtualization

7.10 Concepts in Practice: EMC Isilon and EMC VNX Gateway

EMC Isilon is the scale-out NAS solution. Isilon offers high scalability of both performance and storage capacity. It provides the capability to address big-data challenges.

The VNX Gateway, a member of the EMC VNX family, provides a gateway NAS solution. It provides multiprotocol file access, dynamic expansion of file systems, high availability, and high performance.

For more information on EMC Isilon and VNX Gateway, visit `www.emc.com`.

7.10.1 EMC Isilon

Isilon has a specialized operating system called OneFS that enables the scale-out NAS architecture. OneFS combines the three layers of traditional storage architectures — file system, volume manager, and RAID — into one unified software layer, creating a single file system that spans across all nodes in an Isilon cluster. OneFS enables data protection and automated data balancing. It provides the ability to seamlessly add storage and other resources without system downtime. With OneFS, throughput scales linearly with the number of nodes in a cluster.

OneFS enables different node types to be mixed in a single cluster through the addition of the SmartPools application software. SmartPools enables deploying a single file system to span multiple nodes that have different performance characteristics and capacities. Isilon offers different types of nodes, such as the X-Series, S-Series, NL-Series, and Accelerator. These nodes have different prices, performance levels, and storage capabilities. Each type of node is optimized for handling a specific type of workload.

OneFS enables the storage system administrator to specify the access pattern (random, concurrent, or sequential) on a per-file or per-directory basis. This unique capability enables OneFS to tailor data layout decisions, cache-retention policies, and data prefetch policies to maximize performance of individual workflows.

OneFS constantly monitors the health of all files and disks within a cluster, and if components are at risk, the file system automatically flags the problem components for replacement and transparently relocates those files to healthy components. OneFS also ensures data integrity if the file system has an unexpected failure during a write operation.

When a new storage node is added, the Autobalance feature of OneFS automatically moves data onto this new node via the Infiniband based internal network. This automatic rebalancing ensures that the new node does not become a hot spot for new data. The Autobalance feature is transparent to the clients and can be adjusted to minimize the impact on high-performance workloads.

OneFS includes a core technology, called FlexProtect, to provide data protection. FlexProtect provides protection for up to four simultaneous failures of either nodes or individual drives per stripe. FlexProtect ensures minimal data reconstruction time if a failure occurs. FlexProtect provides file-specific protection capabilities. Different protection levels can be assigned to individual files, directories, or to portions of a file system. These protection levels are aligned based on the importance of data and workflow.

7.10.2 EMC VNX Gateway

The VNX Series Gateway contains one or more NAS heads, called X-Blades, that access external storage arrays, such as Symmetrix, block-based VNX, or CLARiiON storage array, via SAN. X-Blades run the VNX operating environment that is optimized for high-performance and multiprotocol network file system access. Each X-Blade consists of processors, redundant data paths, power supplies, Gigabit Ethernet, and 10-Gigabit Ethernet optical ports. All the X-Blades in a VNX gateway system are managed by Control Station, which provides a single point for configuring VNX Gateway. The VNX Gateway supports both pNFS and EMC patented Multi-Path File System (MPFS) protocols, which further improves the VNX Gateway performance.

VNX Series Gateway offers two models: VG2 and VG8. VG8 supports up to eight X-Blades, whereas VG2 supports up to two. X-Blades may be configured as either primary or standby. A primary X-Blade is the operating NAS head, whereas a standby X-Blade becomes operational if the primary X-Blade fails. The Control Station handles an X-Blade failover. The Control Station also provides other high-availability features, such as fault monitoring, fault reporting, call home, and remote diagnostics.

Summary

Decisions for choosing an appropriate storage infrastructure are based on maintaining the balance between cost and performance. Organizations look for the performance and scalability of SAN combined with the ease of use and lower total cost of ownership of NAS solutions. Both SAN and NAS have enjoyed unique advantages in enterprises, and advances in IP technology have scaled NAS solutions to meet the demands of performance-sensitive applications. With the advancement of storage networking technology, both SAN-based and NAS-based accesses have converged to a single platform.

Although NAS invariably imposes higher protocol overhead, it tends to be the most efficient for file-sharing tasks. NAS performance has significantly improved with the emergence of MPFS and pNFS protocols. These protocols use SAN speed to provide access to file data. They also offload the file-data processing load from the NAS device. NAS can also provide file-level access control to its clients. Organizations can also deploy NAS solutions for their database applications. Scale-out NAS fulfills the need for big-data performance and big-data capacity. Applications generating big data are optimized and more easily managed by using a single-expandable file system. File-level virtualization provides the flexibility to move files across NAS devices without disrupting the access to the files.

NAS devices impose additional latency to the client traffic while converting file I/O to block I/Os and vice versa. Also, nested directory structure and management of permission for individual files and directories add overhead to NAS. The overhead increases as the NAS file system grows. Hence, NAS clients are limited by the performance of the NAS device. Although the use of pNFS and MPFS protocols has considerably improved the NAS performance, these protocols might pose some security challenges. Object-based storage, detailed in the following chapter, addresses the performance and security challenges in the file-serving environment. Unified storage, also detailed in the following chapter, provides a single-storage platform for accessing files, blocks, and objects simultaneously. Unified storage brings ease of management and eliminates the additional cost of deploying separate storage systems for storing file-, block-, and object-based data.

EXERCISES

1. SAN is configured for a backup–to-disk environment, and the storage configuration has additional capacity available. Can you have a NAS gateway configuration use this SAN-attached storage? Discuss the implications of sharing the backup-to-disk SAN environment with NAS.
2. Explain how the performance of NAS can be affected if the TCP window size at the sender and receiver are not synchronized.
3. How does the use of jumbo frames affect the NAS performance?
4. Research the file access and sharing features of pNFS.
5. A NAS implementation configured jumbo frames on the NAS head with 9,000 as its MTU. However, the implementers did not see any performance improvement and actually experienced performance degradation. What could be the cause? Research the end-to-end jumbo frame support requirements in a network.
6. How does file-level virtualization ensure nondisruptive file mobility?

Chapter 8
Object-Based and Unified Storage

KEY CONCEPTS

- Object-Based Storage
- Content Addressed Storage
- Unified Storage

Recent studies have shown that more than 90 percent of data generated is unstructured. This growth of unstructured data has posed new challenges to IT administrators and storage managers. With this growth, traditional NAS, which is a dominant solution for storing unstructured data, has become inefficient. Data growth adds high overhead to the network-attached storage (NAS) in terms of managing a large number of permissions and nested directories. In an enterprise environment, NAS also manages large amounts of metadata generated by hosts, storage systems, and individual applications. Typically this metadata is stored as part of the file and distributed throughout the environment. This adds to the complexity and latency in searching and retrieving files. These challenges demand a smarter approach to manage unstructured data based on its content rather than metadata about its name, location, and so on. *Object-based storage* is a way to store file data in the form of objects based on its content and other attributes rather than the name and location.

Due to varied application requirements, organizations have been deploying storage area networks (SANs), NAS, and object-based storage devices (OSDs) in their data centers. Deploying these disparate storage solutions adds management complexity, cost and environmental overhead. An ideal solution would be to have an integrated storage solution that supports block, file, and object access. Unified storage has emerged as a solution that consolidates block, file, and object-based access within one unified platform. It supports multiple protocols for data access and can be managed using a single management interface.

This chapter details object-based storage, its components, and operation. It also details *content addressed storage* (CAS), a special type of OSD. Further, this chapter covers the components and data access method in unified storage.